Elementary Analysis II

— ***EYNTKA*** **Series** —

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Introduction

This book lifts analysis off the real line. The single-variable theory developed in *Elementary Analysis* [1] is stated for functions on intervals, and almost every argument in it turns on two features of $\mathbb{R}$: that it carries a distance, and that it is complete with respect to that distance. Neither feature is special to the real line. Once they are isolated as axioms, the same arguments run on spaces of sequences, spaces of functions, and spaces of far stranger objects, and they run without change.

That abstraction is the subject here. The book develops metric spaces and completeness, the contraction principle and its fixed points, the several inequivalent-looking notions of compactness that coincide for metric spaces, and uniform convergence. It then turns to the space $C(K)$ of continuous functions on a compact space, where the payoff comes: Stone-Weierstrass, which says polynomials are enough; Arzela-Ascoli, which says equicontinuity and boundedness are enough to extract convergent subsequences; and the Baire category theorem, which manufactures objects nobody can write down.

These are the theorems the later analysis volumes use rather than reprove, and this book is the series' single home for them.

Part I

Metric Spaces

Single-variable analysis proves its theorems about $\mathbb{R}$, but it rarely uses more of $\mathbb{R}$ than two facts: that any two points have a distance between them, and that a sequence whose terms crowd together has somewhere to converge to. This Part promotes those two facts to axioms and asks how much of the theory survives.

The answer is: nearly all of the vocabulary, and rather less of the substance. Chapter 1 installs the vocabulary and finds that open sets, closure, convergence and continuity all transfer without a word of change, and that the three competing definitions of continuity stay equivalent. Chapter 2 isolates the second fact as completeness and shows it is a strictly metric notion, not a topological one. **??** extracts the sharpest single consequence of completeness, the contraction principle, which converts a shrinking map into an existence-and-uniqueness theorem. **??** confronts the one notion that does not transfer intact: in $\mathbb{R}^n$ compactness means closed and bounded, in a general metric space it does not, and sorting out what it does mean occupies the chapter.

Throughout, four spaces recur and are worth becoming familiar with: $\mathbb{R}^n$, the space $C[0,1]$ of continuous functions under the distance $\sup_x |f(x) - g(x)|$, the sequence space ℓ^2, and the Cantor set. Each new definition is worth testing against all four.

1

Metric Spaces and Their Topology

The ϵ-δ definition of continuity mentions the real numbers only through the quantity $|x - y|$. So does the definition of a convergent sequence, of an open interval, of a limit point, and of a Cauchy sequence. Replacing $|x - y|$ by an abstract distance $d(x, y)$ therefore changes nothing in those definitions, and this chapter carries out the replacement and checks what it costs in the theorems.

The four sections install the vocabulary in order. Section 1.1 gives the axioms and a stock of examples, several of which are spaces whose points are functions or sequences rather than numbers. Section 1.2 builds open and closed sets, closure and convergence out of the distance alone. Section 1.3 proves that the ϵ-δ, sequential and preimage-of-open definitions of continuity remain equivalent, and separates continuity from its uniform and Lipschitz strengthenings. Section 1.4 handles the three constructions every later chapter uses without comment: passing to a subspace, forming a product, and replacing a metric by an equivalent one.

1.1 Metrics and Examples

Definition 1.1.1: Metric Space

A *metric* on a set X is a function $d\colon X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

(i) $d(x, y) = 0$ if and only if $x = y$;

(ii) $d(x, y) = d(y, x)$;

(iii) $d(x, z) \leq d(x, y) + d(y, z)$.

The pair (X, d) is a *metric space*, and the elements of X are its *points*. Where the metric is clear from context, X alone names the space.

Nothing in the definition requires that distances be positive, because they cannot help being so.

Lemma 1.1.2: Distances Are Nonnegative

If d is a metric on X then $d(x, y) \geq 0$ for all $x, y \in X$.

***Proof*:**
Apply the triangle inequality with $z = x$ and then symmetry:

$$0 = d(x, x) \leq d(x, y) + d(y, x) = 2d(x, y)$$

so $d(x, y) \geq 0$, as we sought to show.

The first axiom is the one that is easy to weaken by accident, and the cost of weakening it is severe. Requiring only $d(x, x) = 0$, rather than the equivalence, admits the function $d \equiv 0$ on any set whatsoever: it is symmetric, it satisfies the triangle inequality, and every point is at distance zero from itself. Such a function is a *pseudometric*. Under it every sequence converges to every point at once, so limits are not unique (proposition 1.2.6), and the entire theory of section 1.2 collapses. The "only if" is what forces distinct points to be separated by some positive distance, and every result in this book uses it.

The examples below are the stock this book draws on. The first two are the familiar ones; the rest are spaces whose points are not numbers, and they are the reason the abstraction is worth making.

Example 1.1.3: The Standard Metrics on $\mathbb{R}^n$

On $\mathbb{R}^n$ define, for $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$,

$$d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i| \qquad d_2(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i|^2 \right)^{1/2} \qquad d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$$

the *taxicab*, *Euclidean* and *sup* metrics. For d_1 and d_∞ the axioms are immediate from the corresponding facts about $|\cdot|$ on $\mathbb{R}$: symmetry and the vanishing condition hold coordinatewise, and for the triangle inequality,

$$|x_i - z_i| \leq |x_i - y_i| + |y_i - z_i|$$

summing gives the claim for d_1, while taking the maximum over i of the right-hand side, which dominates each summand separately, gives it for d_∞. The triangle inequality for d_2 is the case $p = 2$, n finite of theorem 1.1.8 below. When $n = 1$ all three agree with $d(x, y) = |x - y|$, the metric that makes this book a generalization of the last one.

Example 1.1.4: The Sup Metric on $C[a, b]$

Let $C[a, b]$ denote the set of continuous functions $f\colon [a, b] \to \mathbb{R}$, and set

$$d_\infty(f, g) = \sup_{x \in [a,b]} |f(x) - g(x)|$$

The supremum is finite because a continuous function on a closed bounded interval is bounded [1], so $f - g$ is bounded. If $d_\infty(f, g) = 0$ then $f(x) = g(x)$ at every x, so $f = g$ as functions; symmetry is clear; and for the triangle inequality, for each fixed x,

$$|f(x) - h(x)| \le |f(x) - g(x)| + |g(x) - h(x)| \le d_\infty(f, g) + d_\infty(g, h)$$

so the right-hand side is an upper bound for the set whose supremum is $d_\infty(f, h)$.

A point of this space is an entire function, and two functions are close when their graphs lie in a thin horizontal band around one another. This is the space **??** is about, and d_∞ is the distance in which uniform convergence is ordinary convergence (**??**).

Example 1.1.5: The Discrete Metric

On any set X define $d(x, y) = 0$ if $x = y$ and $d(x, y) = 1$ otherwise. The first two axioms are immediate. For the triangle inequality, if $x = z$ the left side is 0; otherwise the left side is 1 and at least one of $d(x, y)$, $d(y, z)$ must be 1, since y cannot equal both x and z.

Every set carries a metric, then, so the existence of a metric says nothing about a set on its own. The discrete metric is the standard source of counterexamples: it is the extreme case in which no two distinct points are near each other, and it recurs below whenever a plausible-looking implication needs refuting.

The next example needs an inequality, and the inequality needs a smaller one. All three are stated for sequences, and specialize to finite sums by padding with zeros.

Proposition 1.1.6: Young's Inequality

Let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then for all real $a, b \ge 0$,

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}$$

***Proof*:**

If $a = 0$ or $b = 0$ the left side is 0 and the right side is nonnegative. Assume both are positive. The logarithm is concave, so for the weights $\frac{1}{p}$ and $\frac{1}{q}$, which are positive and sum to 1,

$$\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \ge \frac{1}{p}\log(a^p) + \frac{1}{q}\log(b^q) = \log a + \log b = \log(ab)$$

Exponentiating, and using that the exponential is increasing, gives the claim, as we sought to show.

Theorem 1.1.7: Hölder's Inequality

Let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$, and let $x = (x_k)_{k\geq 1}$ and $y = (y_k)_{k\geq 1}$ be real sequences. Then

$$\sum_{k=1}^{\infty} |x_k y_k| \leq \left(\sum_{k=1}^{\infty} |x_k|^p\right)^{1/p} \left(\sum_{k=1}^{\infty} |y_k|^q\right)^{1/q}$$

whenever the two factors on the right are finite.

***Proof*:**

Write $A = \left(\sum_k |x_k|^p\right)^{1/p}$ and $B = \left(\sum_k |y_k|^q\right)^{1/q}$. If $A = 0$ then every x_k is 0 and both sides vanish; likewise if $B = 0$. So assume $A, B > 0$ and set

$$a_k = \frac{|x_k|}{A} \qquad b_k = \frac{|y_k|}{B}$$

which satisfy $\sum_k a_k^p = 1$ and $\sum_k b_k^q = 1$. Applying proposition 1.1.6 termwise and summing,

$$\sum_{k=1}^{\infty} a_k b_k \leq \sum_{k=1}^{\infty} \left(\frac{a_k^p}{p} + \frac{b_k^q}{q}\right) = \frac{1}{p} + \frac{1}{q} = 1$$

Multiplying through by AB recovers the stated inequality, completing the proof.

Theorem 1.1.8: Minkowski's Inequality

Let $1 \leq p < \infty$ and let x, y be real sequences with $\sum_k |x_k|^p$ and $\sum_k |y_k|^p$ both finite. Then $\sum_k |x_k + y_k|^p$ is finite and

$$\left(\sum_{k=1}^{\infty} |x_k + y_k|^p\right)^{1/p} \leq \left(\sum_{k=1}^{\infty} |x_k|^p\right)^{1/p} + \left(\sum_{k=1}^{\infty} |y_k|^p\right)^{1/p}$$

***Proof*:**

Write $\|z\|_p = \left(\sum_k |z_k|^p\right)^{1/p}$ throughout.

Step 1: the left side is finite. For each k, $|x_k + y_k| \leq 2\max(|x_k|, |y_k|)$, so $|x_k + y_k|^p \leq 2^p\left(|x_k|^p + |y_k|^p\right)$, and summing shows $\|x + y\|_p < \infty$.

Step 2: the case $p = 1$. Here the inequality is the triangle inequality for $|\cdot|$ applied termwise and summed.

Step 3: the case $p > 1$. Let $q = \frac{p}{p-1}$, so that $\frac{1}{p} + \frac{1}{q} = 1$ and $(p-1)q = p$. Splitting one factor off

each term,

$$\|x+y\|_p^p = \sum_k |x_k+y_k|\,|x_k+y_k|^{p-1} \le \sum_k |x_k|\,|x_k+y_k|^{p-1} + \sum_k |y_k|\,|x_k+y_k|^{p-1}$$

using $|x_k+y_k| \le |x_k|+|y_k|$. Apply theorem 1.1.7 to each sum on the right, pairing (x_k) or (y_k) with $\big(|x_k+y_k|^{p-1}\big)$:

$$\|x+y\|_p^p \le \big(\|x\|_p+\|y\|_p\big)\left(\sum_k |x_k+y_k|^{(p-1)q}\right)^{1/q} = \big(\|x\|_p+\|y\|_p\big)\|x+y\|_p^{p/q}$$

where the last equality uses $(p-1)q = p$. If $\|x+y\|_p = 0$ the claimed inequality is immediate, so assume it is positive and divide by $\|x+y\|_p^{p/q}$, which Step 1 guarantees is a finite positive number. Since $p - \frac{p}{q} = p\big(1-\frac{1}{q}\big) = p\cdot\frac{1}{p} = 1$, the left side becomes $\|x+y\|_p$, completing the proof.

Example 1.1.9: The Sequence Spaces ℓ^p

For $1 \le p < \infty$ let

$$\ell^p = \left\{ x = (x_k)_{k\ge 1} \;\middle|\; x_k \in \mathbb{R} \text{ and } \sum_{k=1}^{\infty} |x_k|^p < \infty \right\}$$

and let ℓ^∞ be the set of bounded real sequences. Define

$$d_p(x,y) = \left(\sum_{k=1}^{\infty} |x_k-y_k|^p\right)^{1/p} \qquad \text{and} \qquad d_\infty(x,y) = \sup_{k\ge 1} |x_k-y_k|$$

Theorem 1.1.8 shows first that $x-y \in \ell^p$ whenever $x, y \in \ell^p$, so d_p is defined and finite, and then, applied to $x-y$ and $y-z$, that

$$d_p(x,z) = \|(x-y)+(y-z)\|_p \le \|x-y\|_p + \|y-z\|_p = d_p(x,y) + d_p(y,z)$$

The other two axioms are immediate, as they are for d_∞, whose triangle inequality follows from the argument of example 1.1.3 with the maximum replaced by a supremum.

Concretely, the sequence $x = (1, \frac{1}{2}, \frac{1}{3}, \dots)$ lies in ℓ^2, since $\sum k^{-2}$ converges, but not in ℓ^1, since the harmonic series diverges, so the inclusion $\ell^1 \subseteq \ell^2$ is strict. Exercise 1.1.1 (3) establishes the whole chain $\ell^p \subsetneq \ell^r$ for $p < r$, with a witness for each strictness. The space ℓ^2 is one of this book's four running examples: it is the smallest space in which the difference between "closed and bounded" and "compact" becomes visible (section 3.1.1).

Every metric so far has been built from a notion of size that adds. The next one is built from divisibility, and it satisfies a triangle inequality much stronger than the one required.

Definition 1.1.10: Ultrametric

A metric d on X is an *ultrametric* if it satisfies the stronger inequality

$$d(x,z) \le \max\big(d(x,y),\, d(y,z)\big) \qquad \text{for all } x, y, z \in X$$

Example 1.1.11: The p-adic Metric on $\mathbb{Q}$

Fix a prime p. Every nonzero rational x factors uniquely as $x = p^v \frac{a}{b}$ with $v \in \mathbb{Z}$ and a, b nonzero integers coprime to p; write $v_p(x) = v$. Define

$$|x|_p = p^{-v_p(x)} \quad \text{for } x \neq 0, \qquad |0|_p = 0, \qquad d_p(x, y) = |x - y|_p$$

A rational is small in this metric when it is divisible by a high power of p. So $|3|_3 = \frac{1}{3}$ and $|\frac{1}{9}|_3 = 9$, and

$$d_3(1, 10) = |-9|_3 = 3^{-2} = \tfrac{1}{9}$$

makes 1 and 10 closer than 1 and 2, whose distance is $|-1|_3 = 1$.

This is an ultrametric. For x, y nonzero with $x + y \neq 0$, writing $x = p^{v_p(x)}u$ and $y = p^{v_p(y)}w$ with u, w having no factor of p, the common factor $p^{\min(v_p(x), v_p(y))}$ can be extracted from $x + y$, so $v_p(x + y) \geq \min(v_p(x), v_p(y))$; negating the exponent reverses the inequality and gives $|x + y|_p \leq \max(|x|_p, |y|_p)$. Applying this to $x - z = (x - y) + (y - z)$ gives the ultrametric inequality, and the ordinary triangle inequality follows since a maximum is at most a sum of nonnegative terms. The arithmetic this metric opens up, and the field $\mathbb{Q}_p$ obtained by completing $\mathbb{Q}$ with respect to it (section 2.4), belong to *Local Field Theory* [2]; here it serves as an example of how little a metric needs to resemble $|x - y|$.

Proposition 1.1.12: Every Triangle in an Ultrametric Space Is Isosceles

Let (X, d) be an ultrametric space and let $x, y, z \in X$. If $d(x, y) \neq d(y, z)$ then $d(x, z) = \max\big(d(x, y), d(y, z)\big)$.

***Proof*:**

By symmetry of the hypothesis assume $d(x, y) < d(y, z)$, so the maximum is $d(y, z)$. The ultrametric inequality gives $d(x, z) \leq \max(d(x, y), d(y, z)) = d(y, z)$ directly. For the reverse, apply the ultrametric inequality to the pair (y, x), (x, z):

$$d(y, z) \leq \max\big(d(y, x),\, d(x, z)\big)$$

The first entry is $d(x, y)$, which is strictly less than $d(y, z)$, so it cannot be the maximum; hence the maximum is $d(x, z)$ and $d(y, z) \leq d(x, z)$. The two inequalities give equality, as we sought to show.

The conclusion is genuinely strange: in $\mathbb{Q}$ with d_3, no three points form a scalene triangle. Ultrametric spaces are the standard warning against reading metric statements through Euclidean pictures, and example 1.1.13 is a second such warning, this time inside the plane itself.

Example 1.1.13: The French Railway Metric

On $\mathbb{R}^2$, write $\|x\|$ for the Euclidean length and fix the origin O. Define

$$d(x, y) = \begin{cases} \|x - y\| & \text{if } x, y, O \text{ are collinear} \\ \|x\| + \|y\| & \text{otherwise} \end{cases}$$

Travel between two points is possible only along lines through the origin, so any journey not already radial must pass through O; the name records a rail network in which every route runs

through the capital.

The first two axioms are clear. For the triangle inequality $d(x,z) \le d(x,y) + d(y,z)$, consider two cases. If x, z, O are collinear and y lies on that same line, all distances involved are Euclidean distances along a line and the inequality is the usual one. If x, z, O are collinear but y is not on their line, then no pair among $\{x,y\}$ and $\{y,z\}$ is radial, since y radial with x would put y on the line through O and x, which is the line through O and z. Hence

$$d(x,y) + d(y,z) = \|x\| + 2\|y\| + \|z\| \ge \|x\| + \|z\| \ge \|x - z\| = d(x,z)$$

the last step being the Euclidean triangle inequality applied to x, O, z. If instead x, z, O are not collinear then $d(x,z) = \|x\| + \|z\|$, and y cannot be radial with both x and z unless $y = O$, in which case $d(x,y) + d(y,z) = \|x\| + \|z\|$ with equality. Otherwise at least one of the two pairs is non-radial and contributes $\|x\| + \|y\|$ or $\|y\| + \|z\|$ in full, while the other contributes at least $\|z\| - \|y\|$ or $\|x\| - \|y\|$ by the Euclidean triangle inequality; adding gives at least $\|x\| + \|z\|$.

Exercise 1.1.1

1. Show that axiom (iii) of definition 1.1.1 cannot be dropped by exhibiting a function $d\colon X \times X \to \mathbb{R}_{\ge 0}$ on a three-point set satisfying (i) and (ii) but not (iii).
2. Let (X,d) be a metric space and define $d'(x,y) = \min\big(d(x,y), 1\big)$. Show that d' is a metric on X. Show also that $d''(x,y) = d(x,y)^2$ need not be one.
3. Show that $\ell^p \subseteq \ell^r$ whenever $1 \le p < r \le \infty$, and that the inclusion is strict. (For the inclusion, first reduce to the case $\|x\|_p = 1$ and observe what that forces about each $|x_k|$.)
4. In $\mathbb{Q}$ with the 5-adic metric, compute $d_5(1, 26)$, $d_5(\frac{1}{5}, 0)$ and $d_5(3, 3 + 5^4)$. Then show that the sequence $x_n = 5^n$ converges to 0, and that $x_n = n!$ does too.
5. Let (X,d) be an ultrametric space. Show that if $d(x,y) < r$ then every point z with $d(x,z) < r$ also satisfies $d(y,z) < r$; conclude that any two balls of radius r are either disjoint or equal, so that every point of a ball is a centre of it.
6. In the French railway metric of example 1.1.13, describe the set of points at distance less than r from a point $x \ne O$, distinguishing the cases $r \le \|x\|$ and $r > \|x\|$.

1.2 Open and Closed Sets

A metric assigns a number to each pair of points, which is more information than most arguments need. What they need is which points are near a given one, and that is captured by the balls: sets of the form "everything within r of x". This section builds the standard vocabulary of nearness out of the balls alone, and in doing so isolates the part of a metric space that survives replacing the metric by a different but comparable one, a distinction section 1.4 makes precise.

Definition 1.2.1: Balls and Spheres

Let (X, d) be a metric space, $x \in X$ and $r > 0$. The *open ball* and *closed ball* of radius r about x are

$$B_r(x) = \{y \in X \mid d(x, y) < r\} \qquad \overline{B}_r(x) = \{y \in X \mid d(x, y) \leq r\}$$

and the *sphere* is $S_r(x) = \{y \in X \mid d(x, y) = r\}$. The point x is a *centre* of the ball and r a *radius*.

The indefinite article in "a centre" is deliberate. In $\mathbb{R}$ a ball determines its centre and radius, but exercise 1.1.1 (5) showed that in an ultrametric space every point of a ball is a centre of it, and in the discrete metric $B_2(x) = X$ for every x. Nothing below assumes a ball remembers how it was specified.

Definition 1.2.2: Open and Closed Sets

A subset $U \subseteq X$ is *open* if for every $x \in U$ there is an $r > 0$ with $B_r(x) \subseteq U$. A subset $A \subseteq X$ is *closed* if its complement $X \setminus A$ is open.

Open and closed are not opposites and not exhaustive. In $\mathbb{R}$ the interval $[0, 1)$ is neither, while $\emptyset$ and $\mathbb{R}$ are both. The definition is a condition at every point of the set, which is why the first thing to check is that the balls themselves satisfy it.

Proposition 1.2.3: Open Balls Are Open and Closed Balls Are Closed

Every open ball in a metric space is an open set, and every closed ball is a closed set.

***Proof*:**
Let $y \in B_r(x)$ and put $s = r - d(x, y)$, which is positive by the definition of the ball. If $d(y, z) < s$ then the triangle inequality gives

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + s = r$$

so $B_s(y) \subseteq B_r(x)$, which is the required condition at y.

For the closed ball, let $y \notin \overline{B}_r(x)$, so $d(x, y) > r$, and put $s = d(x, y) - r > 0$. If $d(y, z) < s$ then, rearranging the triangle inequality $d(x, y) \leq d(x, z) + d(z, y)$,

$$d(x, z) \geq d(x, y) - d(y, z) > d(x, y) - s = r$$

so $z \notin \overline{B}_r(x)$. Hence $B_s(y)$ misses $\overline{B}_r(x)$ entirely, the complement is open, and $\overline{B}_r(x)$ is closed, as we sought to show.

Theorem 1.2.4: The Metric Topology

Let (X, d) be a metric space and let $\mathcal{T}$ denote its collection of open sets. Then

(i) $\emptyset$ and X lie in $\mathcal{T}$;

(ii) any union of members of $\mathcal{T}$ lies in $\mathcal{T}$;

(iii) any *finite* intersection of members of $\mathcal{T}$ lies in $\mathcal{T}$.

***Proof*:**

(i) The condition on $\emptyset$ is vacuous, and for X any radius works since every ball is contained in X.

(ii) Let $\{U_\alpha\}_{\alpha \in A} \subseteq \mathcal{T}$ and let $x \in \bigcup_\alpha U_\alpha$. Then $x \in U_\beta$ for some β, and openness of U_β gives an $r > 0$ with $B_r(x) \subseteq U_\beta \subseteq \bigcup_\alpha U_\alpha$.

(iii) Let $U_1, \ldots, U_n \in \mathcal{T}$ and let $x \in \bigcap_i U_i$. Each U_i gives an $r_i > 0$ with $B_{r_i}(x) \subseteq U_i$. Put $r = \min_i r_i$, which is positive because the collection is finite. Then $B_r(x) \subseteq B_{r_i}(x) \subseteq U_i$ for every i, so $B_r(x) \subseteq \bigcap_i U_i$.

The finiteness in (iii) is not an artifact of the proof. In $\mathbb{R}$ the sets $B_{1/n}(0) = (-\frac{1}{n}, \frac{1}{n})$ are open for every n, but

$$\bigcap_{n=1}^{\infty} B_{1/n}(0) = \{0\}$$

which contains no ball at all. The failure is exactly that an infinite collection of positive radii can have infimum zero, which is what the proof's min avoids.

A collection satisfying the three conditions of theorem 1.2.4 is a *topology*, and the open sets of a metric space are its *metric topology*. This book needs the general notion only once, in **??**, and *Topology* [4] develops it properly. Here the point of the theorem is that the balls do behave as the intervals of $\mathbb{R}$ do.

Definition 1.2.5: Convergent Sequence

A sequence $(x_n)_{n \geq 1}$ in a metric space X *converges* to $x \in X$ if for every $\epsilon > 0$ there is an N such that $d(x_n, x) < \epsilon$ for all $n \geq N$. Equivalently, every ball about x contains all but finitely many terms. Write $x_n \to x$, and call x a *limit* of the sequence.

Proposition 1.2.6: Limits Are Unique

A sequence in a metric space has at most one limit.

***Proof*:**

Suppose $x_n \to x$ and $x_n \to y$, and let $\epsilon > 0$. Choose N_1 with $d(x_n, x) < \epsilon/2$ for $n \geq N_1$ and N_2

with $d(x_n, y) < \epsilon/2$ for $n \geq N_2$. For any $n \geq \max(N_1, N_2)$ the triangle inequality gives

$$d(x, y) \leq d(x, x_n) + d(x_n, y) < \epsilon$$

Since $\epsilon > 0$ was arbitrary and $d(x, y)$ does not depend on it, $d(x, y) = 0$, and the first metric axiom forces $x = y$, completing the proof.

This is the promise made when the "only if" of definition 1.1.1 was insisted on. Under the pseudometric $d \equiv 0$ every sequence converges to every point, and the proof above breaks at exactly the last step: $d(x, y) = 0$ no longer says $x = y$.

Definition 1.2.7: Interior, Closure and Boundary

Let $A \subseteq X$. The *interior* A° is the union of all open sets contained in A; the *closure* $\overline{A}$ is the intersection of all closed sets containing A; the *boundary* is $\partial A = \overline{A} \setminus A^\circ$.

By theorem 1.2.4 the interior is open, being a union of open sets, and it is the largest open subset of A. The closure is closed, since an intersection of closed sets has open complement by de Morgan and part (ii) of the theorem, and it is the smallest closed superset of A. Neither description says which points belong, and for that the metric returns.

Definition 1.2.8: Limit Point

A point $x \in X$ is a *limit point* of $A \subseteq X$ if every ball about x meets $A \setminus \{x\}$. The set of limit points of A is written A'.

Theorem 1.2.9: Characterizations of the Closure

Let $A \subseteq X$ and $x \in X$. The following are equivalent:

(i) $x \in \overline{A}$;

(ii) every ball about x meets A;

(iii) x is the limit of some sequence of points of A.

Consequently $\overline{A} = A \cup A'$.

***Proof*:**

Step 1: (i) implies (ii). Suppose some ball $B_r(x)$ misses A. Then $X \setminus B_r(x)$ is a closed set containing A, so $\overline{A} \subseteq X \setminus B_r(x)$, and since $x \in B_r(x)$ this gives $x \notin \overline{A}$.

Step 2: (ii) implies (iii). For each $n \geq 1$ choose $a_n \in B_{1/n}(x) \cap A$, which is nonempty by hypothesis. Then $d(a_n, x) < 1/n$, so given $\epsilon > 0$ any N with $1/N < \epsilon$ works, and $a_n \to x$.

Step 3: (iii) implies (i). Let $a_n \to x$ with every $a_n \in A$, and let C be any closed set containing A. If $x \notin C$ then $X \setminus C$ is open and contains x, so it contains some $B_r(x)$, which therefore contains all a_n with n large. But those a_n lie in $A \subseteq C$, a contradiction. Hence x lies in every closed set containing A, that is, in $\overline{A}$.

For the last claim, if $x \in \overline{A}$ then by (ii) every ball about x meets A; either that intersection

always contains a point other than x, making $x \in A'$, or some ball meets A only in x itself, forcing $x \in A$. Conversely $A \subseteq \overline{A}$ always, and a limit point of A satisfies (ii), completing the proof.

The third characterization is the one used most often, and it is worth naming the consequence for closed sets, since it converts a statement about complements into a statement about sequences.

Corollary 1.2.10: Closed Sets Contain Their Limits

A subset $A \subseteq X$ is closed if and only if $\overline{A} = A$, and if and only if the limit of every convergent sequence of points of A lies in A.

***Proof*:**
If A is closed then A is one of the closed sets containing A, so $\overline{A} \subseteq A$; the reverse inclusion always holds. If $\overline{A} = A$ then A is closed, being an intersection of closed sets. The equivalence with the sequential condition is theorem 1.2.9(iii): the limits of sequences in A are exactly the points of $\overline{A}$, and requiring them all to lie in A says exactly $\overline{A} \subseteq A$, as we sought to show.

Definition 1.2.11: Dense and Separable

A subset $D \subseteq X$ is *dense* if $\overline{D} = X$. The space X is *separable* if it has a countable dense subset.

By theorem 1.2.9, D is dense exactly when every ball in X contains a point of D: no point of the space can be isolated from D. Separability says that a countable list of points suffices to approximate every point of the space to any accuracy, and it is the hypothesis under which an uncountable space can still be handled by a sequence of choices.

Example 1.2.12: Separability of the Standard Spaces

(i) $\mathbb{R}^n$ is separable: the points with rational coordinates form a countable set, and given x and $\epsilon > 0$, choosing each coordinate within ϵ/n of x_i puts the resulting point q within ϵ of x in d_1. Since each of $d_\infty(x, q)$ and $d_2(x, q)$ is at most $d_1(x, q)$, the same q works for all three metrics of example 1.1.3.

(ii) ℓ^p is separable for $1 \leq p < \infty$. Take D to be the set of sequences with rational entries and only finitely many nonzero. Given $x \in \ell^p$ and $\epsilon > 0$, convergence of $\sum_k |x_k|^p$ gives an N with $\sum_{k>N} |x_k|^p < (\epsilon/2)^p$; truncating at N and then perturbing the first N entries to rationals within $\epsilon/(2N^{1/p})$ each puts a member of D within ϵ of x.

(iii) ℓ^∞ is *not* separable. For each subset $S \subseteq \mathbb{N}$ let χ_S be its indicator sequence. These are $2^{\aleph_0}$ many points of ℓ^∞, and $d_\infty(\chi_S, \chi_T) = 1$ whenever $S \neq T$, since the two differ by 1 at any index in exactly one of them. The balls $B_{1/2}(\chi_S)$ are therefore pairwise disjoint and uncountable in number, so any dense set must contain a point of each and cannot be countable.

That $C[0, 1]$ is separable is true but not elementary: the natural countable candidate is the set of polynomials with rational coefficients, and showing it is dense is the Weierstrass approximation

theorem, proved in section 3.6.

The last example of this section is the fourth of the book's running spaces, and it is the standard source of a set that is closed, uncountable, and yet contains no ball whatsoever.

Example 1.2.13: The Cantor Set
Let $C_0 = [0,1]$ and obtain C_{n+1} from C_n by deleting the open middle third of each of its constituent intervals, so that

$$C_1 = [0,\tfrac{1}{3}] \cup [\tfrac{2}{3},1] \qquad C_2 = [0,\tfrac{1}{9}] \cup [\tfrac{2}{9},\tfrac{1}{3}] \cup [\tfrac{2}{3},\tfrac{7}{9}] \cup [\tfrac{8}{9},1]$$

and C_n is a union of 2^n closed intervals of length 3^{-n}. The *Cantor set* is $C = \bigcap_{n\geq 0} C_n$, taken with the metric inherited from $\mathbb{R}$.

It is closed, being an intersection of closed sets. It has empty interior: any ball $B_r(x) \subseteq C$ would be an interval of positive length contained in every C_n, but the intervals making up C_n have length $3^{-n} \to 0$. And it is nonempty, since every endpoint of every interval of every C_n survives all later stages; indeed C consists exactly of the reals in $[0,1]$ admitting a base-3 expansion using only the digits 0 and 2, which is why it is uncountable. Section 2.3 returns to it once completeness is available, where it becomes the standard example of a nonempty perfect set.

Exercise 1.2.1

1. Show that $\overline{B_r(x)} \subseteq \overline{B}_r(x)$ in any metric space, and give a metric space in which the inclusion is strict. (Try a space with very few points.)
2. Prove that $X \setminus A^\circ = \overline{X \setminus A}$ and $X \setminus \overline{A} = (X \setminus A)^\circ$ for every $A \subseteq X$.
3. Show that $\overline{A \cup B} = \overline{A} \cup \overline{B}$ for any two subsets, and that $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$ with the inclusion possibly strict. Show also that the first identity can fail for infinite unions.
4. Describe the open sets, the closed sets and the convergent sequences of a set carrying the discrete metric of example 1.1.5. Which subsets are dense, and for which sets is the space separable?
5. Let X be separable and $Y \subseteq X$. Show that Y is separable in the inherited metric. (A countable dense subset of X need not lie in Y; for each pair (q,n) with q in the countable dense set, pick a point of $Y \cap B_{1/n}(q)$ when there is one.)
6. Show that the Cantor set contains no interval of positive length by computing the total length removed at all stages, and deduce that C has empty interior a second way.

1.3 Continuity in Three Forms

Single-variable analysis offers three descriptions of a continuous function: the ϵ-δ condition, the preservation of convergent sequences, and the condition that preimages of open sets be open. Each is stated using $|x-y|$ and nothing else, so each transcribes to a metric space unchanged. What is not automatic is that the three remain equivalent, and the first theorem of this section checks that they do.

The rest of the section separates continuity from two strengthenings that coincide with it on $\mathbb{R}$ often enough to be confused with it. Both differences matter later: uniform continuity is what section 3.1.2 shows to follow from continuity on a compact domain, and the Lipschitz condition with constant less than 1 is the hypothesis of **??**.

Definition 1.3.1: Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f\colon X \to Y$ is *continuous at* $x \in X$ if for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$d_X(x,t) < \delta \quad \Rightarrow \quad d_Y\big(f(x), f(t)\big) < \epsilon$$

and *continuous* if it is continuous at every point of X.

Theorem 1.3.2: Three Characterizations of Continuity

For $f\colon X \to Y$ between metric spaces the following are equivalent:

(i) f is continuous;

(ii) f preserves convergence: whenever $x_n \to x$ in X, $f(x_n) \to f(x)$ in Y;

(iii) $f^{-1}(V)$ is open in X for every open $V \subseteq Y$.

***Proof*:**

Step 1: (i) implies (ii). Let $x_n \to x$ and let $\epsilon > 0$. Choose δ as in definition 1.3.1 for this ϵ at the point x, and then N with $d_X(x_n, x) < \delta$ for $n \geq N$. For such n, $d_Y(f(x_n), f(x)) < \epsilon$.

Step 2: (ii) implies (i). Argue by contraposition. If f fails to be continuous at some x, there is an $\epsilon > 0$ for which no δ works; in particular $\delta = 1/n$ fails for each n, giving a point x_n with $d_X(x, x_n) < 1/n$ but $d_Y(f(x), f(x_n)) \geq \epsilon$. Then $x_n \to x$ while $f(x_n) \not\to f(x)$, so (ii) fails.

Step 3: (i) implies (iii). Let $V \subseteq Y$ be open and let $x \in f^{-1}(V)$. Openness of V gives an $\epsilon > 0$ with $B_\epsilon(f(x)) \subseteq V$, and continuity at x gives a $\delta > 0$ with $f\big(B_\delta(x)\big) \subseteq B_\epsilon(f(x)) \subseteq V$. Hence $B_\delta(x) \subseteq f^{-1}(V)$.

Step 4: (iii) implies (i). Fix $x \in X$ and $\epsilon > 0$. The ball $B_\epsilon(f(x))$ is open in Y by proposition 1.2.3, so $f^{-1}\big(B_\epsilon(f(x))\big)$ is open in X and contains x; it therefore contains some $B_\delta(x)$, and that δ satisfies the requirement of definition 1.3.1.

The third condition mentions no distances, only open sets, which is what makes it the definition topology adopts. The first two do mention distances, and the practical consequence is that a property defined through open sets alone cannot distinguish two metrics with the same open sets; section 1.4 takes this up, and chapter 2 exhibits a property that is *not* of this kind.

Proposition 1.3.3: Composition of Continuous Maps

If $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous then so is $g \circ f$.

***Proof*:**
Let $W \subseteq Z$ be open. Then $(g \circ f)^{-1}(W) = f^{-1}\big(g^{-1}(W)\big)$, and $g^{-1}(W)$ is open by theorem 1.3.2(iii) applied to g, so its preimage under f is open by the same criterion applied to f. Hence $g \circ f$ is continuous, as we sought to show.

In definition 1.3.1 the number δ is allowed to depend on both ϵ and the point x. Forbidding the second dependence gives a strictly stronger condition, and forbidding it in a quantitative way gives a stronger one still.

Definition 1.3.4: Uniform Continuity

A function $f\colon X \to Y$ is *uniformly continuous* if for every $\epsilon > 0$ there is a $\delta > 0$, depending on ϵ alone, such that $d_X(x,t) < \delta$ implies $d_Y(f(x), f(t)) < \epsilon$ for *all* $x, t \in X$.

Definition 1.3.5: Lipschitz Map

A function $f\colon X \to Y$ is *Lipschitz* with constant $L \geq 0$ if

$$d_Y\big(f(x), f(t)\big) \leq L\, d_X(x,t) \qquad \text{for all } x, t \in X$$

A Lipschitz map with $L = 1$ is 1-*Lipschitz* or *nonexpanding*.

Proposition 1.3.6: The Continuity Hierarchy

Every Lipschitz map is uniformly continuous, and every uniformly continuous map is continuous.

***Proof*:**
Let f be Lipschitz with constant L and let $\epsilon > 0$. If $L = 0$ then f is constant and any δ serves; otherwise $\delta = \epsilon / L$ gives $d_Y(f(x), f(t)) \leq L\, d_X(x,t) < L\delta = \epsilon$, and δ depends only on ϵ. The second implication is immediate, since a δ valid at every point is in particular valid at each one, completing the proof.

Example 1.3.7: Separating the Three Notions

(i) $f(x) = x^2$ on $\mathbb{R}$ is continuous but not uniformly continuous. Given $\delta > 0$, take $x = 1/\delta$ and $t = x + \delta/2$; then $|x - t| < \delta$ while

$$|f(t) - f(x)| = |t - x|\,|t + x| = \frac{\delta}{2}\left(\frac{2}{\delta} + \frac{\delta}{2}\right) > 1$$

so no δ works for $\epsilon = 1$. The obstruction is that the function's steepness is unbounded, not any failure at a single point.

(ii) $f(x) = \sqrt{x}$ on $[0,1]$ is uniformly continuous but not Lipschitz. For uniform continuity, $|\sqrt{x} - \sqrt{t}|^2 \leq |\sqrt{x} - \sqrt{t}|\,|\sqrt{x} + \sqrt{t}| = |x - t|$, so $\delta = \epsilon^2$ works everywhere. It is not Lipschitz

because

$$\frac{|f(x)-f(0)|}{|x-0|} = \frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \to \infty \quad \text{as } x \to 0^+$$

so no single constant L bounds the ratio.

(iii) Any map out of a discrete metric space is Lipschitz with L equal to the diameter of its image when that is finite, since distinct points are at distance 1. Continuity is therefore no restriction at all on such a space.

Two maps built from the metric itself are Lipschitz, and both are used repeatedly. The first says the metric is continuous in its two arguments jointly.

Proposition 1.3.8: The Metric Is Continuous

For any metric space (X, d) and any $x, y, x', y' \in X$,

$$\big|d(x,y) - d(x',y')\big| \le d(x,x') + d(y,y')$$

Consequently, if $x_n \to x$ and $y_n \to y$ then $d(x_n, y_n) \to d(x, y)$.

***Proof*:**
Two applications of the triangle inequality give

$$d(x,y) \le d(x,x') + d(x',y') + d(y',y)$$

so $d(x,y) - d(x',y') \le d(x,x') + d(y,y')$ after using symmetry on the last term. Exchanging the roles of the primed and unprimed pairs gives the same bound for $d(x',y') - d(x,y)$, and the two together bound the absolute value. For the consequence, the right-hand side tends to 0, completing the proof.

Definition 1.3.9: Distance to a Set

For $\emptyset \neq A \subseteq X$ and $x \in X$, the *distance from x to A* is

$$d(x, A) = \inf_{a \in A} d(x, a)$$

Proposition 1.3.10: Distance to a Set Is 1-Lipschitz

Let $\emptyset \neq A \subseteq X$. Then $|d(x,A) - d(y,A)| \le d(x,y)$ for all $x, y \in X$, and

$$\overline{A} = \{x \in X \mid d(x,A) = 0\}$$

***Proof*:**
For any $a \in A$, the triangle inequality gives $d(x,A) \le d(x,a) \le d(x,y) + d(y,a)$. The left side does not involve a, so taking the infimum over $a \in A$ on the right yields $d(x,A) \le d(x,y) + d(y,A)$, that is, $d(x,A) - d(y,A) \le d(x,y)$. Exchanging x and y gives the other inequality, and the two bound the absolute value.

For the description of the closure, $d(x, A) = 0$ says that for every $\epsilon > 0$ some $a \in A$ has $d(x, a) < \epsilon$, that is, every ball about x meets A. By theorem 1.2.9(ii) this is exactly the condition $x \in \overline{A}$, as we sought to show.

Isomorphism in this subject comes in two strengths, and the gap between them is the subject of chapter 2.

Definition 1.3.11: Homeomorphism and Isometry

A bijection $f\colon X \to Y$ is a *homeomorphism* if both f and f^{-1} are continuous, and X and Y are then *homeomorphic*. It is an *isometry* if

$$d_Y\big(f(x), f(t)\big) = d_X(x, t) \qquad \text{for all } x, t \in X$$

and X and Y are then *isometric*.

An isometry is a homeomorphism: it is 1-Lipschitz, hence continuous by proposition 1.3.6, and its inverse satisfies the same identity, hence is continuous too. A homeomorphism preserves exactly the notions theorem 1.3.2(iii) can express, which is to say open sets, closures, convergence and continuity itself. An isometry preserves the metric outright, so it preserves in addition every quantity computed from distances: diameters, boundedness, and completeness.

Example 1.3.12: Homeomorphic but Not Isometric

The map $f\colon \mathbb{R} \to (-1, 1)$ given by

$$f(x) = \frac{x}{1 + |x|}$$

is a bijection with inverse $f^{-1}(y) = \frac{y}{1-|y|}$, and both are continuous as quotients of continuous functions with nonvanishing denominator, so $\mathbb{R}$ and $(-1, 1)$ are homeomorphic in their usual metrics. They are not isometric: $(-1, 1)$ has diameter 2 while $\mathbb{R}$ has infinite diameter, and an isometry preserves the set of distances realized.

The two spaces are therefore indistinguishable by every property expressible in open sets, yet one of them is complete and the other is not (section 2.1). Completeness is a property of the metric, not of the topology, and this pair is the standard example.

Exercise 1.3.1

1. Show that $f\colon X \to Y$ is continuous if and only if $f^{-1}(C)$ is closed in X for every closed $C \subseteq Y$. Show also that continuity does *not* imply that $f(U)$ is open for every open U, by considering a constant map.

2. Let $f, g\colon X \to \mathbb{R}$ be continuous. Show that $f + g$, $f - g$, fg and $\max(f, g)$ are continuous, and that f/g is continuous on $\{x \in X \mid g(x) \neq 0\}$. Deduce that every polynomial function $\mathbb{R}^n \to \mathbb{R}$ is continuous.

3. Show that $f(\overline{A}) \subseteq \overline{f(A)}$ for every continuous f and every $A \subseteq X$, and give an example where the inclusion is strict.

4. Call a sequence (x_n) *Cauchy* when for every $\epsilon > 0$ there is an N with $d(x_n, x_m) < \epsilon$ for all $n, m \geq N$. Show that a uniformly continuous map sends Cauchy sequences to Cauchy

sequences, and that a merely continuous map need not. (Use $f(x) = 1/x$ on $(0, 1)$.)

5. Let A, B be disjoint nonempty closed subsets of X. Show that
$$g(x) = \frac{d(x, A)}{d(x, A) + d(x, B)}$$
is a well-defined continuous function $X \to [0, 1]$ with $g \equiv 0$ on A and $g \equiv 1$ on B.

6. Show that every isometry is injective, and that a surjective 1-Lipschitz map need not be an isometry.

1.4 Subspaces, Products, and Equivalent Metrics

Three constructions appear in every later chapter, usually without comment. A subset of a metric space is a metric space; a product of two is a metric space; and a single space usually carries several metrics that one would like to treat as interchangeable. Each construction raises the same question, which is what survives it, and the third gives the language for answering all three: two metrics are interchangeable for topological purposes exactly when they have the same open sets, and interchangeable for metric purposes when each bounds a constant multiple of the other.

Definition 1.4.1: Subspace Metric

Let (X, d) be a metric space and $Y \subseteq X$. The *subspace metric* on Y is the restriction $d|_{Y \times Y}$, and Y with this metric is a *subspace* of X.

The axioms are inherited verbatim, so the only content is in how the open sets of Y relate to those of X. They are not the same sets, and the difference is a standard source of confusion.

Proposition 1.4.2: Open Sets of a Subspace

Let $Y \subseteq X$ carry the subspace metric. A set $U \subseteq Y$ is open in Y if and only if $U = V \cap Y$ for some V open in X. The balls satisfy $B_r^Y(y) = B_r^X(y) \cap Y$ for every $y \in Y$ and $r > 0$.

***Proof*:**
The statement about balls is immediate from the definition, since the condition $d(y, z) < r$ is the same whether z ranges over Y or over X.

Suppose $U = V \cap Y$ with V open in X, and let $y \in U$. Then $y \in V$, so some $B_r^X(y) \subseteq V$, whence $B_r^Y(y) = B_r^X(y) \cap Y \subseteq V \cap Y = U$. So U is open in Y.

Conversely let U be open in Y. For each $y \in U$ choose $r_y > 0$ with $B_{r_y}^Y(y) \subseteq U$, and put $V = \bigcup_{y \in U} B_{r_y}^X(y)$, which is open in X by theorem 1.2.4. Then

$$V \cap Y = \bigcup_{y \in U} \left(B_{r_y}^X(y) \cap Y \right) = \bigcup_{y \in U} B_{r_y}^Y(y)$$

and this union contains every $y \in U$ while being contained in U, so it equals U, as we sought to show.

So $[0, \frac{1}{2})$ is open in the subspace $[0,1]$ of $\mathbb{R}$, being $(-\frac{1}{2}, \frac{1}{2}) \cap [0,1]$, though it is not open in $\mathbb{R}$. Openness is always relative to an ambient space, and in this book the ambient space is named whenever it is not the obvious one.

Definition 1.4.3: Product Metrics

Let (X, d_X) and (Y, d_Y) be metric spaces. For $m = (x, y)$ and $m' = (x', y')$ in $X \times Y$ define

$$d_E(m, m') = \sqrt{d_X(x, x')^2 + d_Y(y, y')^2}$$
$$d_{\max}(m, m') = \max\left\{d_X(x, x'),\, d_Y(y, y')\right\}$$
$$d_{\text{sum}}(m, m') = d_X(x, x') + d_Y(y, y')$$

Each is a metric. For d_{sum} and $d_{\max}$ the verification is the coordinatewise argument of example 1.1.3; for d_E it is the case $p = 2$, $n = 2$ of theorem 1.1.8 applied to the pair of nonnegative numbers $\big(d_X(x, x'), d_Y(y, y')\big)$ and $\big(d_X(x', x''), d_Y(y', y'')\big)$, together with the fact that each coordinate distance already obeys a triangle inequality and $t \mapsto t^2$ is increasing on $[0, \infty)$.

Rather than choose between the three, the next result shows the choice does not matter, in a sense the rest of the section makes precise.

Proposition 1.4.4: Comparison of the Product Metrics

On $X \times Y$,

$$d_{\max} \le d_E \le d_{\text{sum}} \le 2\, d_{\max}$$

pointwise.

***Proof*:**
Write $a = d_X(x, x')$ and $b = d_Y(y, y')$, both nonnegative.

For the first inequality, $\max(a, b)^2$ is either a^2 or b^2, and each is at most $a^2 + b^2$; taking square roots, which preserves order on $[0, \infty)$, gives $\max(a, b) \le \sqrt{a^2 + b^2}$.

For the second, $(a + b)^2 = a^2 + 2ab + b^2 \ge a^2 + b^2$ since $ab \ge 0$, and taking square roots gives $\sqrt{a^2 + b^2} \le a + b$.

For the third, both a and b are at most $\max(a, b)$, so their sum is at most $2 \max(a, b)$, completing the proof.

Definition 1.4.5: Equivalent Metrics

Two metrics d and d' on the same set X are *topologically equivalent* if they have the same open sets. They are *strongly equivalent* if there are constants $c_1, c_2 > 0$ with

$$c_1\, d(x, y) \le d'(x, y) \le c_2\, d(x, y) \qquad \text{for all } x, y \in X$$

Proposition 1.4.6: Strong Equivalence Implies Topological

Strongly equivalent metrics are topologically equivalent.

***Proof*:**
Let U be open for d and let $x \in U$, so $B_r^d(x) \subseteq U$ for some $r > 0$. If $d'(x,z) < c_1 r$ then $c_1 d(x,z) \leq d'(x,z) < c_1 r$, so $d(x,z) < r$ and $z \in U$. Hence $B_{c_1 r}^{d'}(x) \subseteq U$ and U is open for d'. The reverse inclusion follows by the symmetric argument using c_2, completing the proof.

Corollary 1.4.7: The Three Product Metrics Are Interchangeable

The metrics d_E, $d_{\max}$ and d_{sum} on $X \times Y$ are strongly equivalent, hence topologically equivalent. On $\mathbb{R}^n$ the three metrics of example 1.1.3 satisfy

$$d_\infty \leq d_2 \leq d_1 \leq n\, d_\infty$$

and are therefore strongly equivalent as well.

***Proof*:**
Proposition 1.4.4 gives $d_{\max} \leq d_E \leq d_{\text{sum}} \leq 2d_{\max}$, so each pair among the three is bracketed by constant multiples of the other, and proposition 1.4.6 applies. On $\mathbb{R}^n$ the same argument with n coordinates in place of two gives $d_\infty \leq d_2 \leq d_1 \leq n\, d_\infty$, completing the proof.

Whenever $\mathbb{R}^n$ or a finite product appears below, no metric is specified: by corollary 1.4.7 every statement about open sets, convergence, continuity or closure is independent of the choice.

The converse of proposition 1.4.6 fails, and the failure is the reason the two notions have separate names.

Example 1.4.8: Topologically but Not Strongly Equivalent
On $\mathbb{R}$ let $d(x,y) = |x-y|$ and $d'(x,y) = \min\big(|x-y|, 1\big)$, which is a metric by exercise 1.1.1 (2). They are topologically equivalent: any d'-ball of radius $r \leq 1$ is the d-ball of the same radius, and balls of radius at most 1 already witness openness for both metrics.

They are not strongly equivalent. Strong equivalence would give a $c_1 > 0$ with $c_1|x-y| \leq \min(|x-y|, 1)$ for all x, y, and taking $y = 0$ and $x \to \infty$ makes the left side unbounded while the right side never exceeds 1.

The two metrics therefore have identical open sets, identical convergent sequences and identical continuous functions, yet differ on every question involving actual distances: $\mathbb{R}$ has infinite diameter under d and diameter 1 under d', and a set can be d'-bounded without being d-bounded.

Proposition 1.4.9: Convergence in a Product Is Coordinatewise

A sequence $\big((x_n, y_n)\big)$ in $X \times Y$ converges to (x,y) for any one of the three product metrics if and only if $x_n \to x$ in X and $y_n \to y$ in Y.

***Proof*:**
By corollary 1.4.7 it suffices to argue for $d_{\max}$. If $d_{\max}\big((x_n, y_n), (x,y)\big) < \epsilon$ then both $d_X(x_n, x) < \epsilon$ and $d_Y(y_n, y) < \epsilon$, since each is at most the maximum, so convergence in the product forces convergence in each coordinate. Conversely, given $\epsilon > 0$ choose N_1 and N_2 beyond which the two coordinate distances are less than ϵ; for $n \geq \max(N_1, N_2)$ the maximum is less than ϵ, as

we sought to show.

Products of finitely many factors are handled by any of the three formulas. For countably many factors all three fail: a supremum of infinitely many distances can be infinite, and a sum of infinitely many can diverge. Rescaling each factor to have diameter at most 1 and then damping the terms repairs both problems at once.

Definition 1.4.10: Countable Product Metric

Let $(X_n, d_n)_{n\geq 1}$ be metric spaces. On $\prod_{n\geq 1} X_n$ define

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \min\left(d_n(x_n, y_n),\, 1\right)$$

Proposition 1.4.11: The Countable Product Metric

The function d of definition 1.4.10 is a metric, and $x^{(k)} \to x$ with respect to it if and only if $x_n^{(k)} \to x_n$ in X_n for every n.

***Proof*:**

The series converges, being dominated termwise by $\sum 2^{-n} = 1$. Each summand is a metric on X_n by exercise 1.1.1 (2), so symmetry and the triangle inequality hold termwise and are preserved by the (convergent) sum. If $d(x,y) = 0$ then every summand vanishes, forcing $x_n = y_n$ for all n.

For the convergence claim, suppose first $d(x^{(k)}, x) \to 0$ and fix n. Since the nth summand is at most the whole sum,

$$2^{-n} \min\left(d_n(x_n^{(k)}, x_n), 1\right) \leq d(x^{(k)}, x) \to 0$$

so $\min(d_n(x_n^{(k)}, x_n), 1) \to 0$, which forces $d_n(x_n^{(k)}, x_n) \to 0$ because the minimum equals the distance once the distance drops below 1.

Conversely suppose every coordinate converges and let $\epsilon > 0$. Choose N with $2^{-N} < \epsilon/2$, so the tail $\sum_{n>N} 2^{-n} \min(\cdot, 1) \leq \sum_{n>N} 2^{-n} = 2^{-N} < \epsilon/2$ regardless of the points involved. For each of the finitely many $n \leq N$, choose K_n with $d_n(x_n^{(k)}, x_n) < \epsilon/2$ for $k \geq K_n$. For $k \geq \max_{n\leq N} K_n$,

$$d(x^{(k)}, x) < \sum_{n=1}^{N} 2^{-n} \frac{\epsilon}{2} + \frac{\epsilon}{2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

completing the proof.

The damping factors 2^{-n} are one choice among many; any positive summable sequence produces a metric with the same convergent sequences. What the construction shows is that countable products stay metrizable, which finite products achieve more cheaply and uncountable products do not achieve at all.

Example 1.4.12: The Hilbert Cube

Taking every factor to be $[0,1]$ with its usual metric gives the *Hilbert cube* $[0,1]^{\mathbb{N}}$, whose points are sequences in $[0,1]$ and whose metric is

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n}|x_n - y_n|$$

the minimum being unnecessary since all distances are already at most 1. By proposition 1.4.11 convergence here is coordinatewise, so the sequence $x^{(k)} = (0,\dots,0,1,0,\dots)$ with the 1 in position k converges to the zero sequence, each coordinate being eventually 0. In ℓ^∞, by contrast, these same points are pairwise at distance 1 (example 1.2.12), so they converge to nothing. Coordinatewise convergence and uniform convergence are different notions, and **??** is about the difference.

Exercise 1.4.1

1. Let $Y \subseteq X$ be a subspace. Show that $A \subseteq Y$ is closed in Y if and only if $A = C \cap Y$ for some C closed in X. Show that if Y is itself closed in X, then a set closed in Y is closed in X, and give an example showing this fails without that hypothesis.
2. Show that the projections $\pi_X \colon X \times Y \to X$ and $\pi_Y \colon X \times Y \to Y$ are 1-Lipschitz for $d_{\max}$, and that a map $f \colon Z \to X \times Y$ is continuous if and only if $\pi_X \circ f$ and $\pi_Y \circ f$ both are.
3. Show that strongly equivalent metrics have the same Cauchy sequences and the same bounded sets, and that topologically equivalent metrics need not, using example 1.4.8.
4. Verify that topological equivalence and strong equivalence are each equivalence relations on the set of metrics on a fixed set X.
5. Let X be an infinite set with the discrete metric d, and let $d'(x,y) = \min(d(x,y), \frac{1}{2})$ for $x \neq y$. Determine whether d and d' are topologically equivalent, and whether they are strongly equivalent.
6. Compute the diameter of the Hilbert cube under the metric of example 1.4.12, and exhibit two points realizing it.

2

Completeness

Convergence, as definition 1.2.5 states it, requires a limit to be named in advance. That is an awkward requirement, because the sequences analysis produces are usually built by a procedure, the procedure produces terms rather than a limit, and whether a limit exists is the question being asked. What is available instead is the observation that the terms crowd together, and the whole content of this chapter is that in the right spaces this observation is enough.

Spaces where it is enough are the complete ones, and the property turns out to be sharper than it first appears. Section 2.1 defines it and shows that it is a property of the metric and not of the open sets: two spaces can be homeomorphic with one complete and the other not, so no argument phrased in open sets alone can establish it. Section 2.2 verifies it for the spaces this book actually uses, and the verification for $C_b(X)$ is the one **??** rests on. Section 2.3 extracts what completeness gives, in the form of a nested-set principle that produces points nobody has constructed, and uses it to show that a nonempty complete space with no isolated points is uncountable. Section 2.4 completes the chapter by showing that an incomplete space can always be repaired, in exactly one way.

2.1 Cauchy Sequences and Complete Spaces

Convergence, as definition 1.2.5 states it, is a relation between a sequence and a point. This section isolates the condition that remains when the point is struck out of the statement, and names the spaces in which the two conditions agree.

To test a sequence against definition 1.2.5 one must already hold the limit in hand, since the limit occurs in the condition being tested. That is a real restriction. A sequence produced by an iteration, an algorithm or a series comes with no candidate limit attached, and whether such a point exists is exactly the question at issue. What can be tested without a candidate is whether the terms of the sequence eventually crowd together.

Definition 2.1.1: Cauchy Sequence

A sequence $(x_n)_{n\geq 1}$ in a metric space (X, d) is *Cauchy* if for every $\epsilon > 0$ there is an N such that

$$d(x_n, x_m) < \epsilon \qquad \text{for all } n, m \geq N$$

The difference from definition 1.2.5 is one symbol. Convergence bounds $d(x_n, x)$, where x is a point of X named in advance; the Cauchy condition bounds $d(x_n, x_m)$, where both entries are terms of the sequence. No point outside the sequence appears in definition 2.1.1, so the condition can be checked by someone who knows the terms and nothing else about the space.

A computation makes the difference visible. In $\mathbb{R}$ let $x_n = \sum_{k=1}^{n} 1/k^2$. For $m > n$ the terms $1/k^2$ with $k \geq 2$ satisfy $1/k^2 \leq 1/\big(k(k-1)\big)$, and that bound telescopes:

$$|x_m - x_n| = \sum_{k=n+1}^{m} \frac{1}{k^2} \leq \sum_{k=n+1}^{m} \frac{1}{k(k-1)} = \sum_{k=n+1}^{m} \left(\frac{1}{k-1} - \frac{1}{k}\right) = \frac{1}{n} - \frac{1}{m} < \frac{1}{n}$$

So given $\epsilon > 0$, any $N > 1/\epsilon$ forces $|x_m - x_n| < 1/N < \epsilon$ for all $m > n \geq N$, and the case $m = n$ is trivial. The sequence is Cauchy. Nothing in the computation refers to a limit, and nothing in it produces one: the value $\pi^2/6$ is invisible to the estimate.

The two conditions are not independent. Bounding the distance from every term to a common point bounds the distances between terms, by the triangle inequality, and that is the whole of one implication.

Proposition 2.1.2: Convergent Sequences Are Cauchy

Every convergent sequence in a metric space is Cauchy.

***Proof*:**
Let $x_n \to x$ in (X, d) and let $\epsilon > 0$. By definition 1.2.5 there is an N with $d(x_n, x) < \epsilon/2$ for all $n \geq N$. For any $n, m \geq N$, the triangle inequality and symmetry give

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

as we sought to show.

The converse is false, and each failure of it is a failure of the space rather than of the sequence: the terms crowd together and the space gives no point for them to crowd around. Two properties hold of every Cauchy sequence regardless, and both are proved without a limit. The first is the metric form of the statement that a convergent real sequence is bounded.

Proposition 2.1.3: Cauchy Sequences Are Bounded

Let $(x_n)_{n\geq 1}$ be a Cauchy sequence in a metric space X. Then there is an $R > 0$ with $x_n \in B_R(x_1)$ for every n; that is, the set of terms $\{x_n \mid n \geq 1\}$ is bounded.

***Proof*:**
Apply definition 2.1.1 with $\epsilon = 1$: there is an N with $d(x_n, x_m) < 1$ for all $n, m \geq N$. Put

$$R = 1 + \max_{1 \leq j \leq N} d(x_j, x_1)$$

which is a maximum over a finite nonempty index set, hence a real number, and which satisfies $R \geq 1 > 0$ because the maximum is at least $d(x_1, x_1) = 0$.

For $n \leq N$ the term $d(x_n, x_1)$ is one of those entering the maximum, so $d(x_n, x_1) \leq R - 1 < R$. For $n > N$ both indices n and N are at least N, so the Cauchy estimate applies to the pair and

$$d(x_n, x_1) \leq d(x_n, x_N) + d(x_N, x_1) < 1 + d(x_N, x_1) \leq R$$

In either case $x_n \in B_R(x_1)$, completing the proof.

Boundedness is necessary and not sufficient. In $\mathbb{R}$ the sequence $x_n = (-1)^n$ has all its terms in $B_2(0)$, yet $|x_n - x_{n+1}| = 2$ for every n, so no N meets the requirement of definition 2.1.1 for $\epsilon = 1$. The Cauchy condition asks the terms to accumulate at a single location, and boundedness only confines them to a bounded region.

The second property matters whenever the limit is obtained by extraction rather than built outright. Some completeness proofs construct the candidate limit directly, coordinate by coordinate or pointwise. Others, and in particular those routed through compactness, produce a limit only for a subsequence obtained by a construction such as repeated bisection of an interval, or a diagonal choice across a countable family of sequences; **??** carries out both. Extraction needs a name.

Definition 2.1.4: Subsequence

A *subsequence* of a sequence $(x_n)_{n \geq 1}$ in X is a sequence $(x_{n_k})_{k \geq 1}$ determined by a strictly increasing choice of indices $n_1 < n_2 < n_3 < \cdots$ in $\mathbb{N}$.

Such a choice satisfies $n_k \geq k$ for every k: the indices are positive integers, so $n_1 \geq 1$, and $n_{k+1} > n_k \geq k$ forces $n_{k+1} \geq k + 1$. That inequality is what lets an estimate valid beyond index N be applied to the kth term of any subsequence, and it is used without comment below. For a Cauchy sequence the extraction loses nothing.

Proposition 2.1.5: A Cauchy Sequence with a Convergent Subsequence Converges

Let $(x_n)_{n \geq 1}$ be a Cauchy sequence in a metric space X, and suppose some subsequence $(x_{n_k})_{k \geq 1}$ converges to $x \in X$. Then $x_n \to x$.

***Proof*:**
Let $\epsilon > 0$. Since (x_n) is Cauchy there is an N with $d(x_n, x_m) < \epsilon/2$ for all $n, m \geq N$, and since $x_{n_k} \to x$ there is a K with $d(x_{n_k}, x) < \epsilon/2$ for all $k \geq K$.

Fix one index $k \geq \max(K, N)$. Then $n_k \geq k \geq N$, so for every $n \geq N$ the pair (n, n_k) falls under the Cauchy estimate, and the triangle inequality gives

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

The index k was fixed after ϵ, and the resulting bound holds for all $n \geq N$ with N depending on ϵ alone. Hence $x_n \to x$, as we sought to show.

Neither hypothesis can be dropped. A Cauchy sequence need not converge, as the examples below show, and a sequence with a convergent subsequence need not converge either: $x_n = (-1)^n$ in $\mathbb{R}$ has the constant subsequence $x_{2k} = 1$ converging to 1, while (x_n) itself converges to nothing, since a convergent sequence is Cauchy by proposition 2.1.2 and this one was seen above not to be. It is the conjunction that forces convergence. **??** produces convergent subsequences from hypotheses of an entirely different kind, and proposition 2.1.5 is what converts each such extraction into a statement about the original sequence.

For sequences of real numbers the Cauchy condition and convergence agree, and that agreement is one of the standard formulations of the completeness of $\mathbb{R}$ [1]. In a general metric space it is not a theorem but a hypothesis.

Definition 2.1.6: Complete Metric Space

A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a point of X. A subset $A \subseteq X$ is *complete* if it is complete as a metric space under the subspace metric of definition 1.4.1.

By proposition 2.1.2 the convergent sequences of a metric space always sit among its Cauchy sequences, so completeness is the assertion that the two classes coincide. The content is an existence statement: a description of a point by successive approximations, given without naming the point, always picks out an actual point of the space. Failure of the property is therefore always the same phenomenon, a description with no object, and the two standard examples exhibit it by deleting the object from a space that had it.

Example 2.1.7: Two Incomplete Spaces

(i) $\mathbb{Q}$ with the metric $d(x, y) = |x - y|$. Define a sequence of rationals by

$$x_1 = 2 \qquad x_{n+1} = \frac{x_n}{2} + \frac{1}{x_n} = \frac{x_n^2 + 2}{2x_n}$$

Each x_n is a positive rational: this holds for x_1, and if x_n is a positive rational then so is $(x_n^2 + 2)/(2x_n)$. The first four terms are 2, $\frac{3}{2}$, $\frac{17}{12}$ and $\frac{577}{408}$.

Write $e_n = x_n^2 - 2$. Squaring the recursion,

$$e_{n+1} = \frac{(x_n^2 + 2)^2 - 8x_n^2}{4x_n^2} = \frac{x_n^4 - 4x_n^2 + 4}{4x_n^2} = \frac{(x_n^2 - 2)^2}{4x_n^2} = \frac{e_n^2}{4x_n^2}$$

so $e_n \geq 0$ for every $n \geq 2$, and $e_1 = 2 \geq 0$ as well. Hence $x_n^2 \geq 2$ for all n, and since each x_n is positive this gives $x_n > 1$ and $4x_n^2 \geq 8$. Therefore $e_{n+1} \leq e_n^2/8$, and induction gives $e_n \leq 2$ for all n: it holds at $n = 1$, and $e_n \leq 2$ forces $e_{n+1} \leq 4/8 < 2$. Feeding $e_n \leq 2$ back in,

$$e_{n+1} \leq \frac{e_n \cdot e_n}{8} \leq \frac{2e_n}{8} = \frac{e_n}{4}$$

so $e_n \leq 2 \cdot 4^{1-n}$ for every n.

The sequence decreases, since $x_n - x_{n+1} = \frac{x_n}{2} - \frac{1}{x_n} = \frac{e_n}{2x_n} \geq 0$. So for $m \geq n$ both $x_n \geq x_m$ and $x_n + x_m > 2$ hold, and $x_m^2 \geq 2$ gives

$$0 \leq x_n - x_m = \frac{x_n^2 - x_m^2}{x_n + x_m} \leq \frac{x_n^2 - 2}{x_n + x_m} \leq \frac{e_n}{2} \leq 4^{1-n}$$

Given $\epsilon > 0$, choosing N with $4^{1-N} < \epsilon$ makes $|x_n - x_m| < \epsilon$ for all $n, m \geq N$, so (x_n) is Cauchy in $\mathbb{Q}$.

It has no limit in $\mathbb{Q}$. Suppose $x_n \to q$ with $q \in \mathbb{Q}$. Every x_n lies in $(1, 2]$, so

$$|q^2 - 2| \leq |q^2 - x_n^2| + |x_n^2 - 2| = |q - x_n|\,|q + x_n| + e_n \leq \big(|q| + 2\big)|q - x_n| + 2 \cdot 4^{1-n}$$

Both terms on the right tend to 0, and the left side does not depend on n, so $q^2 = 2$. No rational has square 2: writing $q = a/b$ in lowest terms, $a^2 = 2b^2$ makes a even, say $a = 2c$, whence $b^2 = 2c^2$ makes b even, contradicting the choice of lowest terms. So $\mathbb{Q}$ is not complete.

Inside $\mathbb{R}$ the same sequence converges, to $\sqrt{2}$. The terms are not defective; the space is missing the point they approximate.

(ii) The interval $(0, 1)$ with the metric inherited from $\mathbb{R}$, and $x_n = 1/n$ for $n \geq 2$. For $m \geq n \geq N$,

$$|x_n - x_m| = \left|\frac{1}{n} - \frac{1}{m}\right| \leq \frac{1}{n} \leq \frac{1}{N}$$

so any $N > 1/\epsilon$ witnesses the Cauchy condition for ϵ. Suppose $1/n \to x$ with $x \in (0, 1)$. Then $x > 0$, and every $n \geq 2/x$ satisfies $1/n \leq x/2$, hence

$$|x_n - x| = x - \frac{1}{n} \geq x - \frac{x}{2} = \frac{x}{2}$$

So no N works for $\epsilon = x/2$, and the sequence has no limit in $(0, 1)$.

Both failures have the same shape. Each space sits inside a larger one where the sequence does converge, and each fails to contain the limit: $\sqrt{2} \notin \mathbb{Q}$ and $0 \notin (0, 1)$. In both cases the offending subset is not closed in the larger space, $\overline{\mathbb{Q}} = \mathbb{R}$ and $\overline{(0, 1)} = [0, 1]$, and that observation is the whole content of the next theorem, which turns completeness of a subspace into a question about closure.

Theorem 2.1.8: Closed Subspaces Are Complete and Complete Subspaces Are Closed

Let (X, d) be a metric space and let $A \subseteq X$ carry the subspace metric.

(i) If X is complete and A is closed in X, then A is complete.

(ii) If A is complete, then A is closed in X, whether or not X is complete.

***Proof*:**
Throughout, the subspace metric on A is the restriction of d (definition 1.4.1), so for points of A the quantity $d(a, a')$ is the same computed in A or in X. A sequence in A is therefore Cauchy in A if and only if it is Cauchy in X, and it converges in A to a point $a \in A$ if and only if it

converges to a in X.

(i) Let (a_n) be a Cauchy sequence in A. By the previous paragraph it is Cauchy in X, so completeness of X gives an $x \in X$ with $a_n \to x$. Every term lies in A and A is closed, so $x \in A$ by corollary 1.2.10. Convergence of (a_n) to a point of A in X is convergence in A, so (a_n) converges in A. Hence A is complete.

(ii) By corollary 1.2.10 it suffices to show that the limit of every convergent sequence of points of A lies in A. So let $a_n \in A$ with $a_n \to x$ in X. The sequence is convergent in X, hence Cauchy in X by proposition 2.1.2, hence Cauchy in A. Completeness of A gives an $a \in A$ with $a_n \to a$ in A, and therefore $a_n \to a$ in X as well. A sequence in X has at most one limit by proposition 1.2.6, so $x = a \in A$, completing the proof.

The two halves have unequal hypotheses. Statement (ii) asks nothing of the ambient space: a complete subset of a metric space is closed in it, whether or not the ambient space is itself complete. Statement (i) does need X complete, and without that hypothesis it fails at once: $\mathbb{Q}$ is closed in $\mathbb{Q}$ and is not complete.

Inside a complete space the two conditions collapse into one, and that collapse is how completeness gets checked for every subspace of a space already known to be complete. The Cantor set C of example 1.2.13 is an intersection of closed subsets of $\mathbb{R}$, hence closed, hence complete. The interval $[0, 1]$ is complete for the same reason, while $(0, 1)$ is not closed in $\mathbb{R}$ and example 2.1.7 confirms that it is not complete. The criterion also recovers the first example: $\mathbb{Q}$ is dense in $\mathbb{R}$ (the case $n = 1$ of example 1.2.12), so $\overline{\mathbb{Q}} = \mathbb{R} \neq \mathbb{Q}$ and $\mathbb{Q}$ is not closed.

That criterion is stated relative to an ambient space, and it invites the question of whether completeness might, like openness or convergence, be a matter of the open sets alone. It is not, and the following example is the standard witness.

Example 2.1.9: Completeness Is Not Preserved by Homeomorphism

The map $f\colon \mathbb{R} \to (-1, 1)$ of example 1.3.12, given by $f(x) = x/(1 + |x|)$, is a homeomorphism. The line $\mathbb{R}$ is complete [1]. The interval $(-1, 1)$ is not: the points $x_n = 1 - 1/n$ all lie in $(-1, 1)$, and for $m \geq n \geq N$

$$|x_n - x_m| = \left|\frac{1}{m} - \frac{1}{n}\right| \leq \frac{1}{N}$$

so taking $N > 1/\epsilon$ makes (x_n) Cauchy. A limit $x \in (-1, 1)$ would also be a limit in $\mathbb{R}$, where the sequence converges to 1, so uniqueness of limits (proposition 1.2.6) would force $x = 1 \notin (-1, 1)$.

So two homeomorphic spaces differ on completeness. The difference can be arranged without changing the underlying set at all. On $\mathbb{R}$ define

$$\rho(x, y) = |f(x) - f(y)|$$

This is a metric: symmetry and the triangle inequality are inherited from the metric of $\mathbb{R}$ applied to the values $f(x)$, and $\rho(x, y) = 0$ gives $f(x) = f(y)$, hence $x = y$ since f is injective.

The metrics ρ and $d(x, y) = |x - y|$ are topologically equivalent in the sense of definition 1.4.5. Regarded as a map $(\mathbb{R}, \rho) \to (-1, 1)$, f is a surjective isometry by the definition of ρ, hence a homeomorphism (definition 1.3.11); regarded as a map $(\mathbb{R}, d) \to (-1, 1)$ it is a homeomorphism by example 1.3.12. The identity map $(\mathbb{R}, d) \to (\mathbb{R}, \rho)$ is the composite of the second with the inverse of the first, hence a homeomorphism by proposition 1.3.3, and a homeomorphism which is the identity on the underlying set says exactly that the two metrics have the same open sets.

Yet $(\mathbb{R}, \rho)$ is not complete. Take $x_n = n$. Then

$$\rho(n,m) = \left|\frac{n}{1+n} - \frac{m}{1+m}\right| = \frac{|n-m|}{(1+n)(1+m)} \leq \frac{1}{1+\min(n,m)}$$

where the last step uses $|n-m| \leq \max(n,m) < 1+\max(n,m)$ together with the factorization $(1+n)(1+m) = \big(1+\max(n,m)\big)\big(1+\min(n,m)\big)$. Given $\epsilon > 0$, any $N > 1/\epsilon$ makes $\rho(n,m) < \epsilon$ for all $n, m \geq N$, so $(n)_{n\geq 1}$ is ρ-Cauchy. It has no ρ-limit: the condition $\rho(n,x) \to 0$ says $f(n) \to f(x)$ in $\mathbb{R}$, and $f(n) = n/(1+n) \to 1$, so proposition 1.2.6 would give $f(x) = 1$, which no real x satisfies since $|f(x)| = |x|/(1+|x|) < 1$.

The same sequence shows ρ and d are not strongly equivalent, and shows it without any computation of constants: strongly equivalent metrics have the same Cauchy sequences, because $c_1 d \leq \rho \leq c_2 d$ converts each Cauchy estimate into the other with ϵ rescaled, and $(n)_{n\geq 1}$ is ρ-Cauchy but not d-Cauchy.

A property that every homeomorphism preserves is what the word "topological" means, and completeness is not one. The mechanism is theorem 1.3.2(iii): continuity is expressible in open sets alone, so a homeomorphism carries the open sets of one space bijectively onto those of the other, and every notion built from open sets alone (closure, interior, boundary, density, convergence, continuity of maps into and out of the space) transfers along with them. Completeness does not transfer, so completeness is not built from open sets alone. It is a property of the metric, and example 2.1.9 shows the metric can be exchanged for another with the same open sets and the opposite verdict.

The consequence constrains every later completeness proof. Open sets, closures, preimages and continuity may still appear in one, but only alongside a hypothesis carrying metric information: theorem 2.1.8(i) reasons through closedness, and it also assumes the ambient space complete, which is not data the open sets record. Each completeness proof in this book therefore either estimates distances directly, producing an N from an ϵ, or imports the estimate from a space already known to be complete. What does transfer is isometry, which preserves every quantity computed from distances (definition 1.3.11) and so preserves Cauchy sequences, convergent sequences and completeness alike. The gap example 1.3.12 opened between the two notions of isomorphism is exactly what example 2.1.9 measures.

Exercise 2.1.1

1. A uniformly continuous map carries Cauchy sequences to Cauchy sequences: the δ that definition 1.3.4 gives for a given ϵ converts a Cauchy estimate for δ into one for ϵ. Show that such a map need not carry completeness to completeness: let $\mathbb{N} = \{1, 2, 3, \dots\} \subseteq \mathbb{R}$ and $Y = \{1/n \mid n \geq 1\} \subseteq \mathbb{R}$ carry the metrics inherited from $\mathbb{R}$, and consider $g\colon \mathbb{N} \to Y$ with $g(n) = 1/n$. Show that $\mathbb{N}$ is complete, that g is a Lipschitz bijection, and that Y is not complete. Then explain why no Cauchy sequence of $\mathbb{N}$ witnesses the failure.

2. Let $(x_n)_{n\geq 1}$ be a sequence in a metric space with $\sum_{n=1}^{\infty} d(x_n, x_{n+1}) < \infty$. Show that (x_n) is Cauchy. Show that the weaker hypothesis $d(x_n, x_{n+1}) \to 0$ does not suffice, using the partial sums $x_n = \sum_{k=1}^{n} 1/k$ in $\mathbb{R}$.

3. Let X carry the discrete metric of example 1.1.5. Show that a sequence in X is Cauchy if and only if it is eventually constant, and deduce that X is complete. Show also that every subset of X is closed, and check that this agrees with theorem 2.1.8(ii).

4. Let $d(x,y) = |x-y|$ and $d'(x,y) = \min(|x-y|, 1)$ on $\mathbb{R}$, the pair of example 1.4.8, which is topologically but not strongly equivalent. Show that d and d' nevertheless have exactly the same Cauchy sequences, and that $(\mathbb{R}, d')$ is complete. Conclude that strong equivalence is sufficient but not necessary for two metrics to share their Cauchy sequences.

5. Show that the Cantor set C of example 1.2.13 is complete. Verify that $t_n = \sum_{j=1}^{n} 2 \cdot 3^{1-2j}$ lies in C for every n, compute $\lim_n t_n$, and check directly that the limit lies in C. Then explain why $\mathbb{Q}$, which like C has empty interior in $\mathbb{R}$, fails to be complete.

6. Show that a metric space X is complete if and only if every Cauchy sequence in X has a convergent subsequence. Deduce that a metric space in which *every* sequence has a convergent subsequence is complete.

2.2 The Complete Spaces of Analysis

Section 2.1 settled what completeness says; which spaces satisfy it is a separate question, and one that has to be answered space by space. This section answers it for the spaces the rest of the book runs on: $\mathbb{R}^n$, the bounded functions on a set and the bounded continuous functions on a metric space under the sup metric, and the sequence spaces ℓ^p. It closes with a space that fails, and with the reading of that failure: one set of functions, complete under one metric and not under another.

The first case costs almost nothing, because the completeness of $\mathbb{R}$ is available from *Elementary Analysis* [1] and a point of $\mathbb{R}^n$ is a finite list of reals. What has to be checked is that the reduction to coordinates works for Cauchy sequences and not just for convergent ones, and section 1.4 gave both halves of that reduction.

Theorem 2.2.1: Completeness of Finite Products and of $\mathbb{R}^n$

Let (X, d_X) and (Y, d_Y) be complete metric spaces. Then $X \times Y$ is complete under each of the three product metrics of definition 1.4.3. Consequently $\mathbb{R}^n$ is complete under each of d_1, d_2 and d_∞.

***Proof*:**
By proposition 1.4.4 each of the three product metrics is bounded above and below by a constant multiple of $d_{\max}$, so rescaling the tolerance ϵ by that constant converts a Cauchy condition for one of them into a Cauchy condition for another; and by proposition 1.4.9 the three have the same convergent sequences. It is therefore enough to argue for $d_{\max}$.

Let $(m_k)_{k\geq 1}$ be Cauchy in $X \times Y$ for $d_{\max}$, and write $m_k = (x_k, y_k)$. Since

$$d_X(x_k, x_l) \leq d_{\max}(m_k, m_l) \qquad \text{and} \qquad d_Y(y_k, y_l) \leq d_{\max}(m_k, m_l)$$

every N witnessing the Cauchy condition for (m_k) witnesses it for (x_k) in X and for (y_k) in Y as well. Completeness of the factors gives limits $x_k \to x$ and $y_k \to y$, and proposition 1.4.9 converts the pair of coordinate convergences back into $m_k \to (x, y)$ in $X \times Y$.

For $\mathbb{R}^n$, induct on n. The case $n = 1$ is the completeness of $\mathbb{R}$ [1]. For the inductive step, identify $\mathbb{R}^n$ with $\mathbb{R}^{n-1} \times \mathbb{R}$. Under that identification the metric d_∞ of example 1.1.3 on $\mathbb{R}^n$ is exactly $d_{\max}$ built from d_∞ on $\mathbb{R}^{n-1}$ and $|\cdot|$ on $\mathbb{R}$, since a maximum over n numbers is the maximum

of the first $n-1$ and the last. The first part of the theorem and the inductive hypothesis give completeness for d_∞, and the comparison $d_\infty \le d_2 \le d_1 \le n\, d_\infty$ recorded in corollary 1.4.7 transfers it to d_1 and d_2 by the same rescaling, completing the proof.

No new analysis entered that argument: the completeness of $\mathbb{R}^n$ is the completeness of $\mathbb{R}$ used n times. Two conveniences made it cheap, and both are consequences of the number of coordinates being finite. The candidate limit was assembled from finitely many separate limits, so no care was needed about uniformity in the coordinate; and once assembled it was automatically a point of the space, since every n-tuple of reals lies in $\mathbb{R}^n$. Neither convenience persists in the passage to infinitely many coordinates, and the proofs below are longer precisely where they fail.

Combining the theorem with theorem 2.1.8 produces many complete spaces at once: every closed subset of $\mathbb{R}^n$ is complete in the subspace metric. The closed unit ball, any sphere $S_r(x)$, and the Cantor set C of example 1.2.13, which is closed as an intersection of closed sets, are all complete metric spaces in their own right. The rationals $\mathbb{Q}^n$ are not, being dense in $\mathbb{R}^n$ without being closed.

$\mathbb{R}^n$ was cheap because a point of it is a finite list. The spaces that follow have points that are functions, and there the data attached to a single point is already infinite. The largest such space requires nothing of its functions but boundedness, and it is proved complete first because the two spaces that matter both sit inside it: the bounded continuous functions as a closed subspace, and ℓ^∞ as an equality.

For a nonempty set X, let $B(X)$ denote the set of bounded functions $f\colon X \to \mathbb{R}$ and put

$$d_\infty(f,g) = \sup_{x\in X} \big|f(x) - g(x)\big|$$

The supremum is finite because $f-g$ is bounded, and the verification that d_∞ is a metric is the argument of example 1.1.4 with $[a,b]$ replaced by X: that argument used only boundedness of the difference, never continuity or any structure on the domain. When X is a nonempty metric space, $C_b(X) \subseteq B(X)$ denotes the set of bounded continuous functions $X \to \mathbb{R}$, carrying the subspace metric of definition 1.4.1, which is again d_∞.

Theorem 2.2.2: Completeness of the Bounded Functions

Let X be a nonempty set. Then $\big(B(X), d_\infty\big)$ is complete.

***Proof*:**

Let $(f_n)_{n\ge1}$ be Cauchy in $B(X)$.

Step 1: manufacturing the candidate limit. Fix $x \in X$. Since $|f_n(x) - f_m(x)|$ is one of the numbers whose supremum is $d_\infty(f_n, f_m)$,

$$\big|f_n(x) - f_m(x)\big| \le d_\infty(f_n, f_m)$$

so the real sequence $\big(f_n(x)\big)_{n\ge1}$ is Cauchy, and by the completeness of $\mathbb{R}$ [1] it converges. Define

$$f(x) = \lim_{n\to\infty} f_n(x)$$

which specifies a function $f\colon X \to \mathbb{R}$. It is the only possible limit: if $g \in B(X)$ satisfied $d_\infty(f_n, g) \to 0$ then the displayed inequality with g in place of f_m would force $f_n(x) \to g(x)$ at every x, and limits of real sequences are unique.

Step 2: the candidate lies in the space. Apply the Cauchy condition with $\epsilon = 1$ to get an N with $d_\infty(f_n, f_N) \leq 1$ for all $n \geq N$. Writing $M = \sup_x |f_N(x)|$, which is finite because f_N is bounded, every $x \in X$ and every $n \geq N$ satisfy

$$\big|f_n(x)\big| \leq \big|f_n(x) - f_N(x)\big| + \big|f_N(x)\big| \leq 1 + M$$

Letting $n \to \infty$ at the fixed point x preserves the non-strict inequality, so $|f(x)| \leq 1 + M$ for every $x \in X$. Hence f is bounded and $f \in B(X)$.

Step 3: convergence in the metric. This is the step where the order in which the quantifiers are chosen decides the argument. Let $\epsilon > 0$ and choose N so that

$$d_\infty(f_n, f_m) \leq \frac{\epsilon}{2} \qquad \text{for all } n, m \geq N$$

This N is fixed now, before any point of X is named. Fix $n \geq N$ and $x \in X$. For every $m \geq N$,

$$\big|f_n(x) - f_m(x)\big| \leq d_\infty(f_n, f_m) \leq \frac{\epsilon}{2}$$

As $m \to \infty$ the left-hand side converges to $|f_n(x) - f(x)|$, by Step 1 and the continuity of the metric on $\mathbb{R}$ (proposition 1.3.8), and a non-strict inequality survives that limit:

$$\big|f_n(x) - f(x)\big| \leq \frac{\epsilon}{2}$$

The point x was arbitrary and N did not depend on it, so this bound holds at every point of X simultaneously and $\epsilon/2$ is an upper bound for the set whose supremum is $d_\infty(f_n, f)$. Therefore $d_\infty(f_n, f) \leq \epsilon/2 < \epsilon$ for all $n \geq N$, which says $f_n \to f$ in $B(X)$, completing the proof.

The three steps are the template for every completeness proof about a space of functions or sequences, and the later ones in this section follow it without deviation. Step 1 uses nothing about the space beyond the completeness of $\mathbb{R}$: the limit is built one point at a time, and at that stage it is only a function, with no claim on membership. Step 2 checks the defining property of the space, here boundedness, and it is the step that fails when the space is chosen badly. Step 3 upgrades convergence at each point separately to convergence in the metric, and the mechanism of the upgrade is that N is chosen from the Cauchy condition before x is named. Choosing N after x would give a bound at each point and no bound on the supremum, which is exactly the difference between a sequence of functions converging at every point and converging in d_∞. **??** makes that difference its subject.

The space of bounded real sequences is an instance rather than an analogue. A sequence is a function $\mathbb{N} \to \mathbb{R}$; it is bounded as a sequence exactly when it is bounded as a function; and the metric $d_\infty(x, y) = \sup_k |x_k - y_k|$ of example 1.1.9 is the sup metric of $B(\mathbb{N})$ written in sequence notation. So $\ell^\infty = B(\mathbb{N})$ as metric spaces, and theorem 2.2.2 already proves ℓ^∞ complete.

The set $B(X)$ is large and has no analytic content of its own, since it requires nothing of its functions except that their values not run away. The space analysis works in is the continuous part of it, and the passage from one to the other is handled by theorem 2.1.8: rather than running the three steps again, it is enough to show that $C_b(X)$ is closed in $B(X)$. That closedness is a statement with a name of its own, namely that a limit in the sup metric of continuous functions is continuous, and the proof is the standard splitting of one estimate into three.

Theorem 2.2.3: Completeness of the Bounded Continuous Functions

Let X be a nonempty metric space. Then $C_b(X)$ is a closed subset of $\big(B(X), d_\infty\big)$, and $\big(C_b(X), d_\infty\big)$ is complete. In particular $C[a,b]$ with the metric of example 1.1.4 is complete.

***Proof*:**

By corollary 1.2.10 it suffices to show that whenever $f_n \in C_b(X)$ and $f_n \to f$ in $B(X)$, the limit f is continuous, since f is then bounded and continuous and so lies in $C_b(X)$.

Fix $x_0 \in X$ and $\epsilon > 0$. Choose n with $d_\infty(f_n, f) < \epsilon/3$, which is possible because $d_\infty(f_n, f) \to 0$, and hold that n fixed. Continuity of f_n at x_0 gives a $\delta > 0$ such that $d(x, x_0) < \delta$ implies $|f_n(x) - f_n(x_0)| < \epsilon/3$. For any such x, inserting f_n twice and applying the triangle inequality in $\mathbb{R}$ gives

$$\big|f(x) - f(x_0)\big| \le \big|f(x) - f_n(x)\big| + \big|f_n(x) - f_n(x_0)\big| + \big|f_n(x_0) - f(x_0)\big|$$

The first and third terms are each at most $d_\infty(f, f_n) < \epsilon/3$, and the middle term is less than $\epsilon/3$ by the choice of δ, so the sum is less than ϵ. Hence f is continuous at x_0, and since x_0 was arbitrary, $f \in C_b(X)$. So $C_b(X)$ is closed in $B(X)$, and theorem 2.2.2 together with theorem 2.1.8 makes it complete.

For the last claim, take $X = [a,b]$. Every continuous function on a closed bounded interval is bounded [1], so $C[a,b] = C_b([a,b])$ as sets, and the metric of example 1.1.4 is d_∞, completing the proof.

The two outer terms in the displayed estimate are controlled at the point x and at the point x_0 by the same number $d_\infty(f, f_n)$, and that is the whole content of the argument. A bound valid at each point separately would not do: the first term is evaluated at a point x that is only produced after n has been chosen, so any control over it has to be uniform in the point. This is why the theorem is false for pointwise limits. On $[0,1]$ the functions $f_n(x) = x^n$ are continuous and converge at every point, to 0 for $x < 1$ and to 1 at $x = 1$, and the limit is not continuous. Consistently with theorem 2.2.3, the sequence (x^n) is not Cauchy in $C[0,1]$: for any n, substituting $t = x^n$, which ranges over all of $[0,1]$ as x does,

$$d_\infty(f_n, f_{2n}) = \sup_{x \in [0,1]} \big|x^n - x^{2n}\big| = \max_{t \in [0,1]} (t - t^2) = \frac{1}{4}$$

attained at $t = \frac{1}{2}$, that is, at $x = 2^{-1/n}$. A Cauchy sequence in $C[0,1]$ cannot have two terms at distance $\frac{1}{4}$ arbitrarily far out.

Also complete, by the same theorem, is every closed subset of $C[a,b]$: the functions vanishing at a prescribed point, the functions with values in $[0,1]$, the functions with a prescribed integral. Each is complete because it is closed, and each is closed because it is cut out by conditions preserved under convergence in d_∞.

The sequence spaces ℓ^p with p finite are the remaining running example, and they are where the two conveniences of theorem 2.2.1 both fail. The coordinatewise argument still produces a candidate limit, but that candidate need not lie in the space: the points $x^{(k)} = (1, \dots, 1, 0, 0, \dots)$ with k ones all lie in ℓ^1 and converge in each coordinate to $(1, 1, 1, \dots)$, which lies in no ℓ^p with p finite. What rules such behaviour out for a *Cauchy* sequence is a bound on the tails that is uniform in k, and producing that bound is the content of the next proof.

Theorem 2.2.4: Completeness of the Sequence Spaces

Let $1 \leq p < \infty$. Then (ℓ^p, d_p) is complete, and (ℓ^∞, d_∞) is complete.

***Proof*:**
The case of ℓ^∞ is theorem 2.2.2 in different notation: $\ell^\infty = B(\mathbb{N})$ as metric spaces, since a bounded sequence is a bounded function $\mathbb{N} \to \mathbb{R}$ and the two metrics are the same supremum.

Fix $1 \leq p < \infty$ for the remainder, and let $\left(x^{(k)}\right)_{k\geq 1}$ be Cauchy in ℓ^p, with $x^{(k)} = \left(x_1^{(k)}, x_2^{(k)}, \dots\right)$.

Step 1: the coordinatewise limit. Fix an index j. A single term of a sum of nonnegative terms is at most the whole sum, so

$$\left|x_j^{(k)} - x_j^{(l)}\right| = \left(\left|x_j^{(k)} - x_j^{(l)}\right|^p\right)^{1/p} \leq \left(\sum_{i=1}^{\infty} \left|x_i^{(k)} - x_i^{(l)}\right|^p\right)^{1/p} = d_p\left(x^{(k)}, x^{(l)}\right)$$

Hence $\left(x_j^{(k)}\right)_{k\geq 1}$ is Cauchy in $\mathbb{R}$ and converges; set $x_j = \lim_k x_j^{(k)}$ and $x = (x_j)_{j\geq 1}$.

Step 2: the tail estimate, and membership in the space. Let $\epsilon > 0$ and choose N with $d_p\left(x^{(k)}, x^{(l)}\right) \leq \epsilon$ for all $k, l \geq N$. Fix $k \geq N$ and a finite $J \geq 1$. For every $l \geq N$, discarding the terms beyond J from a sum of nonnegative terms gives

$$\sum_{j=1}^{J} \left|x_j^{(k)} - x_j^{(l)}\right|^p \leq d_p\left(x^{(k)}, x^{(l)}\right)^p \leq \epsilon^p \tag{2.1}$$

The left-hand side of (2.1) is a sum of finitely many terms, each of which converges as $l \to \infty$ by Step 1, so letting $l \to \infty$ preserves the bound:

$$\sum_{j=1}^{J} \left|x_j^{(k)} - x_j\right|^p \leq \epsilon^p$$

This holds for every J, and the left-hand side is nondecreasing in J, so the series converges and its sum obeys the same bound:

$$\sum_{j=1}^{\infty} \left|x_j^{(k)} - x_j\right|^p \leq \epsilon^p \qquad \text{for all } k \geq N \tag{2.2}$$

In particular the sequence $x^{(k)} - x$ lies in ℓ^p. Since $x^{(k)} \in \ell^p$ and ℓ^p is closed under differences by theorem 1.1.8, the identity $x = x^{(k)} - \left(x^{(k)} - x\right)$ places x in ℓ^p.

Step 3: convergence in the metric. Taking pth roots in (2.2) reads $d_p\left(x^{(k)}, x\right) \leq \epsilon$ for all $k \geq N$. Since $\epsilon > 0$ was arbitrary and N was chosen from it alone, $x^{(k)} \to x$ in ℓ^p, completing the proof.

The proof is theorem 2.2.2 again with "bounded" replaced by "p-summable": a limit built one coordinate at a time, a check that it lies in the space, and an upgrade from coordinatewise convergence to convergence in the metric. The device carrying Step 2 is worth isolating, since it recurs whenever a statement about an infinite sum has to be extracted from statements about its terms. A bound on

the full sum is inherited by each finite truncation; a finite truncation involves finitely many limits and so passes to the limit termwise; and the resulting bound, being independent of where the truncation was made, passes back to the full sum by monotone convergence of the partial sums. No interchange of two infinite limits is ever performed, which is what makes the argument elementary.

Steps 2 and 3 also came out of a single computation, (2.2), where in theorem 2.2.2 they were separate. That is a feature of the norm rather than a shortcut: the quantity that has to be small for convergence, and the quantity that has to be finite for membership, are the same sum.

Every space in this section so far has been complete. The next one is not, and it differs from the incomplete spaces of example 2.1.7 in the respect that matters here: its underlying set is $C[0,1]$, exactly the set that theorem 2.2.3 has just proved complete, and the only thing changed is the metric.

Example 2.2.5: Incompleteness of $C[0,1]$ in the L^1 Metric

On $C[0,1]$ define

$$d_1(f,g) = \int_0^1 \big|f(x) - g(x)\big|\,dx$$

The subscript repeats the one carried by the ℓ^1 metric of example 1.1.9 and by the taxicab metric on $\mathbb{R}^n$, the pattern in each case being a sum or an integral of first powers; the ambient space says which is meant. The integral exists because $|f-g|$ is continuous, hence Riemann integrable [1]. Symmetry is clear, the triangle inequality follows from $|f-h| \le |f-g| + |g-h|$ pointwise and monotonicity of the integral, and the separation axiom is the following observation, used twice more below: if h is continuous on an interval $[a,b]$ with $a<b$ and $\int_a^b |h| = 0$, then h vanishes identically there. Otherwise $|h(x_0)| = c > 0$ for some $x_0 \in [a,b]$, and continuity gives an $\eta > 0$ with $|h| > c/2$ on an interval $J \subseteq [a,b]$ of length at least η containing x_0, whence $\int_a^b |h| \ge \int_J |h| \ge c\eta/2 > 0$.

For $n \ge 2$ let $f_n \colon [0,1] \to \mathbb{R}$ be the ramp of width $1/n$ that climbs from 0 to 1 just to the right of $\frac{1}{2}$:

$$f_n(x) = \begin{cases} 0 & 0 \le x \le \frac{1}{2} \\ n\big(x - \frac{1}{2}\big) & \frac{1}{2} \le x \le \frac{1}{2} + \frac{1}{n} \\ 1 & \frac{1}{2} + \frac{1}{n} \le x \le 1 \end{cases}$$

The three formulas agree at the two joins, giving 0 at $x = \frac{1}{2}$ and 1 at $x = \frac{1}{2} + \frac{1}{n}$, so each f_n is continuous.

The sequence is Cauchy. For $m \ge n \ge 2$ the functions f_n and f_m agree on $[0, \frac{1}{2}]$, where both vanish, and on $[\frac{1}{2} + \frac{1}{n}, 1]$, where both equal 1. On the remaining interval, of length $1/n$, both take values in $[0,1]$, so their difference is at most 1 in absolute value. Hence

$$d_1(f_n, f_m) = \int_{1/2}^{1/2+1/n} \big|f_n - f_m\big| \le \frac{1}{n}$$

and given $\epsilon > 0$ any $N > 1/\epsilon$ witnesses the Cauchy condition.

No continuous function is its limit. Suppose $f \in C[0,1]$ satisfied $d_1(f_n, f) \to 0$. Since f_n vanishes on $[0, \frac{1}{2}]$,

$$\int_0^{1/2} |f| = \int_0^{1/2} \big|f - f_n\big| \le d_1(f_n, f)$$

for every n; the left-hand side does not depend on n, so it is 0, and by the observation above f vanishes identically on $[0, \frac{1}{2}]$. Now fix $\eta \in (0, \frac{1}{2})$. For every $n > 1/\eta$ the function f_n equals 1 on

$[\frac{1}{2}+\eta, 1]$, so the same argument gives

$$\int_{1/2+\eta}^{1} |f-1| = \int_{1/2+\eta}^{1} |f-f_n| \le d_1(f_n, f)$$

and hence $f \equiv 1$ on $[\frac{1}{2}+\eta, 1]$. As η was arbitrary in $(0, \frac{1}{2})$, $f \equiv 1$ on $(\frac{1}{2}, 1]$. But then $f(\frac{1}{2}) = 0$ while $f(\frac{1}{2}+\frac{1}{k}) = 1$ for all large k, so f does not preserve the convergence $\frac{1}{2}+\frac{1}{k} \to \frac{1}{2}$ and is not continuous at $\frac{1}{2}$ by theorem 1.3.2(ii). This contradiction shows that $\big(C[0,1], d_1\big)$ is not complete.

The sequence is not failing to converge for want of trying. Let g be the step function equal to 0 on $[0, \frac{1}{2}]$ and to 1 on $(\frac{1}{2}, 1]$, which is bounded with a single discontinuity and so is Riemann integrable. The same computation as above gives $\int_0^1 |f_n - g| \le 1/n$, so in any space of integrable functions containing g the ramps do converge, and they converge to g. The limit exists; it is the space that is too small to hold it.

Two conclusions follow. The first is about completeness itself. The set $C[0,1]$ is complete under d_∞ by theorem 2.2.3 and incomplete under d_1 by example 2.2.5, with not a single function added or removed. Completeness is therefore a property of the pair (X, d) and not of the underlying set: nothing about a set of points, on its own, determines whether its Cauchy sequences converge. That the topology does not determine it either is the separate content of example 2.1.9, where the two metrics on $\mathbb{R}$ induce the same open sets and still disagree about completeness. The two metrics compared here do not induce the same open sets, so the present example does not reach that far; what it adds is that the phenomenon already occurs on the space analysis works in, between the two metrics that space is most often given.

The second is about what to do with a space that fails. There are two repairs. One is abstract and applies to every metric space at once: adjoin the missing limits formally, which is the completion constructed in section 2.4. The other is to identify the completion concretely, and for d_1 that means enlarging $C[0,1]$ to a space of functions closed under the limits the metric requires. The enlargement cannot be carried out with the Riemann integral, whose class of integrable functions is not closed under such limits either; building an integral whose function space is complete is the business of measure theory, and the space that results is called $L^1[0,1]$.[1] The failure exhibited here is a large part of why that theory exists.

Exercise 2.2.1

1. Show that a subset $A \subseteq \mathbb{R}^n$ is complete in the subspace metric if and only if it is closed in $\mathbb{R}^n$. Which of $\mathbb{Q} \cap [0,1]$, the Cantor set, and $\{x \in \mathbb{R}^2 \mid \|x\| < 1\}$ are complete?

2. For $n \ge 0$ let $f_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}$, a polynomial and hence an element of $C[0,1]$. Compute $d_\infty(f_n, f_m)$ for $m > n$ exactly, show that (f_n) is Cauchy in $\big(C[0,1], d_\infty\big)$, and identify its limit.

3. Let X be a nonempty set and $a \in X$. Show that $\{f \in B(X) \mid f(a) = 0\}$ is complete under d_∞. Show that $\{f \in B(X) \mid \sup_{x \in X} |f(x)| < 1\}$ is not.

4. In ℓ^1, exhibit a sequence of points converging in every coordinate to a point of ℓ^1 but not converging in d_1, and a sequence of points converging in every coordinate to a sequence lying in no ℓ^p with p finite. Verify directly that neither sequence is Cauchy in d_1, as theorem 2.2.4

[1] *Measure Theory* [7] constructs the Lebesgue integral and proves L^1 complete; that volume depends on this one, so the pointer runs forward and nothing below relies on it.

requires.

5. Regard ℓ^1 as a subset of ℓ^∞ and give it the metric d_∞. Show that $\left(\ell^1, d_\infty\right)$ is not complete, using the points $y^{(k)} = \left(1, \frac{1}{2}, \ldots, \frac{1}{k}, 0, 0, \ldots\right)$. Reconcile this with theorem 2.2.4.

6. Show that $d_1(f,g) \leq d_\infty(f,g)$ for all $f, g \in C[0,1]$, where d_1 is the metric of example 2.2.5. Deduce from example 2.2.5 and theorem 2.2.3 that d_1 and d_∞ are not strongly equivalent. Then show they are not topologically equivalent either, by proving that $\{f \in C[0,1] \mid d_\infty(f,0) < 1\}$ contains no d_1-ball about the zero function.

2.3 Cantor's Intersection Theorem

Completeness has so far been something to verify: a property a space either has or lacks, checked case by case. This section spends it. The mechanism is a way of pinning down a point by a sequence of ever-tighter constraints, no one of which names it, and then letting completeness give the point the constraints describe. Cantor's intersection theorem is that mechanism in its bare form, and two consequences follow from it: a characterization of completeness in terms of nested sets, and the fact that a complete space with no isolated points is too large to be listed.

The constraints will be sets, and "ever tighter" has to be measured. In a bare metric space the only measurement available is the largest distance between two points of the set.

Definition 2.3.1: Diameter of a Set

Let (X,d) be a metric space and $A \subseteq X$. The *diameter* of A is

$$\operatorname{diam}(A) = \sup\{d(a,b) \mid a, b \in A\}$$

where $\operatorname{diam}(A) = \infty$ if the set of distances is unbounded, and $\operatorname{diam}(\emptyset) = 0$.

Two properties are immediate from the definition and are used constantly below. The first is monotonicity: if $A \subseteq B$ then the supremum defining $\operatorname{diam}(A)$ is taken over a subset of the set defining $\operatorname{diam}(B)$, so $\operatorname{diam}(A) \leq \operatorname{diam}(B)$. The second is that $\operatorname{diam}(A) < \infty$ says exactly that a nonempty A fits inside a single ball, the empty set being contained in every ball already. Indeed, if $\operatorname{diam}(A) = D < \infty$ and $a_0 \in A$ is any point, then every $a \in A$ satisfies $d(a, a_0) \leq D$, so $A \subseteq \overline{B}_D(a_0)$; conversely if $A \subseteq \overline{B}_r(x)$ then any $a, b \in A$ satisfy $d(a,b) \leq d(a,x) + d(x,b) \leq 2r$, so $\operatorname{diam}(A) \leq 2r$, and the same bound holds for $A \subseteq B_r(x)$ because $B_r(x) \subseteq \overline{B}_r(x)$. Taking A to be the closed ball itself records the fact that

$$\operatorname{diam}\left(\overline{B}_r(x)\right) \leq 2r$$

in every metric space. Diameter is thus a quantitative refinement of boundedness, and the theorem below needs the quantity, not just the qualitative statement.

Example 2.3.2: Diameters in the Standard Spaces

(i) In $\mathbb{R}$ with $d(x,y) = |x-y|$, the interval $[a,b]$ has diameter $b-a$, attained at the endpoints, and so does (a,b), where the supremum is not attained. The closed ball $\overline{B}_r(x) = [x-r, x+r]$ has diameter exactly $2r$, and so does the open ball $B_r(x) = (x-r, x+r)$, so the bound of $2r$ recorded above is sharp on the real line.

(ii) Under the discrete metric of example 1.1.5, $\operatorname{diam}(A) = 1$ for every A with at least two points and $\operatorname{diam}(A) = 0$ otherwise. An uncountable set and a two-point set have the same diameter: the quantity measures separation, not size.

(iii) In $C[0,1]$ with the sup metric of example 1.1.4, the set $A = \{f \in C[0,1] \mid \sup_x |f(x)| \leq 1\}$ has diameter 2. It is at most 2 because $d_\infty(f,g) \leq \sup_x |f(x)| + \sup_x |g(x)| \leq 2$, and it is at least 2 because the constant functions 1 and -1 both lie in A and are at distance 2.

(iv) In the Cantor set of example 1.2.13, each of the 2^n constituent intervals of C_n has diameter 3^{-n}, while $\operatorname{diam}(C) = 1$ because 0 and 1 both survive every stage. The diameters that tend to zero are the pieces, not the set.

Diameter interacts with closure in the simplest possible way, and the next lemma is what lets a hypothesis about closed sets be arranged rather than assumed.

Lemma 2.3.3: Diameter of a Closure

For every $A \subseteq X$, $\operatorname{diam}(\overline{A}) = \operatorname{diam}(A)$.

***Proof*:**
Since $A \subseteq \overline{A}$, monotonicity gives $\operatorname{diam}(A) \leq \operatorname{diam}(\overline{A})$, and if $\operatorname{diam}(A) = \infty$ the two are equal. So assume $\operatorname{diam}(A) < \infty$ and let $x, y \in \overline{A}$ and $\epsilon > 0$. By theorem 1.2.9(ii) the balls $B_{\epsilon/2}(x)$ and $B_{\epsilon/2}(y)$ each meet A, say in a and b respectively. The triangle inequality applied twice gives

$$d(x,y) \leq d(x,a) + d(a,b) + d(b,y) < \frac{\epsilon}{2} + \operatorname{diam}(A) + \frac{\epsilon}{2}$$

Since $\epsilon > 0$ was arbitrary and $d(x,y)$ does not depend on it, $d(x,y) \leq \operatorname{diam}(A)$. Taking the supremum over $x, y \in \overline{A}$ gives $\operatorname{diam}(\overline{A}) \leq \operatorname{diam}(A)$, as we sought to show.

Passing to the closure therefore changes nothing metrically: a nested family of sets with diameters tending to zero can always be replaced by the family of their closures, which is still nested (closure preserves inclusion) and still has diameters tending to zero. What the replacement does not preserve is the intersection, and that discrepancy is exactly the content of the closedness hypothesis in the theorem below.

The theorem itself answers a question that single-variable analysis answers for intervals. A nested sequence of closed intervals in $\mathbb{R}$ whose lengths shrink to zero has exactly one common point [1], and that point is often the only description available of the number being constructed: it is how a decimal expansion specifies a real number, one digit narrowing the interval at a time. Nothing in that argument uses intervals, or $\mathbb{R}$, beyond the fact that Cauchy sequences converge.

Theorem 2.3.4: Cantor's Intersection Theorem

Let (X, d) be a complete metric space and let

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$$

be nonempty closed subsets of X with $\operatorname{diam}(A_n) \to 0$ as $n \to \infty$. Then $\bigcap_{n\geq 1} A_n$ consists of exactly one point.

***Proof*:**

Step 1: a sequence attached to the nest. Each A_n is nonempty, so choose a point $x_n \in A_n$ for every $n \geq 1$.

Step 2: the sequence is Cauchy. Let $\epsilon > 0$ and choose N with $\operatorname{diam}(A_N) < \epsilon$, which is possible because the diameters tend to zero. If $m, n \geq N$ then $A_m \subseteq A_N$ and $A_n \subseteq A_N$ by nesting, so both x_m and x_n lie in A_N and therefore

$$d(x_m, x_n) \leq \operatorname{diam}(A_N) < \epsilon$$

which is the Cauchy condition of definition 2.1.1.

Step 3: the limit lies in every set. By completeness there is an $x \in X$ with $x_n \to x$. Fix n. For every $m \geq n$ the point x_m lies in $A_m \subseteq A_n$, so the tail $(x_m)_{m\geq n}$ is a sequence in A_n converging to x. Since A_n is closed, corollary 1.2.10 puts $x \in A_n$. As n was arbitrary, $x \in \bigcap_n A_n$.

Step 4: there is no second point. Suppose $x, y \in \bigcap_n A_n$. For every n both points lie in A_n, so $d(x, y) \leq \operatorname{diam}(A_n)$, and the right side tends to 0. Hence $d(x, y) = 0$ and $x = y$ by the first metric axiom of definition 1.1.1, completing the proof.

Every hypothesis is used at an identifiable place. Nonemptiness gives the sequence in Step 1; nesting and the shrinking diameters make it Cauchy in Step 2; completeness converts the Cauchy sequence into an actual point in Step 3; closedness keeps that point inside each A_n in Step 3; and the shrinking diameters return in Step 4 to force uniqueness. The theorem is worth reading as a manufacturing process rather than a statement about intersections: it takes a descending list of constraints and returns a point meeting all of them, and the point need not be given by any formula. **??** manufactures a point in exactly this spirit, drawing on completeness directly rather than on the theorem, and what it produces is a fixed point.

Since each hypothesis has a job, dropping any one of them should break the conclusion, and it does.

Example 2.3.5: Nested Closed Sets with Empty Intersection

(i) *Without completeness.* Work in $\mathbb{Q}$ with the metric inherited from $\mathbb{R}$ and set

$$A_n = \left\{ q \in \mathbb{Q} \;\middle|\; |q - \sqrt{2}| \leq \tfrac{1}{n} \right\}$$

Each A_n is nonempty, since $\mathbb{Q}$ is dense in $\mathbb{R}$ [1]; each is closed in $\mathbb{Q}$, being the intersection with $\mathbb{Q}$ of a closed subset of $\mathbb{R}$ (proposition 1.4.2 applied to complements); the family is nested; and $\operatorname{diam}(A_n) \leq 2/n \to 0$. But a rational lying in every A_n would satisfy $|q - \sqrt{2}| \leq 1/n$ for all n, hence $q = \sqrt{2}$, which is not rational. So $\bigcap_n A_n = \emptyset$.

(ii) *Without closedness.* In $\mathbb{R}$, which is complete by theorem 2.2.1, put $A_n = (0, \frac{1}{n})$. These are nonempty, nested, and $\operatorname{diam}(A_n) = 1/n \to 0$, but a point x of the intersection would satisfy $0 < x < 1/n$ for every n, which is impossible. The closures $\overline{A}_n = [0, \frac{1}{n}]$ satisfy every hypothesis, and by lemma 2.3.3 they have the same diameters; their intersection is $\{0\}$, a point that belonged to none of the original sets.

(iii) *Without shrinking diameters.* In $\mathbb{R}$ put $A_n = [n, \infty)$. These are nonempty, closed and nested, with empty intersection, and every one has infinite diameter.

(iv) *Without shrinking diameters, second version.* Finite diameter is not enough either. In ℓ^2 let e_k denote the sequence with a 1 in position k and zeros elsewhere, and put $A_n = \{e_k \mid k \geq n\}$. Any two distinct members are at distance $\sqrt{2}$, so a convergent sequence drawn from A_n is Cauchy and hence eventually constant, which puts its limit in A_n; by corollary 1.2.10 each A_n is closed. The family is nested and nonempty, and $\operatorname{diam}(A_n) = \sqrt{2}$ for every n. Yet $e_k \notin A_{k+1}$, so no point survives all stages and the intersection is empty.

Item (iii) shows that the diameter condition cannot simply be dropped, and item (iv) shows more, that it cannot even be relaxed to boundedness in a complete space: the sets of unit vectors in ℓ^2 are closed, nested, nonempty and of constant diameter $\sqrt{2}$, and they close in on nothing.

The theorem used completeness, and the failure in $\mathbb{Q}$ shows that completeness is not decoration there. In fact the nesting property is strong enough to recover completeness on its own, which turns theorem 2.3.4 from a consequence of completeness into an equivalent formulation of it.

Corollary 2.3.6: Completeness from Nested Sets

A metric space X is complete if and only if every nested sequence $A_1 \supseteq A_2 \supseteq \cdots$ of nonempty closed subsets of X with $\operatorname{diam}(A_n) \to 0$ has nonempty intersection.

***Proof*:**
If X is complete then theorem 2.3.4 gives the intersection property, with the stronger conclusion that the intersection is a single point.

For the converse, assume the intersection property and let $(x_n)_{n\geq 1}$ be a Cauchy sequence in X. Define the tails

$$T_n = \{x_m \mid m \geq n\} \qquad A_n = \overline{T}_n$$

Each A_n is closed and nonempty, and $T_{n+1} \subseteq T_n$ gives $A_{n+1} \subseteq A_n$, so the family is nested. For the diameters, let $\epsilon > 0$ and use the Cauchy condition to choose N with $d(x_m, x_k) < \epsilon/2$ for all $m, k \geq N$. Then $\operatorname{diam}(T_N) \leq \epsilon/2$, and lemma 2.3.3 gives $\operatorname{diam}(A_N) = \operatorname{diam}(T_N) \leq \epsilon/2 < \epsilon$; by monotonicity $\operatorname{diam}(A_n) \leq \operatorname{diam}(A_N) < \epsilon$ for every $n \geq N$. Hence $\operatorname{diam}(A_n) \to 0$.

The hypothesis now gives a point $x \in \bigcap_n A_n$, and this x is the limit of the sequence. Given $\epsilon > 0$, choose N with $\operatorname{diam}(A_N) < \epsilon$. For $n \geq N$ both $x_n \in T_N \subseteq A_N$ and $x \in A_N$, so $d(x_n, x) \leq \operatorname{diam}(A_N) < \epsilon$. Thus every Cauchy sequence converges and X is complete, completing the proof.

The converse is the direction that gets used. Verifying completeness of a space reduces to exhibiting, for each shrinking nest of closed sets, one point common to all of them, with no Cauchy sequence mentioned. The proof also explains the choice of sets: the closed tails of a Cauchy sequence are the canonical shrinking nest, and lemma 2.3.3 is what makes their diameters shrink at the same rate as

the Cauchy estimate.

The second consequence of theorem 2.3.4 is a statement about size. A countable space can be listed, and a list is a weak kind of description; the theorem manufactures points that no list anticipates, provided the space has enough room to keep dodging. The condition that gives the room is that no point of the space is separated from the rest.

Definition 2.3.7: Isolated Points and Perfect Spaces

A point x of a metric space X is *isolated* if $B_r(x) = \{x\}$ for some $r > 0$. The space X is *perfect* if it is nonempty and has no isolated point.
A subset $A \subseteq X$ is *perfect* if it is closed in X and perfect as a subspace (definition 1.4.1).

A point x fails to be isolated exactly when every ball about it meets $X \setminus \{x\}$, which by definition 1.2.8 says that x is a limit point of X. So a space is perfect precisely when it is nonempty and $X = X'$, every one of its points being approachable from within the space. For a subset the two readings agree: by proposition 1.4.2 the ball $B_r^A(a)$ is $B_r(a) \cap A$, so a is a limit point of A in the subspace exactly when it is one in X, and a closed set A with $A \subseteq A'$ satisfies $A = A'$ by theorem 1.2.9.

Example 2.3.8: Spaces With and Without Isolated Points

(i) $\mathbb{R}$ is perfect, since for any x and any $r > 0$ the point $x + r/2$ lies in $B_r(x)$ and differs from x. So is any interval $[a, b]$ with $a < b$: the two points $x + \frac{1}{2}\min(r, b - x)$ and $x - \frac{1}{2}\min(r, x - a)$ both lie in $[a, b] \cap B_r(x)$, and at least one differs from x, since x cannot equal both endpoints.

(ii) $\mathbb{Q}$ is perfect. Every ball about a rational contains a second rational, since $\mathbb{Q}$ is dense in $\mathbb{R}$ [1]. Yet $\mathbb{Q}$ is countable, so perfectness alone cannot force a space to be large.

(iii) $\mathbb{Z}$ with the metric inherited from $\mathbb{R}$ has every point isolated, since $B_{1/2}(n) = \{n\}$, so it is not perfect. It is complete, being closed in $\mathbb{R}$ (theorem 2.1.8 with theorem 2.2.1), so completeness alone cannot force a space to be large either.

(iv) ℓ^2 is perfect: given x and $r > 0$, the point $x + \frac{r}{2}e_1$, where e_1 is the sequence with a 1 in the first position and zeros elsewhere, lies in $B_r(x)$ and differs from x.

The rational and integer examples show that the theorem below needs both of its hypotheses, and it needs them for different reasons: perfectness is what gives a second point at each stage of the construction, and completeness is what turns the construction into a point of the space.

The construction runs on closed balls, which are closed sets by proposition 1.2.3. The distinction between $\overline{B}_r(y)$ and $B_r(y)$ matters in the proof: the sets fed to theorem 2.3.4 must be closed, while the room needed to fit the next ball inside the current one comes from the open ball.

Theorem 2.3.9: Complete Perfect Spaces Are Uncountable

Every complete perfect metric space is uncountable.

***Proof*:**
Suppose, for contradiction, that X is complete, perfect and countable, and let $x_1, x_2, x_3, \dots$ be a list containing every point of X, with repetitions allowed.

Step 1: X has at least two points. It is nonempty, so fix $p \in X$. Since p is not isolated, $B_1(p)$ contains a point other than p.

Step 2: a shrinking nest of closed balls that dodges the list. Points $y_n \in X$ and radii r_n with $0 < r_n \leq 2^{-n}$ are constructed recursively so that for every $n \geq 1$,

$$x_n \notin \overline{B}_{r_n}(y_n) \qquad \text{and} \qquad \overline{B}_{r_{n+1}}(y_{n+1}) \subseteq B_{r_n}(y_n) \tag{2.3}$$

For $n = 1$, use Step 1 to pick $y_1 \in X$ with $y_1 \neq x_1$, and set $r_1 = \min\left(\frac{1}{2}, \frac{1}{2}d(y_1, x_1)\right)$, which is positive. Then $d(y_1, x_1) \geq 2r_1 > r_1$, so $x_1 \notin \overline{B}_{r_1}(y_1)$.

Suppose y_n and $r_n > 0$ have been chosen. The point y_n is not isolated, so $B_{r_n}(y_n)$ contains a point other than y_n and therefore contains at least two points; pick $y_{n+1} \in B_{r_n}(y_n)$ with $y_{n+1} \neq x_{n+1}$. Set

$$\delta = r_n - d(y_n, y_{n+1}) > 0 \qquad r_{n+1} = \min\left(2^{-(n+1)}, \frac{\delta}{2}, \frac{d(y_{n+1}, x_{n+1})}{2}\right)$$

All three quantities in the minimum are positive, so $r_{n+1} > 0$, and $r_{n+1} \leq 2^{-(n+1)}$. Since $d(y_{n+1}, x_{n+1}) \geq 2r_{n+1} > r_{n+1}$, the point x_{n+1} lies outside $\overline{B}_{r_{n+1}}(y_{n+1})$. For the inclusion, let $d(z, y_{n+1}) \leq r_{n+1}$. Then

$$d(z, y_n) \leq d(z, y_{n+1}) + d(y_{n+1}, y_n) \leq \frac{\delta}{2} + (r_n - \delta) < r_n$$

so $z \in B_{r_n}(y_n)$. Both conditions of (2.3) hold.

Step 3: applying the intersection theorem. Put $A_n = \overline{B}_{r_n}(y_n)$. Each is nonempty, containing y_n, and closed by proposition 1.2.3. The second condition of (2.3) gives $A_{n+1} \subseteq B_{r_n}(y_n) \subseteq A_n$, so the family is nested, and the bound recorded after definition 2.3.1 gives $\operatorname{diam}(A_n) \leq 2r_n \leq 2^{1-n} \to 0$. Since X is complete, theorem 2.3.4 produces a point z lying in every A_n.

Step 4: the contradiction. Fix n. Then $z \in A_n$ while $x_n \notin A_n$ by the first condition of (2.3), so $z \neq x_n$. This holds for every n, so z appears nowhere in a list that was supposed to contain every point of X. Hence no such list exists and X is uncountable, completing the proof.

The construction is a diagonal argument carried out with balls instead of digits. Each stage is handed one point of the proposed list and shrinks the current ball so as to exclude it, and perfectness is what makes the shrinking possible: a ball around an isolated point contains only that point, leaving nowhere to retreat to when the list names it. Completeness then does the work that no finite stage can do, converting the infinite sequence of retreats into an actual point. Both hypotheses are unavoidable, by the examples of $\mathbb{Q}$ and $\mathbb{Z}$ in example 2.3.8.

Applied to $\mathbb{R}$, the theorem gives uncountability without any mention of decimal expansions: $\mathbb{R}$ is complete by theorem 2.2.1 and perfect by example 2.3.8. The same two lines apply to ℓ^2, complete by theorem 2.2.4 and perfect by example 2.3.8. The case worth working out in full is a space containing no interval at all.

Example 2.3.10: The Cantor Set Is Complete and Perfect

Let $C = \bigcap_{n\geq 0} C_n$ be the Cantor set of example 1.2.13, with the metric inherited from $\mathbb{R}$.

It is complete. It is closed in $\mathbb{R}$, being an intersection of closed sets, and $\mathbb{R}$ is complete by theorem 2.2.1, so theorem 2.1.8 applies.

It is perfect. It is nonempty, and every endpoint of every constituent interval of every C_n belongs to C: deleting an open middle third removes no endpoint of the interval it is taken from, and the endpoints of a constituent interval of C_n are again endpoints of constituent intervals of C_{n+1}, so they survive all later stages. Now let $x \in C$ and $r > 0$, and choose n with $3^{-n} < r$. The point x lies in one of the constituent intervals I of C_n, and $\operatorname{diam}(I) = 3^{-n}$ as computed in example 2.3.2. The interval I has two distinct endpoints, both in C, so at least one of them, say e, differs from x; and $|x - e| \leq \operatorname{diam}(I) = 3^{-n} < r$. Hence $B_r(x)$ meets $C \setminus \{x\}$, and no point of C is isolated.

By theorem 2.3.9 the Cantor set is uncountable. Together with the fact that C has empty interior (example 1.2.13), this makes C an uncountable subset of $\mathbb{R}$ containing no interval of positive length.

Example 1.2.13 asserted the uncountability of C on the grounds that its points are the reals in $[0, 1]$ with a base-3 expansion using only the digits 0 and 2, and theorem 2.3.4 is what makes that description precise. Given a sequence $a = (a_n)_{n\geq 1}$ with each $a_n \in \{0, 2\}$, define intervals recursively by $I_0(a) = [0, 1]$ and letting $I_n(a)$ be the left third of $I_{n-1}(a)$ when $a_n = 0$ and the right third when $a_n = 2$. Each $I_n(a)$ is a constituent interval of C_n, the family is nested and closed with $\operatorname{diam}(I_n(a)) = 3^{-n} \to 0$, and $\mathbb{R}$ is complete, so the theorem returns a single point

$$\bigcap_{n\geq 0} I_n(a) = \{x(a)\} \qquad x(a) = \sum_{n\geq 1} a_n 3^{-n}$$

which lies in $\bigcap_n C_n = C$. The formula for $x(a)$ is read off from the left endpoint of $I_n(a)$, which is $\sum_{k\leq n} a_k 3^{-k}$. Distinct sequences give distinct points: if a and b first differ at index n, then $I_n(a)$ and $I_n(b)$ are different constituent intervals of C_n and hence disjoint. Every point of C arises this way, since $x \in C$ lies in C_n for each n, hence in exactly one constituent interval of C_n, and those intervals are nested and so record a single sequence a with $x = x(a)$. The assignment $a \mapsto x(a)$ is therefore a bijection from $\{0, 2\}^{\mathbb{N}}$ onto C, which is the base-3 description of example 1.2.13, and theorem 2.3.4 is what converts an infinite sequence of left-or-right choices into a real number.

Exercise 2.3.1

1. Let $A, B \subseteq X$ be nonempty sets with $A \cap B \neq \emptyset$. Show that $\operatorname{diam}(A \cup B) \leq \operatorname{diam}(A) + \operatorname{diam}(B)$, and give two subsets of $\mathbb{R}$ with $A \cap B = \emptyset$ for which the inequality fails.

2. The bound $\operatorname{diam}\left(\overline{B}_r(x)\right) \leq 2r$ holds in every metric space. Show that equality holds in $\mathbb{R}^n$ with the Euclidean metric of example 1.1.3 for every $r > 0$, and that the inequality is strict for the discrete metric of example 1.1.5 when $r = 1$.

3. Let X be complete and let $A_1 \supseteq A_2 \supseteq \cdots$ be nonempty closed sets with $\operatorname{diam}(A_n) \to 0$, so that $\bigcap_n A_n = \{x\}$ by theorem 2.3.4. Show that $\sup_{y\in A_n} d(y, x) \to 0$, and deduce that *every* choice of points $x_n \in A_n$ produces a sequence converging to x.

4. Use corollary 2.3.6 to give a second proof that a closed subspace F of a complete metric space X is complete (theorem 2.1.8). Begin by showing that a subset of F that is closed in the subspace F is closed in X.

5. Let $E \subseteq C$ be the set of all endpoints of all constituent intervals of all the C_n, $n \geq 0$. Show that E is countable and dense in C in the sense of definition 1.2.11, and deduce that C contains points that are not endpoints. Show that $1/4$ is one of them.

6. Show that every nonempty countable complete metric space has an isolated point. Exhibit such a space with exactly one point that is not isolated. Which hypothesis of theorem 2.3.9 fails for $\mathbb{Q}$?

2.4 The Completion

Incompleteness is a defect that can always be repaired. A space that is missing the limits of some of its Cauchy sequences can be enlarged by adjoining those limits, and the enlargement is not a matter of taste: it is determined, up to isometry, by the space it repairs. This section builds the enlargement out of the Cauchy sequences themselves and proves both halves of that claim.

The construction puts X inside a larger space, so the word for the placement cannot be "isometry" in the sense of definition 1.3.11, which requires a bijection. What is wanted is the distance-preserving condition with surjectivity dropped.

Definition 2.4.1: Isometric Embedding

Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f\colon X \to Y$ is an *isometric embedding* if

$$d_Y\big(f(x), f(t)\big) = d_X(x, t) \qquad \text{for all } x, t \in X$$

No surjectivity is required.

An isometric embedding is automatically injective: if $f(x) = f(t)$ then $d_X(x,t) = d_Y(f(x), f(t)) = 0$, and the first axiom of definition 1.1.1 forces $x = t$. It is also continuous, being Lipschitz with constant 1 (definition 1.3.5) and hence uniformly continuous by proposition 1.3.6. Giving the image $f(X)$ the subspace metric (definition 1.4.1), the corestriction $X \to f(X)$ is a bijection preserving distances, so an isometric embedding is exactly an isometry onto its image, and definition 1.3.11 is recovered as the case $f(X) = Y$.

Three instances are already at hand. The inclusion $\mathbb{Q} \hookrightarrow \mathbb{R}$ preserves $|x - y|$ by definition, so it is an isometric embedding with non-surjective image. The map $\mathbb{R} \to \ell^2$ sending x to the sequence $(x, 0, 0, \dots)$ is another, since

$$d_2\big((x,0,0,\dots),(t,0,0,\dots)\big) = \big(|x-t|^2\big)^{1/2} = |x - t|$$

and its image is the set of sequences vanishing after the first entry. The inclusion $C[0,1] \hookrightarrow B([0,1])$ of the continuous functions into all bounded functions is a third: both sides carry $d_\infty(f,g) = \sup_x |f(x) - g(x)|$, and the formula does not change when the ambient set grows. Injectivity alone is not enough for any of this: the map $\mathbb{R} \to \mathbb{R}$, $x \mapsto 2x$, is a continuous bijection onto its image and doubles every distance, so it is a homeomorphism but not an isometric embedding.

With the vocabulary in place the repair can be stated. The requirement on $\widehat{X}$ is that it be complete, that it contain an isometrically embedded copy of X, and that it contain nothing further than what the limits of X's Cauchy sequences require. That last requirement is density: every point of $\widehat{X}$ is approachable from the copy of X.

Theorem 2.4.2: Existence of the Completion

Let (X, d) be a metric space. There exist a complete metric space $(\widehat{X}, \widehat{d})$ and an isometric embedding $\iota\colon X \to \widehat{X}$ whose image $\iota(X)$ is dense in $\widehat{X}$. Such a pair $(\widehat{X}, \iota)$ is a *completion* of X.

***Proof*:**

Step 1: the raw material. Let $\mathcal{C}$ be the set of all Cauchy sequences in X (definition 2.1.1). Every constant sequence is Cauchy, so $\mathcal{C}$ is nonempty whenever X is; in the degenerate case $X = \emptyset$ both $\mathcal{C}$ and the space built from it below are empty, and the theorem holds vacuously. A Cauchy sequence is a specification of a location by successive approximation, and $\mathcal{C}$ collects every such specification, whether or not X contains a point answering to it.

Step 2: identifying specifications of the same location. Two Cauchy sequences should name the same point of the enlarged space exactly when they crowd together. For $a = (a_n)_{n\geq 1}$ and $b = (b_n)_{n\geq 1}$ in $\mathcal{C}$ write

$$a \sim b \quad \text{if and only if} \quad d(a_n, b_n) \to 0$$

This is an equivalence relation on $\mathcal{C}$. Reflexivity holds because $d(a_n, a_n) = 0$ for every n, and symmetry because d is symmetric. For transitivity, suppose $a \sim b$ and $b \sim c$; the triangle inequality gives

$$0 \leq d(a_n, c_n) \leq d(a_n, b_n) + d(b_n, c_n)$$

and the right-hand side is a sum of two null sequences, hence null, so $d(a_n, c_n) \to 0$ and $a \sim c$. Write $[a]$ for the equivalence class of a and set $\widehat{X} = \mathcal{C}/\!\sim$.

Step 3: the distance between two classes. The definition to be made is

$$\widehat{d}\big([a], [b]\big) = \lim_{n\to\infty} d(a_n, b_n)$$

and it carries two obligations: that the limit exists, and that it depends only on the two classes and not on the representatives chosen.

For existence, put $t_n = d(a_n, b_n)$ and apply proposition 1.3.8 to the four points a_n, b_n, a_m, b_m:

$$|t_n - t_m| = \big|d(a_n, b_n) - d(a_m, b_m)\big| \leq d(a_n, a_m) + d(b_n, b_m)$$

Given $\epsilon > 0$, choose N_1 with $d(a_n, a_m) < \epsilon/2$ for $n, m \geq N_1$ and N_2 with $d(b_n, b_m) < \epsilon/2$ for $n, m \geq N_2$; for $n, m \geq \max(N_1, N_2)$ the display gives $|t_n - t_m| < \epsilon$. So (t_n) is a Cauchy sequence of real numbers, and $\mathbb{R}$ is complete [1], so $\lim_n t_n$ exists.

For independence of representatives, suppose $a \sim a'$ and $b \sim b'$. The same inequality, now applied to a_n, b_n, a'_n, b'_n, gives

$$\big|d(a_n, b_n) - d(a'_n, b'_n)\big| \leq d(a_n, a'_n) + d(b_n, b'_n)$$

whose right-hand side tends to 0 by hypothesis. Two convergent real sequences whose difference is null have the same limit, so the two candidate values of $\widehat{d}([a], [b])$ agree.

Step 4: $\widehat{d}$ is a metric. Each $d(a_n, b_n)$ is nonnegative by lemma 1.1.2, so the limit is nonnegative; symmetry is inherited term by term; and the triangle inequality follows by taking limits in

$$d(a_n, c_n) \leq d(a_n, b_n) + d(b_n, c_n)$$

all three limits existing by Step 3. That leaves the first axiom of definition 1.1.1, and it is the reason Step 2 passed to classes: $\widehat{d}([a],[b]) = 0$ says $\lim_n d(a_n, b_n) = 0$, which is precisely the relation $a \sim b$, that is, $[a] = [b]$. On $\mathcal{C}$ itself the formula $\lim_n d(a_n, b_n)$ satisfies every axiom except this one, and two distinct Cauchy sequences converging to the same place sit at distance zero from each other; the quotient collapses exactly the pairs the formula cannot separate.

Step 5: the embedding. For $x \in X$ let $\iota(x)$ be the class of the constant sequence at x,

$$\iota(x) = \big[(x, x, x, \dots)\big]$$

which is defined because constant sequences are Cauchy. For $x, y \in X$ the sequence $d(x,y), d(x,y), \dots$ is constant, so

$$\widehat{d}\big(\iota(x), \iota(y)\big) = \lim_{n\to\infty} d(x,y) = d(x,y)$$

and ι is an isometric embedding in the sense of definition 2.4.1.

Step 6: the image is dense. Let $[a] \in \widehat{X}$ with $a = (a_n)_{n\geq 1}$ Cauchy, and let $\epsilon > 0$. Choose N with $d(a_n, a_m) < \epsilon/2$ for all $n, m \geq N$. The single point $a_N \in X$ then satisfies

$$\widehat{d}\big(\iota(a_N), [a]\big) = \lim_{m\to\infty} d(a_N, a_m) \leq \frac{\epsilon}{2} < \epsilon$$

since every term with $m \geq N$ is bounded by $\epsilon/2$ and a limit inherits such a bound. So every ball in $\widehat{X}$ meets $\iota(X)$, and theorem 1.2.9(ii) gives $\overline{\iota(X)} = \widehat{X}$, which is density in the sense of definition 1.2.11.

Step 7: $\widehat{X}$ is complete. Let $(\xi_k)_{k\geq 1}$ be a Cauchy sequence in $\widehat{X}$. The argument produces a candidate limit out of X and then shows the sequence reaches it.

By Step 6, for each k there is a point $y_k \in X$ with

$$\widehat{d}\big(\iota(y_k), \xi_k\big) < \frac{1}{k}$$

Unwinding that step makes the choice explicit: writing ξ_k as the class of a Cauchy sequence $a^{(k)} = (a^{(k)}_n)_{n\geq 1}$, the point y_k may be taken to be $a^{(k)}_{N_k}$ for a suitable index N_k, so that $(y_k)_{k\geq 1}$ is a diagonal choice through the doubly indexed array $a^{(k)}_n$, one entry taken from each of the given sequences.

That diagonal is Cauchy in X. Since ι preserves distances, the triangle inequality applied twice gives

$$\begin{aligned} d(y_k, y_l) &= \widehat{d}\big(\iota(y_k), \iota(y_l)\big) \\ &\leq \widehat{d}\big(\iota(y_k), \xi_k\big) + \widehat{d}(\xi_k, \xi_l) + \widehat{d}\big(\xi_l, \iota(y_l)\big) \\ &< \frac{1}{k} + \widehat{d}(\xi_k, \xi_l) + \frac{1}{l} \end{aligned}$$

Given $\epsilon > 0$, choose K_1 with $\widehat{d}(\xi_k, \xi_l) < \epsilon/3$ for $k, l \geq K_1$, which is possible because (ξ_k) is Cauchy, and K_2 with $1/K_2 < \epsilon/3$. For $k, l \geq \max(K_1, K_2)$ the display is less than ϵ. Hence $y = (y_k)_{k\geq 1}$ lies in $\mathcal{C}$ and $\xi = [y]$ is a point of $\widehat{X}$.

It remains to show $\xi_k \to \xi$. Let $\epsilon > 0$ and choose K with both $1/K < \epsilon/3$ and $d(y_k, y_m) < \epsilon/3$ for all $k, m \geq K$, the latter being available because y is Cauchy. For $k \geq K$, the class ξ is represented by $(y_m)_{m\geq 1}$ and $\iota(y_k)$ by the constant sequence at y_k, so

$$\widehat{d}\big(\iota(y_k), \xi\big) = \lim_{m\to\infty} d(y_k, y_m) \leq \frac{\epsilon}{3}$$

every term with $m \geq K$ obeying that bound. Combining this with the choice of y_k,

$$\widehat{d}(\xi_k, \xi) \leq \widehat{d}\big(\xi_k, \iota(y_k)\big) + \widehat{d}\big(\iota(y_k), \xi\big) < \frac{1}{k} + \frac{\epsilon}{3} < \epsilon$$

for every $k \geq K$. So every Cauchy sequence in $\widehat{X}$ converges, and $\widehat{X}$ is complete in the sense of definition 2.1.6, completing the proof.

The theorem answers the question of existence without saying anything about what the new points are. A point of $\widehat{X}$ is a rule for approaching a location by successive approximation, together with the decision to regard two rules as the same when they approach each other; nothing beyond the metric d goes into the construction, so nothing beyond it can come out. When $X = \mathbb{Q}$ with $d(x, y) = |x - y|$ this is the classical construction of the real numbers from the rationals, and the reader who has seen that construction has seen the whole of the proof above with X specialized.

Density is what keeps $\widehat{X}$ from being larger than the repair requires. A complete space containing an isometric copy of X need not resemble a completion of X at all, since it may contain a great deal besides that copy, and exercise 2.4.1 (6) exhibits the failure. Requiring the copy to be dense forces every point of $\widehat{X}$ to be the limit of points of X. Density by itself says nothing about completeness: $\mathbb{Q}$ contains a dense isometric copy of $\mathbb{Q}$, namely itself, and is not complete (example 2.1.7(i)). What Step 7 adds is that the space produced by the construction, which gives a point for every Cauchy sequence of X and for nothing else, is already complete: the limits demanded by X's Cauchy sequences are all the limits there are to require.

Two different constructions could produce two different-looking complete spaces around X, and for $\mathbb{Q}$ they historically did: Dedekind's cuts and Cantor's Cauchy sequences, both published in 1872, build the real numbers out of different raw material. The next theorem says the difference is immaterial. Any two completions of the same space are isometric, and by an isometry compatible with the two copies of X inside them, which is what permits the phrase "the completion".

Theorem 2.4.3: Uniqueness of the Completion

Let (X, d) be a metric space and let $(\widehat{X}_1, \iota_1)$ and $(\widehat{X}_2, \iota_2)$ be completions of X, with metrics $\widehat{d}_1$ and $\widehat{d}_2$. There is exactly one continuous map $\Phi\colon \widehat{X}_1 \to \widehat{X}_2$ satisfying $\Phi \circ \iota_1 = \iota_2$, and it is an isometry in the sense of definition 1.3.11.

***Proof*:**

Step 1: constructing Φ. Let $\xi \in \widehat{X}_1$. Since $\iota_1(X)$ is dense, theorem 1.2.9(iii) gives a sequence (x_n) in X with $\iota_1(x_n) \to \xi$. A convergent sequence is Cauchy (proposition 2.1.2), and ι_1 preserves distances, so

$$d(x_n, x_m) = \widehat{d}_1\big(\iota_1(x_n), \iota_1(x_m)\big)$$

shows (x_n) is Cauchy in X. Applying ι_2, which also preserves distances, makes $\big(\iota_2(x_n)\big)$ Cauchy in $\widehat{X}_2$, and $\widehat{X}_2$ is complete, so the sequence converges. Define

$$\Phi(\xi) = \lim_{n \to \infty} \iota_2(x_n)$$

Step 2: the value does not depend on the approximating sequence. Suppose (x_n) and (x'_n) both satisfy $\iota_1(x_n) \to \xi$ and $\iota_1(x'_n) \to \xi$. By the consequence recorded in proposition 1.3.8,

$\widehat{d}_1\big(\iota_1(x_n),\iota_1(x'_n)\big) \to \widehat{d}_1(\xi,\xi) = 0$, and both embeddings preserve distances, so

$$\widehat{d}_2\big(\iota_2(x_n),\iota_2(x'_n)\big) = d(x_n,x'_n) = \widehat{d}_1\big(\iota_1(x_n),\iota_1(x'_n)\big) \to 0$$

If $\iota_2(x_n) \to \zeta$ and $\iota_2(x'_n) \to \zeta'$ then proposition 1.3.8 again gives $\widehat{d}_2(\zeta,\zeta') = \lim_n \widehat{d}_2\big(\iota_2(x_n),\iota_2(x'_n)\big) = 0$, so $\zeta = \zeta'$ and Φ is well defined.

Step 3: Φ preserves distances. Let $\xi,\eta \in \widehat{X}_1$ with approximating sequences (x_n) and (y_n), so that $\iota_2(x_n) \to \Phi(\xi)$ and $\iota_2(y_n) \to \Phi(\eta)$. Applying proposition 1.3.8 in $\widehat{X}_2$ and then in $\widehat{X}_1$,

$$\begin{aligned}\widehat{d}_2\big(\Phi(\xi),\Phi(\eta)\big) &= \lim_{n\to\infty} \widehat{d}_2\big(\iota_2(x_n),\iota_2(y_n)\big) = \lim_{n\to\infty} d(x_n,y_n)\\ &= \lim_{n\to\infty} \widehat{d}_1\big(\iota_1(x_n),\iota_1(y_n)\big) = \widehat{d}_1(\xi,\eta)\end{aligned}$$

the two outer equalities using the convergences above and the two inner ones the fact that ι_1 and ι_2 preserve distances. In particular Φ is an isometric embedding, hence injective and continuous.

Step 4: Φ is surjective. Let $\zeta \in \widehat{X}_2$. Density of $\iota_2(X)$ gives (x_n) in X with $\iota_2(x_n) \to \zeta$, and the argument of Step 1, run with the roles of the two completions exchanged, shows (x_n) is Cauchy in X and therefore that $\big(\iota_1(x_n)\big)$ is Cauchy in the complete space $\widehat{X}_1$. Let ξ be its limit. Then (x_n) is an approximating sequence for ξ, so $\Phi(\xi) = \lim_n \iota_2(x_n) = \zeta$.

Step 5: compatibility and uniqueness. For $x \in X$ the constant sequence $x_n = x$ satisfies $\iota_1(x_n) \to \iota_1(x)$, so $\Phi\big(\iota_1(x)\big) = \lim_n \iota_2(x) = \iota_2(x)$, which is the identity $\Phi \circ \iota_1 = \iota_2$. Now let $\Psi\colon \widehat{X}_1 \to \widehat{X}_2$ be any continuous map with $\Psi \circ \iota_1 = \iota_2$, let $\xi \in \widehat{X}_1$, and choose (x_n) with $\iota_1(x_n) \to \xi$. By theorem 1.3.2(ii),

$$\Psi(\xi) = \lim_{n\to\infty} \Psi\big(\iota_1(x_n)\big) = \lim_{n\to\infty} \iota_2(x_n) = \Phi(\xi)$$

so $\Psi = \Phi$. With Steps 3 and 4, Φ is a distance-preserving bijection, as we sought to show.

The mechanism of the proof is worth separating from the statement. A continuous map is determined by its values on a dense set, because every point is a limit of points where the values are already known; that is Step 5. A map defined on a dense set can be extended to the whole space when its target is complete and it does not distort distances too badly, because the Cauchy sequences it is fed have limits waiting for them; that is Step 1. Neither half uses anything particular to completions, and exercise 2.4.1 (4) states the general form: a uniformly continuous map from a dense subset into a complete space extends uniquely to the whole space.

In practice the construction of theorem 2.4.2 is never run. To identify the completion of a given space it suffices to exhibit some complete space in which it sits isometrically and densely, and theorem 2.4.3 then says that space is the completion. The examples below all follow that pattern.

Example 2.4.4: Completions of $\mathbb{Q}$, $(0,1)$ and $C[0,1]$

(i) The completion of $\mathbb{Q}$ with the metric $|x-y|$ is $\mathbb{R}$. The inclusion preserves distances by definition, $\mathbb{R}$ is complete [1], and $\mathbb{Q}$ is dense in $\mathbb{R}$, which is the case $n = 1$ of example 1.2.12(i). So the inclusion is a completion, and theorem 2.4.3 makes every other completion of $\mathbb{Q}$ isometric to $\mathbb{R}$ by an isometry restricting to the identity on $\mathbb{Q}$. The construction of theorem 2.4.2 is not being used here to build $\mathbb{R}$: this book takes $\mathbb{R}$ and its completeness as given, and Step 3

of that proof appealed to the completeness of $\mathbb{R}$ in order to define $\widehat{d}$ at all.

(ii) The completion of $(0,1)$ is $[0,1]$. The inclusion $(0,1) \hookrightarrow [0,1]$ preserves distances; $[0,1]$ is closed in $\mathbb{R}$ and so is complete by theorem 2.1.8; and $(0,1)$ is dense in $[0,1]$, since $\frac{1}{n} \to 0$ and $1-\frac{1}{n} \to 1$ while every other point of $[0,1]$ already lies in $(0,1)$. The completion therefore adds exactly the two missing endpoints. The interval $(0,1)$ is homeomorphic to $\mathbb{R}$, by composing $t \mapsto 2t-1$ with the inverse of the map in example 1.3.12, and $\mathbb{R}$ needs no points added at all; how much a space is missing is a question about its metric, not about its topology (example 2.1.9).

(iii) The completion of $\mathbb{Q}$ with the p-adic metric d_p of example 1.1.11 is the space $\mathbb{Q}_p$ of p-adic numbers, and it is strictly larger than $\mathbb{Q}$. To see this, choose digits $c_k \in \{0,1,\dots,p-1\}$ for $k \geq 0$ and put $s_n = \sum_{k=0}^{n} c_k p^k$. For $m > n$ the difference $s_m - s_n = \sum_{k=n+1}^{m} c_k p^k$ is divisible by p^{n+1}, so $d_p(s_n, s_m) \leq p^{-(n+1)}$ and (s_n) is Cauchy. If a second digit string (c'_k) first differs from (c_k) at index j, then for $n \geq j$

$$s_n - s'_n = p^j \left((c_j - c'_j) + \sum_{k=j+1}^{n} (c_k - c'_k) p^{k-j} \right)$$

and the bracket is congruent to $c_j - c'_j \not\equiv 0$ modulo p, so $v_p(s_n - s'_n) = j$ exactly and $d_p(s_n, s'_n) = p^{-j}$ for every $n \geq j$. The two Cauchy sequences are therefore inequivalent and determine distinct points of the completion. There are uncountably many digit strings and only countably many rationals, so $\mathbb{Q}_p$ is uncountable while the image of $\mathbb{Q}$ inside it is countable: all but countably many points of $\mathbb{Q}_p$ are new. The arithmetic of $\mathbb{Q}_p$, and the theory built on it, belong to *Local Field Theory* [2].

(iv) The completion of $C[0,1]$ under the metric $\int_0^1 |f-g|$ of example 2.2.5 is a space of integrable functions on $[0,1]$, whose points cannot be described as functions without a theory of integration more general than Riemann's, and are left undescribed here.[a]

[a] The theory in question is Lebesgue's: the completion is the space of Lebesgue integrable functions on $[0,1]$, with two functions counted as the same point when they differ only on a set of measure zero. Measure and the Lebesgue integral are not built in this book.

Exercise 2.4.1

1. Let X be complete and let $\iota\colon X \to \widehat{X}$ be as in theorem 2.4.2. Show that ι is surjective, so that a complete space is its own completion.
2. Let X be a complete metric space and let $A \subseteq X$ carry the subspace metric of definition 1.4.1. Show that the inclusion $A \hookrightarrow \overline{A}$ exhibits $\overline{A}$ as a completion of A. Deduce that the completion of $\mathbb{Q} \cap (0,1)$ is $[0,1]$.
3. Let $(\widehat{X}, \iota)$ be a completion of X. Show that $\operatorname{diam} \widehat{X} = \operatorname{diam} X$ for the diameter of definition 2.3.1, so that X is bounded if and only if $\widehat{X}$ is, with the same diameter.
4. Let $D \subseteq X$ be dense, let Y be complete, and let $f\colon D \to Y$ be uniformly continuous for the subspace metric on D. Show that there is exactly one continuous $F\colon X \to Y$ with $F|_D = f$, and that this F is uniformly continuous. (For well-definedness, interleave two approximating

sequences into one.)

5. Suppose d is an ultrametric on X in the sense of definition 1.1.10. Show that $\widehat{d}$ is an ultrametric on $\widehat{X}$, and deduce from proposition 1.1.12 that every triangle in the completion of $(\mathbb{Q}, d_p)$ is isosceles.

6. Exhibit a complete metric space Z and an isometric embedding $j\colon (0,1) \to Z$ for which Z is not isometric to the completion of $(0,1)$. Conclude that the density requirement in theorem 2.4.2 cannot be dropped without losing theorem 2.4.3.

3

Draft Material for Chapters 2 to 7

3.1 Compactness

In metric spaces, we can define the concept of sequential compactness:

Definition 3.1.1: Sequentially Compact

A subset $A \subseteq X$ of a metric space (X, d) is *sequentially compact* if for every sequence in A, there exists a convergent subsequence that converges in A

3.1.1 Compactness in compact metric spaces

Theorem 3.1.2: Generalized Heine-Borel Theorem

A subset of a complete metric space is compact if and only if it is closed and totally bounded.

***Proof*:**
Let (X, d) be complete and let $A \subseteq X$ carry the subspace metric of definition 1.4.1, so that distances, balls and diameters of subsets of A are the same computed in A or in X. A set $S \subseteq X$ is *totally bounded* if for every $\epsilon > 0$ there are finitely many points $s_1, \ldots, s_N \in S$ with $S \subseteq \bigcup_{i \leq N} B_\epsilon(s_i)$, the balls being those of definition 1.2.1. A set $A \subseteq X$ is *compact* if every family of open subsets of X whose union contains A has a finite subfamily whose union contains A. By proposition 1.4.2 the open subsets of the subspace A are exactly the traces $V \cap A$ of open $V \subseteq X$, so this agrees with the finite subcover condition for covers of A by sets open in A. If $A = \emptyset$ then all three properties hold with the empty finite family, so assume A is nonempty.

Step 1: compact implies totally bounded. Let $\epsilon > 0$. The balls $B_\epsilon(a)$ for $a \in A$ are open by proposition 1.2.3, and their union contains A because $d(a,a) = 0$ puts $a \in B_\epsilon(a)$. A finite subfamily whose union contains A is a finite list $a_1, \ldots, a_N \in A$ with $A \subseteq \bigcup_{i \leq N} B_\epsilon(a_i)$, which is the required ϵ-net.

Step 2: compact implies closed in X. Let $x \in X \setminus A$. For each $a \in A$ the point a differs from x, so $d(a,x) > 0$ by definition 1.1.1, and $r_a = \frac{1}{2}d(a,x)$ is positive. The balls $B_{r_a}(a)$ are open and their union contains A, so compactness gives $a_1, \ldots, a_n \in A$ with $A \subseteq \bigcup_{i \leq n} B_{r_{a_i}}(a_i)$. Put $r = \min_{i \leq n} r_{a_i}$, positive as a minimum of finitely many positive numbers. If some $y \in A$ had $d(y,x) < r$, then choosing i with $d(a_i, y) < r_{a_i}$ would give

$$2r_{a_i} = d(a_i, x) \leq d(a_i, y) + d(y, x) < r_{a_i} + r \leq 2r_{a_i}$$

which is impossible. So $B_r(x) \subseteq X \setminus A$, the complement of A is open, and A is closed by definition 1.2.2. Neither Step 1 nor Step 2 used the completeness of X.

Step 3: closed and totally bounded implies compact. Since X is complete and A is closed in X, the subspace A is complete by theorem 2.1.8(i). Suppose towards a contradiction that $\mathcal{U}$ is a family of open subsets of X whose union contains A and that no finite subfamily of $\mathcal{U}$ covers A.

Set $A_0 = A$ and construct sets $A_1 \supseteq A_2 \supseteq \cdots$ recursively. Suppose $n \geq 1$ and that $A_{n-1} \subseteq A$ is covered by no finite subfamily of $\mathcal{U}$, which forces $A_{n-1} \neq \emptyset$ because the empty subfamily covers the empty set. Total boundedness at $\epsilon = 2^{-n-1}$ gives $a_1, \ldots, a_N \in A$ with $A \subseteq \bigcup_{i \leq N} B_{2^{-n-1}}(a_i)$, so

$$A_{n-1} = \bigcup_{i=1}^{N} \left(A_{n-1} \cap B_{2^{-n-1}}(a_i) \right)$$

If each of these N sets were covered by a finite subfamily of $\mathcal{U}$, the union of those N subfamilies would be a finite subfamily covering A_{n-1}. At least one of them is therefore covered by no finite subfamily, and A_n is defined to be such a one. Then $A_n \subseteq A_{n-1}$, and A_n lies inside a ball of radius 2^{-n-1}, so $\operatorname{diam}(A_n) \leq 2^{-n}$ by the bound recorded after definition 2.3.1.

Write $\overline{A_n}$ for the closure of A_n in the metric space A. These sets are closed in A and nonempty, they are nested because closure preserves inclusion, and $\operatorname{diam}(\overline{A_n}) = \operatorname{diam}(A_n) \leq 2^{-n}$ by lemma 2.3.3, so their diameters tend to 0. The space A is complete, so theorem 2.3.4 applied inside A produces a point $p \in A$ with $\bigcap_{n \geq 1} \overline{A_n} = \{p\}$.

Some $U \in \mathcal{U}$ contains p, and U is open in X, so $B_\epsilon(p) \subseteq U$ for some $\epsilon > 0$ by definition 1.2.2. Choose n with $2^{-n} < \epsilon$. Each $y \in A_n$ lies in $\overline{A_n}$ alongside p, so $d(p,y) \leq \operatorname{diam}(\overline{A_n}) \leq 2^{-n} < \epsilon$ and $y \in B_\epsilon(p)$. Hence $A_n \subseteq U$, and the one-element subfamily $\{U\}$ covers A_n, contradicting the choice of A_n. So some finite subfamily of $\mathcal{U}$ covers A, and A is compact.

Sometimes, the following corrollary is known as generalized Heine Borel. In $\mathbb{R}$:

$$\text{Compact} \Rightarrow \text{closed and bounded}$$

and for the converse in metric spaces:

$$\text{Compact} \Leftrightarrow \text{complete and totally bounded}$$

which is the result of his next corrollary

Corollary 3.1.3: Compactness Of Metric Space (Generlized Heine-Borel)

A metric space is compact if and only if it is complete and totally bounded

***Proof*:**
Let (M, d) be a metric space, viewed as a subset of itself. Its complement in itself is empty and hence open, so M is a closed subset of M.

Complete and totally bounded implies compact. Apply theorem 3.1.2 with ambient space $X = M$, complete by hypothesis, and subset $A = M$, closed by the previous paragraph and totally bounded by hypothesis. The conclusion is that M is compact.

Compact implies totally bounded. This is Step 1 of the proof of theorem 3.1.2 taken with $A = M$, and that step used no completeness hypothesis on the ambient space.

Compact implies complete. Let (x_n) be a Cauchy sequence in M, and call $y \in M$ a *cluster point* of (x_n) if every ball $B_\epsilon(y)$ contains x_n for infinitely many indices n. If no point of M were a cluster point, each $y \in M$ would carry an $\epsilon_y > 0$ for which the index set $I_y = \{n \in \mathbb{N} \mid x_n \in B_{\epsilon_y}(y)\}$ is finite. The balls $B_{\epsilon_y}(y)$ are open by proposition 1.2.3 and their union is M, so compactness leaves finitely many of them, centred at $y_1, \ldots, y_m$, already covering M. Every index n has $x_n \in M$ and so lies in one of $I_{y_1}, \ldots, I_{y_m}$, making $\mathbb{N}$ a finite union of finite sets. A cluster point y therefore exists.

Extract a subsequence converging to y. Having chosen $n_1 < \cdots < n_{k-1}$, the ball $B_{1/k}(y)$ contains x_n for infinitely many n, so it contains x_{n_k} for some $n_k > n_{k-1}$. The resulting subsequence (definition 2.1.4) satisfies $d(x_{n_k}, y) < 1/k$ and so converges to y. A Cauchy sequence with a convergent subsequence converges by proposition 2.1.5, so $x_n \to y$. Every Cauchy sequence in M converges, which is completeness in the sense of definition 2.1.6.

Corollary 3.1.4: Closed Bounded Intervals Are Compact

For real numbers $a \leq b$ the interval $[a, b]$ is compact.

***Proof*:**
The interval $[a, b]$ is the closed ball of radius $(b-a)/2$ about $(a+b)/2$ in $\mathbb{R}$, so it is closed by proposition 1.2.3, and $\mathbb{R}$ is complete by [1, The Cauchy Criterion]. Given $\epsilon > 0$, choose an integer $N \geq 1$ with $(b-a)/N < \epsilon$. Every $x \in [a, b]$ lies within distance $(b-a)/N$ of one of the points $a + j(b-a)/N$ with $0 \leq j \leq N$, all of which lie in $[a, b]$, so $[a, b]$ is totally bounded. Hence $[a, b]$ is compact by theorem 3.1.2, completing the proof.

Example 3.1.5: The Closed Unit Ball of ℓ^2 Is Not Compact

The closed unit ball $\overline{B}_1(0)$ of (ℓ^2, d_2) is closed by proposition 1.2.3 and bounded, and ℓ^2 is complete by theorem 2.2.4. For $k \geq 1$ let e_k be the sequence with 1 in place k and 0 elsewhere. Each e_k lies in the ball, and $d_2(e_j, e_k) = \sqrt{2}$ for $j \neq k$. Two points of a ball of radius $\frac{1}{2}$ are less than 1 apart, so such a ball contains at most one e_k, and no finite family of such balls covers $\overline{B}_1(0)$. The ball is therefore not totally bounded, and theorem 3.1.2 shows that it is not compact.

3.1.2 Consequences of compactness

Theorem 3.1.6: Continuity on a Compact Metric Space Is Uniform

Let (X, d_X) be a compact metric space and (Y, d_Y) a metric space. Every continuous $f\colon X \to Y$ is uniformly continuous.

***Proof*:**
If X is empty there is nothing to prove, so assume $X \neq \emptyset$, and let $\epsilon > 0$. For each $x \in X$, continuity at x gives a $\rho_x > 0$ with $d_Y(f(t), f(x)) < \epsilon/2$ whenever $d_X(t, x) < \rho_x$. The balls $B_{\rho_x/2}(x)$ for $x \in X$ are open by proposition 1.2.3 and cover X, so compactness gives $x_1, \ldots, x_n \in X$ with $X = \bigcup_{i \leq n} B_{\rho_{x_i}/2}(x_i)$. Put $\delta = \min_{i \leq n} \rho_{x_i}/2$, which is positive. Let $s, t \in X$ with $d_X(s, t) < \delta$, and choose i with $d_X(s, x_i) < \rho_{x_i}/2$. Then

$$d_X(t, x_i) \leq d_X(t, s) + d_X(s, x_i) < \delta + \frac{\rho_{x_i}}{2} \leq \rho_{x_i}$$

so both s and t lie within ρ_{x_i} of x_i, and

$$d_Y(f(s), f(t)) \leq d_Y(f(s), f(x_i)) + d_Y(f(x_i), f(t)) < \epsilon$$

The number δ depends on ϵ alone, so f is uniformly continuous in the sense of definition 1.3.4, as claimed.

summary

[covering] compactness : every open cover has finite subcover

sequentially compcat: every sequence has a convergent subsequence

In $\mathbb{R}$, Heine Borel provides:

$$\text{compact subset} \Leftrightarrow \text{ closed and bounded subset}$$

In metric space:

compact subset $\Rightarrow$ closed and bounded subset compact subset $\not\Leftarrow$ closed and bounded subset

Or, if we think about it in terms of the entire space:

$$\text{compact space} \Rightarrow \text{ bounded space} \qquad \text{compact space} \not\Leftarrow \text{bounded space}$$

counter example being $\mathbb{Q} \cap [0, 1]$

Example 3.1.7: Example
Take $[0, \frac{1}{\pi} - \frac{1}{n}] \cup [\frac{1}{\pi} + \frac{1}{n}, 1]$ for $n \geq 4$. This covers all of $\mathbb{Q} \cap [0, 1]$, but has no fintie subcover. Note that these sets are open in the subspace topology, but not in $\mathbb{R}$.

However:

$$\text{compact space} \Leftrightarrow \text{ complete and totally bounded}$$

and os in complete metric space:

$$\text{compact space} \Leftrightarrow \text{ totally bounded}$$

and using theorem 3.1.2, we get:

$$\text{compact subset} \Leftrightarrow \text{ closed and totally bounded subset}$$

Note that $C^0([a,b],\mathbb{R})$ is complete (corollary 3.3.10, below). However, the infinite-dimensionality of it means we need another condition for compactness: Using Arzela Ascoli, we get:

$$\text{compact subset} \Leftrightarrow \text{ closed, uniformally bounded, and equicontinuous}$$

3.2 The Contraction Principle

Completeness has a powerful fixed-point consequence: on a complete metric space, any map that uniformly shrinks distances has exactly one fixed point, and it is reached by iterating the map from any starting point. This is the engine behind many existence-and-uniqueness theorems in analysis, from the inverse function theorem to the existence and uniqueness of solutions of ordinary differential equations.

Definition 3.2.1: Fixed Point

A *fixed point* of a map $f : X \to X$ is a point $p \in X$ with $f(p) = p$.

Definition 3.2.2: Contraction

Let (X, d) be a metric space. A map $f : X \to X$ is a *contraction* if there is a constant k with $0 \leq k < 1$ such that

$$d(f(x), f(y)) \leq k\, d(x, y) \qquad \text{for all } x, y \in X.$$

Every contraction is Lipschitz, and hence uniformly continuous: given $\epsilon > 0$, the choice $\delta = \epsilon$ makes $d(x, y) < \delta$ force $d(f(x), f(y)) \leq k\, d(x, y) < \epsilon$.

Theorem 3.2.3: Banach Contraction Principle

Let (X, d) be a nonempty *complete* metric space and let $f : X \to X$ be a contraction with constant k. Then f has a unique fixed point $p \in X$, and for every $x_0 \in X$ the iterates defined by $x_{n+1} = f(x_n)$ converge to p, with

$$d(x_n, p) \leq \frac{k^n}{1-k}\, d(x_0, x_1).$$

***Proof*:**
Fix $x_0 \in X$ and set $x_{n+1} = f(x_n)$. Applying the contraction bound repeatedly,

$$d(x_{n+1}, x_n) = d(f(x_n), f(x_{n-1})) \leq k\, d(x_n, x_{n-1}) \leq \cdots \leq k^n\, d(x_1, x_0).$$

Hence for $m > n$, the triangle inequality and the geometric series give

$$d(x_m, x_n) \leq \sum_{i=n}^{m-1} d(x_{i+1}, x_i) \leq \left(\sum_{i=n}^{m-1} k^i \right) d(x_1, x_0) \leq \frac{k^n}{1-k}\, d(x_1, x_0).$$

Since $0 \leq k < 1$, the right-hand side tends to 0 as $n \to \infty$, so (x_n) is a Cauchy sequence. As X is complete, $x_n \to p$ for some $p \in X$.

This limit is a fixed point: because f is continuous, $f(p) = f\big(\lim_n x_n\big) = \lim_n f(x_n) = \lim_n x_{n+1} = p$. Letting $m \to \infty$ in the bound above (the distance is continuous in each argument) yields the stated error estimate.

The fixed point is unique: if $f(p) = p$ and $f(q) = q$, then $d(p, q) = d(f(p), f(q)) \leq k\, d(p, q)$, so $(1-k)\, d(p, q) \leq 0$. Since $1 - k > 0$, this forces $d(p, q) = 0$, that is, $p = q$.

3.3 Basic Function Spaces

Example 3.3.1: Pointwise Convergence

1. The functions $f_n \colon \mathbb{R} \to \mathbb{R}$, $f_n(x) = x/n$, converge pointwise to the zero function: for each fixed x, $|f_n(x) - 0| = |x|/n \to 0$ as $n \to \infty$.

2. Note that it's possible that a poinwise converging sequence of functions "miss" points. Take

$$f_n(x) : (0, 1) \to \mathbb{R} \qquad f_n(x) = x^n$$

then $f_n \to 0$, the zero-function, since every point is indeed eventually convering to 0. However, it seems like there will be a point which is sort of "up there". We need a strong notion to fix that

Definition 3.3.2: Uniform Convergence

Let M and N be arbitrary metric spaces. A sequence of function $f_n : M \to N$ *converges uniformally* to the limit $f : M \to N$ if for each $\epsilon > 0$ there is an N such that for all $n \geq N$, and all $x \in M$,

$$|f_n(x) - f(x)| < \epsilon$$

The function f is the *uniform limit* of the sequence (f_n) and we write

$$f_n \rightrightarrows f \qquad \text{or} \qquad \operatorname{unif} \lim_{n \to \infty} f_n = f$$

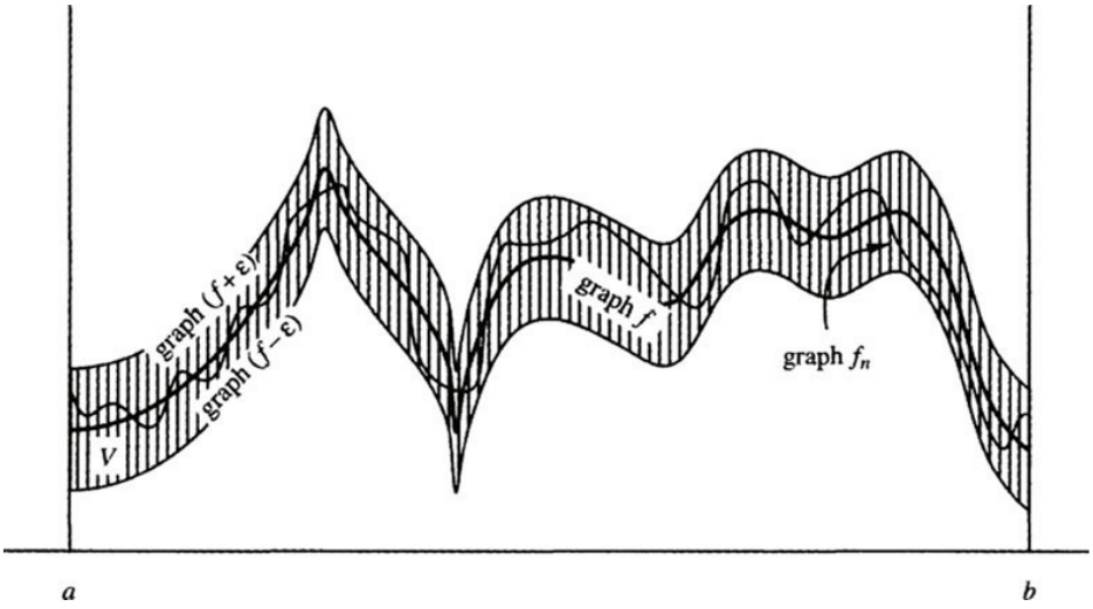


Figure 3.1: uniform convergence visually

Theorem 3.3.3: Uniform Convergence Preserve's Continuity

if $f_n \rightrightarrows f$ and eachf_n is continuous at x_0 then f is continuous at x_0

***Proof*:**
Write d_M and d_N for the metrics of the domain M and the target N, so that the quantity $|f_n(x) - f(x)|$ in definition 3.3.2 is $d_N(f_n(x), f(x))$. Let $\epsilon > 0$. Uniform convergence gives an index n with $d_N(f_n(x), f(x)) < \epsilon/3$ for every $x \in M$, and continuity of this f_n at x_0 gives a $\delta > 0$ with $d_N(f_n(x), f_n(x_0)) < \epsilon/3$ whenever $d_M(x, x_0) < \delta$. For such x, two applications of the triangle inequality give

$$d_N(f(x), f(x_0)) \leq d_N(f(x), f_n(x)) + d_N(f_n(x), f_n(x_0)) + d_N(f_n(x_0), f(x_0)) < \epsilon$$

so f is continuous at x_0 in the sense of definition 1.3.1, as claimed.

So the convergence $x^n \to 0$ on $(0,1)$ is not uniform: $\sup_{x\in(0,1)} x^n = 1$ for every n. On $[0,1]$ the pointwise limit is the discontinuous function

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

so by theorem 3.3.3 the convergence on $[0,1]$ is not uniform either. Note that the converse is not true: if at every point x_0, f_n is continuous and f is continuous at x_0, this does not imply (f_n) is uniformally convergent:

Example 3.3.4: growing steeple
This is called the growing steeple:

$$f_n(x) = \begin{cases} n^2 x & \text{if } 0 \le x \le \frac{1}{n} \\ 2n - n^2 x & \text{if } \frac{1}{n} \le x \le \frac{2}{n} \\ 0 & \text{if } \frac{2}{n} \le x \le 1 \end{cases}$$

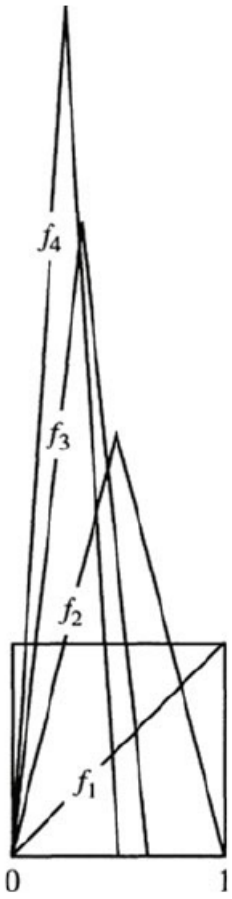


Figure 3.2: growing steeple function

and if you do not like that it goes to infinity, multiply it by $\frac{1}{n}$

Application to function spaces

Definition 3.3.5: Space Of Bounded Functions

Let $C_b([a,b],\mathbb{R})$ represent the set of all bounded functions from $f:[a,b]\to\mathbb{R}$

Then we can define the following norm on this space:

Definition 3.3.6: Sup Norm

Let $f\in C_b$. Then:

$$\|f\| = \sup\{|f(x)||x\in[a,b]\}$$

you should check that this is indeed a norm, and that it will induce a metric as well, and hence a metric topology, and hence the idea of convergence.

In this metric space under this metric, the definition is the same as uniform convergence

Theorem 3.3.7: sup norm convergence if and only if uniformally convergence

Convergence with respect to the sup metric d is equivalent to uniform convergence

***Proof*:**
Let $f_n, f\in C_b$, so that $d(f_n,f)=\|f_n-f\|=\sup_{x\in[a,b]}|f_n(x)-f(x)|$ by definition 3.3.6.

Suppose $d(f_n,f)\to 0$ and let $\epsilon>0$. There is an N with $d(f_n,f)<\epsilon$ for all $n\geq N$, and for such n every $x\in[a,b]$ satisfies $|f_n(x)-f(x)|\leq d(f_n,f)<\epsilon$. This is uniform convergence $f_n\rightrightarrows f$ in the sense of definition 3.3.2.

Conversely, suppose $f_n\rightrightarrows f$ and let $\epsilon>0$. There is an N with $|f_n(x)-f(x)|<\epsilon/2$ for all $n\geq N$ and all $x\in[a,b]$. Then $\epsilon/2$ is an upper bound for the set whose supremum is $d(f_n,f)$, so $d(f_n,f)\leq\epsilon/2<\epsilon$ for $n\geq N$, and $d(f_n,f)\to 0$, as claimed.

Theorem 3.3.8: C_b Is A Complete Metric Space

C_b is a complete metric space

***Proof*:**
Definition 3.3.5 asks nothing of the members of $C_b([a,b],\mathbb{R})$ beyond boundedness, so this space is the set of all bounded functions $[a,b]\to\mathbb{R}$, which is $B([a,b])$ in the notation of chapter 2. The sup norm of definition 3.3.6 induces the metric $d(f,g)=\|f-g\|=\sup_{x\in[a,b]}|f(x)-g(x)|$, which is the metric d_∞ carried by $B([a,b])$. The interval $[a,b]$ is a nonempty set, so theorem 2.2.2 applies with $X=[a,b]$ and gives that $C_b([a,b],\mathbb{R})$ is complete.

With this, we can show that the space of continuous bounded functions is complete too.

Definition 3.3.9: Continuous Bounded Funcitons

Let $C^0([a,b],\mathbb{R})$ denote the set of continuous functions from $f:[a,b]\to\mathbb{R}$. Note that since the domain is compact, the image is compact, and in $\mathbb{R}$ that implies it's bounded, so $C^0\subset C_b$.

Corollary 3.3.10: C^0 Is A Compelte Metric Space

C^0 Is A Compelte Metric Space

***Proof*:**
Let $f_n\in C^0$ converge in C_b to some $f\in C_b$. Then $\sup_{x\in[a,b]}|f_n(x)-f(x)|\to 0$, which is uniform convergence $f_n\rightrightarrows f$ in the sense of definition 3.3.2, so f is continuous by theorem 3.3.3 and lies in C^0. Thus C^0 contains the limits of its convergent sequences and is closed in C_b by corollary 1.2.10. The space C_b is complete by theorem 3.3.8, so theorem 2.1.8(i) makes C^0 complete, as claimed.

3.4 Application to other concepts

We will now explore how uniform convergence works with derivatives, integrals, and series

Let

$$F_n=\sum_{i=1}^{n}f_i(x)$$

Then we can say that (F_n) conveges uniformally just like with (f_n) since it's also a sequence of functions.

If we have

$$F_n=\sum_{i=1}^{\infty}f_i(x)$$

Then we can say it converges uniformally if the sequence of partial sums converges uniformally.

Also, we can say the series of absolutes values $\sum|f_k(x)|$ converges then the series $\sum f_k$ *converges absolutely*

Theorem 3.4.1: Weierstrass M-Test

If $\sum M_k$ is a convergetn sereis of constants and if $f_k\in C_b$ satisfies $\|f_k\|\le M_k$ for all k then $\sum f_k$ converges uniformly and absolutely

***Proof*:**
Write $F_n=\sum_{k=1}^{n}f_k$, which lies in C_b because a finite sum of bounded functions is bounded. Each M_k satisfies $M_k\ge\|f_k\|\ge 0$. For $n>m$ the triangle inequality for d, together with

$d(F_k, F_{k-1}) = \|f_k\|$, gives

$$\begin{aligned} d(F_n, F_m) &\leq d(F_n, F_{n-1}) + \cdots + d(F_{m+1}, F_m) \\ &= \sum_{k=m+1}^{n} \|f_k\| \leq \sum_{k=m+1}^{n} M_k \end{aligned}$$

Let $\epsilon > 0$. The partial sums of $\sum M_k$ converge, so they form a Cauchy sequence [1, The Cauchy Criterion], and there is an N with $\sum_{k=m+1}^{n} M_k < \epsilon$ for all $n > m \geq N$. Hence (F_n) is Cauchy in C_b, which is complete by theorem 3.3.8, so $d(F_n, F) \to 0$ for some $F \in C_b$. By theorem 3.3.7 this is uniform convergence $F_n \rightrightarrows F$, so $\sum f_k$ converges uniformly.

For absolute convergence fix $x \in [a, b]$. Then $|f_k(x)| \leq \|f_k\| \leq M_k$ for every k, so $\sum_k |f_k(x)|$ converges by [1, The Comparison Test], completing the proof.

Two exchanges of a uniform limit with an operation belong to the real line and are proved in *Elementary Analysis* [1], whose Riemann integral and whose derivative on an interval are the objects they speak about. The first says that a uniform limit of Riemann integrable functions on $[a, b]$ is Riemann integrable and that the limit of the integrals is the integral of the limit, with a corollary integrating a uniformly convergent series term by term. The second says that a uniformly convergent sequence of differentiable functions whose derivatives also converge uniformly has a differentiable limit, with derivative the limit of the derivatives. Neither has a metric-space form, since neither the integral nor the derivative is available in a general metric space, so both are cited here and not restated.

Note that since all bounded continuous functions are integrable, but not all integrable bounded functions are continuous, the subset of C_b to which the first of these applies lies between C_b and C^0. Writing $\mathcal{R}$ for the integrable functions,

$$C_b \supset \mathcal{R} \supset C^0$$

Now, we move on to differentiable functions.

Note that this can fail if the derivative does not converge uniformally

Example 3.4.2: derivative not converge uniformally
Take

$$f_n(x) = \sqrt{x^2 + \frac{1}{n}}$$

Then this series converges to $f(x) = |x|$, which is not differentiable! The derivatiev converges point-wise, but not uniformally.

Note that in complex analysis, this will not be the case! the uniform limit of a complex differentiable function is complex differentiable, and the sequence of derivatives will automatically be unifomrally convergent to a limit!

LIke before, infinite sums work well

Corollary 3.4.3: Series Of Infinite Sum And Uniform Convergence

A uniformly convergent sereis of differentaible functiosn can be differentiated term-by-term, provided that the derivative series converges uniformly,

$$\left(\sum_{k=0}^{\infty} f_k(x)\right)' = \sum_{k=0}^{\infty} f_k'(x)$$

***Proof*:**
Apply the term-by-term differentiation theorem for sequences of *Elementary Analysis* [1, Term-by-Term Differentiation] to the partial sums $F_n = \sum_{k\leq n} f_k$. Its two hypotheses become the convergence of $\sum_k f_k$ at a point and the uniform convergence of $\sum_k f_k'$, which are the hypotheses assumed here.

3.5 Compactness and equicontinuity

A Heine-Borel theorem for C^0 would be useful, because closedness and boundedness are easy to check.

In C^0, closed and bounded sets are usually not compact. Take

$$B = \left\{f \in C^0([0,1],\mathbb{R}) \middle| \, \|f\| \leq 1\right\}$$

that is, the set of continuous functions f with $|f| \leq 1$ on $[0,1]$. It is the closed ball of radius 1 about the zero function, so it is closed by proposition 1.2.3, and it is bounded. It is not compact. For $k \geq 1$ let $g_k \in C^0([0,1],\mathbb{R})$ vanish outside $[\frac{1}{k+1}, \frac{1}{k}]$, take the value 1 at the midpoint of that interval, and be linear on each of its two halves. Each g_k lies in B, and two distinct ones are never positive at the same point, so $\|g_j - g_k\| = 1$ for $j \neq k$. Two points of a ball of radius $\frac{1}{2}$ are less than 1 apart, so such a ball contains at most one g_k, no finite family of such balls covers B, and B is not totally bounded. As C^0 is complete (corollary 3.3.10), theorem 3.1.2 shows that B is not compact. The sequence $f_n(x) = x^n$ in B also has no subsequence converging in C^0: it converges at every point to the discontinuous function

$$f(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \end{cases}$$

and a limit in C^0 is in particular a pointwise limit. This comes down to the fact that C^0 is infinite dimensional, whereas the Heine-Borel theorem concerns the finite-dimensional spaces $\mathbb{R}^n$.

The space $C^0([0,1],\mathbb{R})$ is infinite dimensional. The functions g_k above are linearly independent: if $\sum_{k=1}^{N} c_k g_k = 0$, then evaluating at the midpoint m_j of $[\frac{1}{j+1}, \frac{1}{j}]$ gives $c_j = 0$ for each $j \leq N$, because $g_j(m_j) = 1$ while $g_k(m_j) = 0$ for $k \neq j$. So $C^0([0,1],\mathbb{R})$ contains infinitely many linearly independent functions, and it is not finite dimensional.

Now back to finding the appropriate conditions for the Heine-Borel property. To do this, we need a new condition on our functions:

Definition 3.5.1: [Uniform] Equicontinuous

A sequence of functions (f_n) in C^0 is **equicontinuous** if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s-t| < \delta, \forall n \in \mathbb{N} \Rightarrow |f_n(s) - f_n(t)| < \epsilon$$

The functions f_n are *equally continuous*.

Note that the δ depends on teh ϵ, but *not* on the n I put [uniform] in the definition because there are definition of equicontinuity which are not "uniform", that is, you pick a point for which the conditions applies:

Definition 3.5.2: Point-Wise Equicontinuous

A sequence (f_n) in C^0 is *pointwise equicontintuous* if

$$\forall \epsilon > 0 \forall x \in [a,b], \exists \delta > 0 \text{ such that } |x-t| < \delta, n \in \mathbb{N}, |f_n(x) - f_n(t)| < \epsilon$$

From here on out, we will say equicontinuity instead of uniform equicontinuity.

Example 3.5.3: Example

1. The set of function $f : \mathbb{R} \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$ are equicontinuous. However, they are not uniformally continuous.
2. Take $f : [0,1] \to \mathbb{R}$, $f_n(x) = n$. That is, a constant function. Then $\{f_n\}$ is an equicontinuous sequence. However $\{f_n\}$ is not a convergent sequence (and so not uniformally convergent).
3. let $\{f\}$ be a equicontinuous set with a single function. Then f is uniformally continuous by definition. In this sense, equicontinuity is intimately related to uniform continuity.

This definition can also be used to define an equicontinuous set:

Definition 3.5.4: Equicontinuous Set

The set $\mathcal{E} \subset C^0$ is equicontinuous if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s-t| < \delta \text{ and } f \in \mathcal{E} \Rightarrow |f_(s) - f(t)| < \epsilon$$

Essentially, if a function is continuous, than for all $\epsilon > 0$, there exists an open neigbhorhood U such that $d(f(x), f(y)) < \epsilon$ for all $y \in U$. Equicontinuity means that a single neighborhood U can be chosen that will work for all the functions f in the collection $\mathcal{E}$.

You might recall Bolzano-Weierstrass theorem:

Theorem 3.5.5: Bolzano-Weierstrass Theorem

Let $M = \mathbb{R}^n$. Then every bounded sequence has a convergent subsequence.

***Proof*:**
Give $\mathbb{R}^n$ any of the metrics d_1, d_2, d_∞ of example 1.1.3, and write y_i for the ith coordinate of $y \in \mathbb{R}^n$. For $y, z \in \mathbb{R}^n$ and each i,

$$|y_i - z_i| \leq d_\infty(y,z) \leq d_2(y,z) \leq d_1(y,z) \leq n\, d_\infty(y,z)$$

where $d_2 \leq d_1$ holds because squaring $d_1(y,z)$ produces $d_2(y,z)^2$ plus nonnegative cross terms. So a sequence is bounded, or converges to a given point, for one of the three metrics exactly when it does so for the others, and it converges to a in d_1 exactly when each coordinate converges to the corresponding coordinate of a.

Let $(x_k)_{k\geq 1}$ be bounded, say $x_k \in B_R(c)$ for every k. Then $|(x_k)_i - c_i| < R$ for every i and k, so each coordinate sequence $\big((x_k)_i\big)_k$ is a bounded real sequence. Put $k_0(j) = j$. Suppose $1 \leq i \leq n$ and that a strictly increasing sequence of indices $\big(k_{i-1}(j)\big)_j$ has been chosen. The real sequence $\big((x_{k_{i-1}(j)})_i\big)_j$ is bounded, so [1, Bolzano-Weierstrass](1) gives a strictly increasing sequence $(j_l)_l$ and a real number a_i with $(x_{k_{i-1}(j_l)})_i \to a_i$ as $l \to \infty$. Put $k_i(l) = k_{i-1}(j_l)$, again strictly increasing. Along $\big(k_i(l)\big)_l$ the ith coordinate converges to a_i, and each coordinate $i' < i$ still converges to $a_{i'}$, because the sequence $\big((x_{k_i(l)})_{i'}\big)_l$ is a subsequence of $\big((x_{k_{i-1}(j)})_{i'}\big)_j$ and [1, Subsequences of a Sequence with a Limit](1) applies. After step n, the subsequence $\big(x_{k_n(l)}\big)_l$ of definition 2.1.4 has every coordinate converging, so it converges to $a = (a_1, \ldots, a_n)$ in d_1, and hence in all three metrics, completing the proof.

We will be proving the analogous of this with equicontinuous sequences in C^0:

Theorem 3.5.6: Arzela-Ascoli Theorem

Let M be a compact metric space. Every uniformly bounded equicontinuous sequence of functions in $C^0(M, \mathbb{R})$ has a uniformly convergent subsequence

***Proof*:**
Write d for the metric of M, and let $C^0(M, \mathbb{R})$ denote the continuous functions $M \to \mathbb{R}$, which is the notation of definition 3.3.9 with $[a, b]$ replaced by M. Uniform boundedness of the sequence (f_n) means a single constant C with $|f_n(x)| \leq C$ for every n and every $x \in M$. Equicontinuity is definition 3.5.1 with $|s - t|$ read as $d(s, t)$: for every $\epsilon > 0$ there is a $\delta > 0$, not depending on n, such that $d(s, t) < \delta$ forces $|f_n(s) - f_n(t)| < \epsilon$ for every n. If M is empty then every sequence converges uniformly, so assume $M \neq \emptyset$.

Step 1: a countable dense set assembled from finite nets. By corollary 3.1.3 the space M is complete and totally bounded. For each $k \geq 1$ total boundedness gives a finite nonempty $F_k \subseteq M$ with $M = \bigcup_{y \in F_k} B_{1/k}(y)$. Put $D = \bigcup_{k \geq 1} F_k$, a countable union of finite sets and therefore countable, with each F_k a finite subset of D whose $1/k$-balls cover M. Given $x \in M$ and $\epsilon > 0$, choosing k with $1/k < \epsilon$ and $y \in F_k$ with $d(x, y) < 1/k$ shows that every ball about x meets D, so $\overline{D} = M$ by theorem 1.2.9(ii) and D is dense in the sense of definition 1.2.11.

Step 2: a subsequence converging at every point of D. Enumerate $D = \{y_1, y_2, \ldots\}$, repeating terms if D is finite. For each j the real sequence $\big(f_n(y_j)\big)_n$ takes values in $[-C, C]$ and is bounded, so theorem 3.5.5 taken in $\mathbb{R}$ gives it a convergent subsequence. Build index sequences

recursively: let $\big(m(0,i)\big)_i$ be $1,2,3,\ldots$, and given $\big(m(j-1,i)\big)_i$, let $\big(m(j,i)\big)_i$ be a subsequence of it along which the values at y_j converge. Put $m_k = m(k,k)$. Since $\big(m(k,i)\big)_i$ is a subsequence of $\big(m(k-1,i)\big)_i$, its kth term satisfies $m(k,k) \geq m(k-1,k) > m(k-1,k-1)$, so $m_1 < m_2 < \cdots$ and $g_k = f_{m_k}$ is a subsequence of (f_n) in the sense of definition 2.1.4. Fix j. For every $k \geq j$ the index $m_k = m(k,k)$ occurs in $\big(m(j,i)\big)_i$, each stage being a subsequence of the one before, so $\big(g_k(y_j)\big)_{k\geq j}$ is a subsequence of a convergent real sequence and converges. Hence $\big(g_k(x)\big)_k$ converges for every $x \in D$.

Step 3: the subsequence is uniformly Cauchy. Let $\epsilon > 0$ and take $\delta > 0$ from equicontinuity for $\epsilon/3$. Fix k_0 with $1/k_0 < \delta$ and write $F_{k_0} = \{z_1, \ldots, z_p\}$. Each real sequence $\big(g_k(z_i)\big)_k$ converges by Step 2 and is therefore Cauchy, so there is a K_i with $|g_k(z_i) - g_l(z_i)| < \epsilon/3$ for all $k,l \geq K_i$. Put $K = \max_{i\leq p} K_i$. Given $x \in M$, choose i with $x \in B_{1/k_0}(z_i)$, so that $d(x,z_i) < 1/k_0 < \delta$. For $k,l \geq K$,

$$|g_k(x) - g_l(x)| \leq |g_k(x) - g_k(z_i)| + |g_k(z_i) - g_l(z_i)| + |g_l(z_i) - g_l(x)| < \epsilon$$

where the outer two terms are bounded by equicontinuity, which applies to g_k and g_l because each is one of the f_n. The point x was arbitrary and K was chosen before it, so $\sup_{x\in M} |g_k(x) - g_l(x)| \leq \epsilon$ whenever $k,l \geq K$.

Step 4: convergence of the subsequence. Each g_k is continuous and satisfies $|g_k(x)| \leq C$ for all x, so g_k lies in $C_b(M)$, the bounded continuous functions carrying the metric $d_\infty(f,g) = \sup_{x\in M} |f(x) - g(x)|$. Given $\epsilon > 0$, Step 3 applied to $\epsilon/2$ produces a K with $d_\infty(g_k, g_l) \leq \epsilon/2 < \epsilon$ for all $k,l \geq K$, which is the Cauchy condition of definition 2.1.1. The space $C_b(M)$ is complete by theorem 2.2.3, so there is a $g \in C_b(M)$ with $d_\infty(g_k, g) \to 0$. For $\epsilon > 0$ choose K with $d_\infty(g_k, g) < \epsilon$ for all $k \geq K$. Then $|g_k(x) - g(x)| \leq d_\infty(g_k, g) < \epsilon$ for every $x \in M$ and every $k \geq K$, which is uniform convergence $g_k \rightrightarrows g$ in the sense of definition 3.3.2. So (g_k) is the required uniformly convergent subsequence.

Notice how this relates to compactness: if every subsequence of f_n converges, then the closure of $\{f_n | n \in \mathbb{N}\}$ is compact!

The last part of the proof also shows that you can weaken the condition a bit, and say that if you converge on a dense subset, then you are uniformally convergent.

Corollary 3.5.7: Arzela-Ascoli Propagation Theorem

Let M be a compact metric space and let $D \subseteq M$ be dense. If an equicontinuous sequence (f_n) in $C^0(M,\mathbb{R})$ converges pointwise at every point of D, then it converges uniformly on M. Pointwise convergence on a dense subset thus *propagates* to uniform convergence on the whole domain.

***Proof*:**
Write d for the metric of M. Equicontinuity is definition 3.5.1 with $|s-t|$ read as $d(s,t)$, as in the proof of theorem 3.5.6. If M is empty then every sequence of functions on M converges uniformly, so assume $M \neq \emptyset$.

Step 1: the sequence is uniformly Cauchy. Let $\epsilon > 0$ and take $\delta > 0$ from equicontinuity for $\epsilon/3$. The balls $B_{\delta/2}(y)$ for $y \in M$ are open by proposition 1.2.3 and cover M, so compactness

gives $y_1, \ldots, y_p \in M$ with $M = \bigcup_{i \leq p} B_{\delta/2}(y_i)$. Each y_i lies in $\overline{D} = M$, so theorem 1.2.9(ii) gives a point $z_i \in D$ with $d(y_i, z_i) < \delta/2$. Each real sequence $\big(f_n(z_i)\big)_n$ converges by hypothesis and is therefore Cauchy by proposition 2.1.2, so there is a K_i with $|f_n(z_i) - f_m(z_i)| < \epsilon/3$ for all $n, m \geq K_i$. Put $K = \max_{i \leq p} K_i$. Given $x \in M$, choose i with $d(x, y_i) < \delta/2$, so that $d(x, z_i) \leq d(x, y_i) + d(y_i, z_i) < \delta$. For $n, m \geq K$,

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f_n(z_i)| + |f_n(z_i) - f_m(z_i)| + |f_m(z_i) - f_m(x)| < \epsilon$$

where the outer two terms are less than $\epsilon/3$ by equicontinuity. The index K was chosen before x, so it serves every $x \in M$ at once.

Step 2: the uniform limit. For each $x \in M$, Step 1 makes $\big(f_n(x)\big)_n$ a Cauchy sequence of real numbers, which converges by [1, The Cauchy Criterion]. Call its limit $f(x)$. Let $\epsilon > 0$ and take K from Step 1 for $\epsilon/2$. Fix $n \geq K$ and $x \in M$. Every $m \geq K$ satisfies $|f_n(x) - f_m(x)| < \epsilon/2$, and as $m \to \infty$ the left-hand side converges to $|f_n(x) - f(x)|$ by proposition 1.3.8. Limits preserve weak inequalities [1, Limits Preserve Weak Inequalities](2), so $|f_n(x) - f(x)| \leq \epsilon/2 < \epsilon$. Since K does not depend on x, this is uniform convergence $f_n \rightrightarrows f$ in the sense of definition 3.3.2, as claimed.

Corollary 3.5.8: Differentiable Convergent Subsequence

Assume that $f_n : [a, b] \to \mathbb{R}$ is a sequence of differentiable functions whose derivatives are uniformly bounded. If for one point x_0, the sequence $(f_n(x_0))$ is bounded as $n \to \infty$, then the sequence (f_n) has a subsequence that converges uniformly on the whole interval $[a, b]$.

***Proof*:**
Differentiability on $[a, b]$ presupposes $a < b$. Let K be a bound for $|f_n'(x)|$ over all $n \in \mathbb{N}$ and $x \in [a, b]$, and let C be a bound for $|f_n(x_0)|$ over all n. Each f_n is continuous by [1, Differentiability Implies Continuity], so $f_n \in C^0([a, b], \mathbb{R})$. For $s < t$ in $[a, b]$, [1, The Mean Value Theorem] gives a $c \in (s, t)$ with $f_n(t) - f_n(s) = f_n'(c)(t - s)$, so

$$|f_n(s) - f_n(t)| \leq K|s - t| \qquad \text{for all } s, t \in [a, b] \text{ and all } n$$

Given $\epsilon > 0$, put $\delta = \epsilon/(K + 1)$. Whenever $|s - t| < \delta$, every n satisfies $|f_n(s) - f_n(t)| \leq K\delta < \epsilon$, so (f_n) is equicontinuous in the sense of definition 3.5.1. It is uniformly bounded, since every $x \in [a, b]$ and every n satisfy

$$|f_n(x)| \leq |f_n(x) - f_n(x_0)| + |f_n(x_0)| \leq K|x - x_0| + C \leq K(b - a) + C$$

The interval $[a, b]$ is a compact metric space by corollary 3.1.4, so theorem 3.5.6 gives a subsequence of (f_n) that converges uniformly on $[a, b]$, as claimed.

As a quick asside, this theorem has two uses in other domain we will not discuss;

1. A sequecne of solutions to a continuous ordinary differential equation in $\mathbb{R}^M$ has a subsequence that converges to a limit, and that limit is also a solutionof the ODE
2. A sequence of complex analytic functions that converges pointwise, converges uniformly (on

compact subsets of the domain of definition) and the limit is complex analytic.

However, we will be using Arzela-Ascoli for Heine-Borel:

Theorem 3.5.9: Heine-Borel Theroem In A Function Space

A subset $\mathcal{E} \subseteq C^0$ is compact if and only if it is closed, bounded, and equicontinuous

***Proof*:**
Here $C^0 = C^0([a,b],\mathbb{R})$ carries the metric $d(f,g) = \|f-g\|$ of definition 3.3.6, and it is complete by corollary 3.3.10. If $\mathcal{E}$ is empty it has all four properties, so assume $\mathcal{E} \neq \emptyset$.

($\Rightarrow$) Let $\mathcal{E}$ be compact. Since C^0 is complete, theorem 3.1.2 shows that $\mathcal{E}$ is closed and totally bounded. Total boundedness at radius 1 gives $f_1,\dots,f_N \in \mathcal{E}$ with $\mathcal{E} \subseteq \bigcup_{k\leq N} B_1(f_k)$, and then every $f \in \mathcal{E}$ satisfies $d(f,f_1) < 1 + \max_{k\leq N} d(f_k,f_1)$, so $\mathcal{E}$ lies in a single ball and is bounded.

For equicontinuity let $\epsilon > 0$. Total boundedness gives points $f_1,\dots,f_N$ of $\mathcal{E}$ such that the balls $B_{\epsilon/3}(f_k)$ cover $\mathcal{E}$. Each f_k is uniformly continuous, by [1, Continuity on a Closed Bounded Interval Is Uniform] when $a < b$, and because $|s-t| < \delta$ forces $s = t$ when $a = b$. So there is a $\delta_k > 0$ with $|f_k(s) - f_k(t)| < \epsilon/3$ whenever $|s-t| < \delta_k$. Put $\delta = \min_{k\leq N} \delta_k > 0$. Given $f \in \mathcal{E}$, choose k with $d(f,f_k) < \epsilon/3$. For $s,t \in [a,b]$ with $|s-t| < \delta$,

$$|f(s) - f(t)| \leq |f(s) - f_k(s)| + |f_k(s) - f_k(t)| + |f_k(t) - f(t)| < \epsilon$$

since each of the outer two terms is at most $d(f,f_k)$. So $\mathcal{E}$ is equicontinuous in the sense of definition 3.5.4.

($\Leftarrow$) Let $\mathcal{E}$ be closed, bounded and equicontinuous.

Step 1: a uniform bound. Being bounded, $\mathcal{E}$ lies in some ball $B_R(g)$ with $g \in C^0$. The number $\|g\|$ is finite because $g \in C^0 \subseteq C_b$, so every $f \in \mathcal{E}$ and every $x \in [a,b]$ satisfy $|f(x)| \leq |f(x) - g(x)| + |g(x)| < R + \|g\|$.

Step 2: the domain is compact. The interval $[a,b]$ is compact by corollary 3.1.4.

Step 3: every sequence in $\mathcal{E}$ has a subsequence converging to a point of $\mathcal{E}$. Let (f_n) be a sequence in $\mathcal{E}$. It is uniformly bounded by Step 1, and it is equicontinuous in the sense of definition 3.5.1 because each of its terms lies in the equicontinuous set $\mathcal{E}$. By Step 2, theorem 3.5.6 applies with $M = [a,b]$ and gives a subsequence (g_k) and a function $g\colon [a,b] \to \mathbb{R}$ with $g_k \rightrightarrows g$. The function g is continuous by theorem 3.3.3, so $g \in C^0$. Given $\epsilon > 0$, uniform convergence gives a K with $|g_k(x) - g(x)| < \epsilon$ for all $k \geq K$ and all $x \in [a,b]$, so $d(g_k,g) \leq \epsilon$ for $k \geq K$ and $g_k \to g$ in C^0. As $\mathcal{E}$ is closed, $g \in \mathcal{E}$ by corollary 1.2.10.

Step 4: total boundedness. Suppose $\mathcal{E}$ were not totally bounded. Then for some $\epsilon > 0$

no finite family of balls $B_\epsilon(h)$ with centres $h \in \mathcal{E}$ covers $\mathcal{E}$. Choose $h_1 \in \mathcal{E}$, and once $h_1, \ldots, h_{n-1}$ are chosen, choose $h_n \in \mathcal{E}$ outside $B_\epsilon(h_1) \cup \cdots \cup B_\epsilon(h_{n-1})$. Then $d(h_n, h_m) \geq \epsilon$ whenever $n \neq m$. By Step 3 some subsequence of (h_n) converges, so it is Cauchy by proposition 2.1.2, and two of its terms with distinct indices lie within ϵ of each other, which is impossible. So $\mathcal{E}$ is totally bounded.

Step 5: conclusion. The set $\mathcal{E}$ is closed and totally bounded in the complete space C^0, so it is compact by theorem 3.1.2, completing the proof.

3.6 Uniform Approximation in C^0

We now move onto a really interesting part of analysis: Given a continuous but nondifferentiable function f (so, any function in C^0), we we can "arbitrarily" approximate f with a smooth function g! Even better, this smooth function will be a polynomial!!

In order for this to be possible, it would mean that for any $f \in C^0$, there is a polynomial close to it. That is what we are about to prove:

Theorem 3.6.1: Weierstrass Approximation Theorem

The set of polynomials is dense in $C^0([a,b], \mathbb{R})$

Note that since the function is continuous on a compact set, it is uniformally continuous (a fact we will use in the 2nd proof).

Decomposing what density means for a moment: for each $f \in C^0$, and each $\epsilon > 0$, there is a polynomial function $p(x)$ such that for all $x \in [a,b]$

$$|f(x) - p(x)| < \epsilon$$

meaning that we can arbitrarily approximate $f(x)$!!

***Proof*:**
The key idea to building $p(x)$ from f is to sample values of f and recombine them with polynomial weights.

If $a = b$ then every function on $[a,b]$ is a constant polynomial, so assume $a < b$. The affine bijection $\varphi(x) = a + (b-a)x$ carries $[0,1]$ onto $[a,b]$, and composing a polynomial with φ or with φ^{-1} gives a polynomial. The map φ is Lipschitz with constant $b-a$, hence continuous by proposition 1.3.6, so $f \circ \varphi$ is continuous by proposition 1.3.3, and if a polynomial q satisfies $|f(\varphi(x)) - q(x)| < \epsilon$ for all $x \in [0,1]$, then the polynomial $q \circ \varphi^{-1}$ satisfies $|f(y) - q(\varphi^{-1}(y))| < \epsilon$ for all $y \in [a,b]$. So we may assume $[a,b] = [0,1]$. For $f \in C^0([0,1], \mathbb{R})$ and each $n \geq 1$, consider

$$p_n(x) = \sum_{k=0}^{n} \binom{n}{k} c_k x^k (1-x)^{n-k}$$

where $c_k = f\left(\frac{k}{n}\right)$ and $\binom{n}{k}$ is the binomial coefficient. This polynomial is called the *Bernstein polynomial*.

Step 1: three sums. For integers $0 \le k \le m$ and real x write $r_{m,k}(x) = \binom{m}{k} x^k (1-x)^{m-k}$, and read $\binom{m}{k} = 0$, and so $r_{m,k} = 0$, when $k < 0$ or $k > m$. Then $p_n = \sum_{k=0}^{n} f(k/n)\, r_{n,k}$, and $r_{m,k}(x) \ge 0$ for $x \in [0,1]$. First,

$$\sum_{k=0}^{m} r_{m,k}(x) = 1$$

for every $m \ge 0$, by induction on m. For $m = 0$ the sum is $r_{0,0}(x) = 1$. Pascal's rule $\binom{m+1}{k} = \binom{m}{k} + \binom{m}{k-1}$ holds for $1 \le k \le m$, as putting both fractions over $k!\,(m+1-k)!$ shows, and it holds at $k = 0$ and $k = m+1$ with the conventions above, both sides being 1. It gives $r_{m+1,k}(x) = (1-x)\, r_{m,k}(x) + x\, r_{m,k-1}(x)$, and summing over $0 \le k \le m+1$ gives $(1-x)+x = 1$ by the case m.

For $n \ge 1$ and $1 \le k \le n$, the computation $k\binom{n}{k} = \frac{n!}{(k-1)!\,(n-k)!} = n\binom{n-1}{k-1}$ gives $k\, r_{n,k}(x) = nx\, r_{n-1,k-1}(x)$. The term $k = 0$ vanishes, so the first identity at $m = n-1$ gives

$$\sum_{k=0}^{n} k\, r_{n,k}(x) = nx$$

For $n \ge 2$ and $2 \le k \le n$, applying the same computation twice gives $k(k-1)\, r_{n,k}(x) = n(n-1)x^2\, r_{n-2,k-2}(x)$. The terms $k = 0, 1$ vanish, so the first identity at $m = n-2$ gives

$$\sum_{k=0}^{n} k(k-1)\, r_{n,k}(x) = n(n-1)x^2$$

and this also holds for $n = 1$, where both sides are 0. Writing $(k-nx)^2 = k(k-1) + k - 2nxk + n^2x^2$ and combining the three sums,

$$\sum_{k=0}^{n} (k-nx)^2\, r_{n,k}(x) = n(n-1)x^2 + nx - 2n^2x^2 + n^2x^2 = nx(1-x)$$

and dividing by n^2 gives the variance identity

$$\sum_{k=0}^{n} \left(\frac{k}{n} - x\right)^2 r_{n,k}(x) = \frac{x(1-x)}{n}$$

Step 2: the uniform-continuity split. Let $\epsilon > 0$. The function f is uniformly continuous by [1, Continuity on a Closed Bounded Interval Is Uniform], so there is a $\delta > 0$ with $|f(s) - f(t)| < \epsilon/2$ whenever $s, t \in [0,1]$ and $|s - t| < \delta$. The number $M = \sup_{s \in [0,1]} |f(s)|$ is finite by [1, A Continuous Function on a Closed Bounded Interval Is Bounded]. Fix $n \ge 1$ and $x \in [0,1]$. The first identity of Step 1 gives $f(x) = \sum_{k=0}^{n} f(x)\, r_{n,k}(x)$, and $r_{n,k}(x) \ge 0$, so

$$|f(x) - p_n(x)| \le \sum_{k=0}^{n} \left| f(x) - f\left(\frac{k}{n}\right) \right| r_{n,k}(x)$$

Split the indices into $K_1 = \{k \mid |k/n - x| < \delta\}$ and $K_2 = \{k \mid |k/n - x| \ge \delta\}$. Over K_1 each difference is less than $\epsilon/2$, so that part of the sum is at most $\frac{\epsilon}{2} \sum_{k=0}^{n} r_{n,k}(x) = \frac{\epsilon}{2}$. Over K_2 each

difference is at most $|f(x)| + |f(k/n)| \le 2M$ and $1 \le (k/n - x)^2/\delta^2$, so by the variance identity that part is at most

$$\frac{2M}{\delta^2} \sum_{k=0}^{n} \left(\frac{k}{n} - x\right)^2 r_{n,k}(x) = \frac{2M\, x(1-x)}{n\delta^2} \le \frac{M}{2n\delta^2}$$

because $x(1-x) = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2 \le \frac{1}{4}$. Choose an integer $N \ge 1$ with $N > M/(\epsilon\delta^2)$. For $n \ge N$ the last bound is less than $\epsilon/2$, so $|f(x) - p_n(x)| < \epsilon$ for every $x \in [0,1]$. The integer N does not depend on x, so $p_n \rightrightarrows f$, and each p_n is a polynomial, as we sought to show.

***Proof Alternative proof*:**
If $a = b$ there is nothing to prove, since every function on $[a,b]$ is a constant polynomial, so assume $a < b$. The affine bijection $\varphi(x) = a + (b-a)x$ carries $[0,1]$ onto $[a,b]$, and its inverse $\varphi^{-1}(y) = (y-a)/(b-a)$ is affine too. Composing a polynomial with either gives a polynomial again, so a polynomial approximating $f \circ \varphi$ on $[0,1]$ yields, after composing with φ^{-1}, one approximating f on $[a,b]$. A homeomorphism would not serve, since it need not carry polynomials to polynomials. So we may assume $[a,b] = [0,1]$. Let $f \in C^0([0,1], \mathbb{R})$. What we will do is replace f by a function that is easier to approximate, and approximate that function with a polynomial.

Let $g(x) = f(x) - (mx + c)$, where

$$m = \frac{f(1) - f(0)}{1} = f(1) - f(0) \qquad \text{and} \qquad c = f(0)$$

Since we are adding a polynomial to $f(x)$, $g \in C^0$. Notice that

$$g(0) = f(0) - (f(1) - f(0))\, 0 - f(0) = 0 \qquad g(1) = f(1) - f(1) + f(0) - f(0) = 0$$

so $g(0) = 0 = g(1)$. This shows that given any f, it is equivalent to consider f in the "form" of g (the two differing by an affine function), so let $f(0) = 0 = f(1)$. In our construction, we will sometimes need the domain to extend beyond $[0,1]$, so let $f(x) = 0$ for $x \in \mathbb{R} - [0,1]$. Because $f(0) = f(1) = 0$, this extension is continuous on all of $\mathbb{R}$.

Let

$$\beta_n(t) = b_n(1 - t^2)^n \qquad -1 \le t \le 1$$

where the constant b_n is chosen so that $\int_{-1}^{1} \beta_n(t)dt = 1$. For $0 \le x \le 1$, set

$$P_n(x) = \int_{-1}^{1} f(x+t)\beta_n(t)dt$$

This is a weighted average of the value of f using the weight function β_n. We claim that P_n is a polynomial, and $P_n(x) \rightrightarrows f(x)$ as $n \to \infty$.

To check that P_n is a polynomial, we use a change of variables: $u = x + t$. Then for $0 \le x \le 1$ we have

$$P_n(x) = \int_{x-1}^{x+1} f(u)\beta_n(u-x)du = \int_{0}^{1} f(u)\beta_n(u-x)du$$

where the last integral is because $f = 0$ outside $[0,1]$. For $u, x \in [0,1]$ the difference $u - x$ lies in $[-1,1]$, so $\beta_n(u-x) = b_n(1-(u-x)^2)^n$, and expanding gives $\beta_n(u-x) = \sum_{j=0}^{2n} c_j(u)\, x^j$ with

each c_j a polynomial in u. Hence

$$P_n(x) = \sum_{j=0}^{2n} \left(\int_0^1 f(u)\, c_j(u)\, du \right) x^j$$

is a polynomial in x.

To check that $P_n \rightrightarrows f$ as $n \to \infty$, we need to estimate $\beta_n(t)$. We claim that if $\delta > 0$ then

$$\beta_n(t) \rightrightarrows 0 \text{ as } n \to \infty \text{ and } \delta \le |t| \le 1$$

The restriction to $|t| \ge \delta$ cannot be dropped: $(1-t^2)^n \le 1$ on $[-1,1]$ gives $\int_{-1}^1 (1-t^2)^n dt \le 2$, hence $\beta_n(0) = b_n \ge \frac{1}{2}$ for every n.

The normalization reads $b_n \int_{-1}^1 (1-t^2)^n dt = 1$, so a lower bound on that integral is an upper bound on b_n. The integrand is even and nonnegative. Bernoulli's inequality gives $(1-t^2)^n \ge 1 - nt^2$ for $0 \le t \le 1$, and $1 - nt^2 \ge 0$ on $\left[0, n^{-1/2}\right]$, so

$$\int_{-1}^1 (1-t^2)^n dt \ge 2 \int_0^{1/\sqrt{n}} \left(1 - nt^2\right) dt = 2\left(\frac{1}{\sqrt{n}} - \frac{n}{3n^{3/2}} \right) = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}}$$

and therefore $b_n < \sqrt{n}$. Take $0 < \delta < 1$. (For $\delta \ge 1$ the band $\delta \le |t| \le 1$ carries at most $t = \pm 1$, where β_n vanishes.) On that band $0 \le 1 - t^2 \le 1 - \delta^2$, and raising two numbers of $[0,1]$ to the nth power preserves their order, so

$$0 \le \beta_n(t) = b_n (1-t^2)^n \le \sqrt{n} \left(1 - \delta^2\right)^n$$

The bound on the right does not depend on t. Its logarithm is $\frac{1}{2}\log n + n \log(1-\delta^2)$, where the second term is linear in n with a negative coefficient and the first grows like $\log n$, so the logarithm tends to $-\infty$ and $\sqrt{n}(1-\delta^2)^n \to 0$. Hence $\beta_n \rightrightarrows 0$ on $\delta \le |t| \le 1$, which is the claim.

Let $\epsilon > 0$. The extended f is continuous on $[-1,2]$, so by [1, Continuity on a Closed Bounded Interval Is Uniform] there is a $\delta > 0$ such that $|f(s) - f(s')| < \frac{\epsilon}{2}$ whenever $s, s' \in [-1,2]$ and $|s - s'| < \delta$. Shrinking δ preserves this property, so we may take $\delta < 1$. For $x \in [0,1]$ and $|t| < \delta$ both x and $x + t$ lie in $[-1,2]$, so $|f(x+t) - f(x)| < \frac{\epsilon}{2}$. Put $M = \sup_{s \in [0,1]} |f(s)|$, which is finite by [1, A Continuous Function on a Closed Bounded Interval Is Bounded]. Since f vanishes outside $[0,1]$, every real s satisfies $|f(s)| \le M$. Fix $x \in [0,1]$ and $n \ge 1$. Since β_n has integral 1 on $[-1,1]$, writing $f(x) = \int_{-1}^1 f(x)\beta_n(t)dt$ gives

$$\begin{aligned}
|P_n(x) - f(x)| &= \left| \int_{-1}^1 \left(f(x+t) - f(x)\right) \beta_n(t) dt \right| \\
&\le \int_{-1}^1 |f(x+t) - f(x)| \beta_n(t) dt \\
&= \int_{|t|<\delta} |f(x+t) - f(x)| \beta_n(t) dt + \int_{\delta \le |t| \le 1} |f(x+t) - f(x)| \beta_n(t) dt \\
&\le \frac{\epsilon}{2} + 2M \int_{\delta \le |t| \le 1} \beta_n(t) dt
\end{aligned}$$

The first integral is at most $\frac{\epsilon}{2} \int_{-1}^1 \beta_n(t)dt = \frac{\epsilon}{2}$, because $\beta_n \ge 0$ and $|f(x+t) - f(x)| < \frac{\epsilon}{2}$ for $|t| < \delta$. The second is at most $2M \int_{\delta \le |t| \le 1} \beta_n(t)dt$, because $|f(x+t) - f(x)| \le |f(x+t)| + |f(x)| \le 2M$.

By the claim, $\sqrt{n}(1-\delta^2)^n \to 0$, so there is an N with $\sqrt{n}(1-\delta^2)^n < \frac{\epsilon}{8M+1}$ for every $n \geq N$. For such n the bound $0 \leq \beta_n(t) \leq \sqrt{n}(1-\delta^2)^n$ on the band $\delta \leq |t| \leq 1$, whose total length is $2(1-\delta) < 2$, gives

$$2M \int_{\delta \leq |t| \leq 1} \beta_n(t)dt \leq \frac{4M\epsilon}{8M+1} < \frac{\epsilon}{2}$$

and therefore $|P_n(x) - f(x)| < \epsilon$. The integer N depends on ϵ and δ but not on x, so $P_n \rightrightarrows f$ on $[0,1]$.

Undoing the two reductions at the start, by adding back the affine function $mx+c$ and then composing with φ^{-1}, turns the P_n into polynomials that converge uniformly on $[a,b]$ to the original function, as we sought to show.

With this, we can prove the general Stone-Weierstrass theroem on compact metric spaces: that is, change $[0,1]$ into a general comapct metric space.

To do this, we will first define an algebra on the functions. Let $A \subseteq C^0(M,\mathbb{R})$ be nonempty and such that if $f, g \in A$, $c \in \mathbb{R}$, then

$$f+g \in A \qquad cf \in A \qquad f \cdot g \in A$$

where each operation is defined pointwise (ex. $(f+g)(x) = f(x) + g(x)$). Such an A is a *function algebra* in $C^0(M,\mathbb{R})$. For examples, polynoimals form an algebra.

The function algbera *vanishes at point* p if $f(p) = 0$ for all $f \in A$. FOr example, A with polynoimals with zero constatn term vanishes at $x = 0$.

The function algebra *separates points* if for each pair of distinct points $p_1, p_2 \in M$, there is a function $f \in A$ such that $f(p_1) \neq f(p_2)$. For example, the real trigonometric polynomials $a_0 + \sum_{k=1}^{n}(a_k \cos kx + b_k \sin kx)$ form a function algebra, closed under products by the product-to-sum formulas, and it vanishes nowhere because it contains the constant 1. It separates the points of $[0, 2\pi)$, since the point $(\cos x, \sin x)$ of the unit circle determines $x \in [0, 2\pi)$, but $[0, 2\pi)$ is not compact. On the compact interval $[0, 2\pi]$ it does not separate points: every trigonometric polynomial takes the same value at 0 and at 2π.

For compact M every $f \in C^0(M,\mathbb{R})$ is bounded. When M is nonempty, the sets $U_x = f^{-1}\big(B_1(f(x))\big)$ for $x \in M$ are open by proposition 1.2.3 and theorem 1.3.2(iii), and $x \in U_x$, so compactness gives $x_1, \ldots, x_n \in M$ with $M = U_{x_1} \cup \cdots \cup U_{x_n}$. Every $t \in M$ then satisfies $|f(t)| < 1 + \max_{i \leq n} |f(x_i)|$. So for nonempty compact M the set $C^0(M,\mathbb{R})$ is the space $C_b(M)$ of theorem 2.2.3, and density in $C^0(M,\mathbb{R})$ refers to the metric $d_\infty(f,g) = \sup_{x \in M} |f(x) - g(x)|$. Write $\|f\| = \sup_{x \in M} |f(x)|$.

Lemma 3.6.2: Two-Point Interpolation in a Function Algebra

Let A be a function algebra in $C^0(M,\mathbb{R})$ that vanishes nowhere and separates points. For distinct $p, q \in M$ and any $c_1, c_2 \in \mathbb{R}$ there is an $f \in A$ with $f(p) = c_1$ and $f(q) = c_2$.

***Proof*:**
Since A separates points there is a $g \in A$ with $g(p) \neq g(q)$, and since A vanishes nowhere there are $h, k \in A$ with $h(p) \neq 0$ and $k(q) \neq 0$. Put

$$u = gk - g(p)\,k \qquad v = gh - g(q)\,h$$

The products gk and gh lie in A, as do the scalar multiples $-g(p)\,k$ and $-g(q)\,h$, and therefore

so do the sums u and v. Evaluating at p and q,

$$u(p) = 0 \qquad u(q) = \big(g(q) - g(p)\big)k(q) \neq 0 \qquad v(p) = \big(g(p) - g(q)\big)h(p) \neq 0 \qquad v(q) = 0$$

So the function

$$f = \frac{c_1}{v(p)}\, v + \frac{c_2}{u(q)}\, u$$

lies in A and satisfies $f(p) = c_1$ and $f(q) = c_2$, as claimed.

Lemma 3.6.3: The Closure of a Function Algebra Is Closed Under Max and Min

Let M be a nonempty compact metric space, let A be a function algebra in $C^0(M, \mathbb{R})$, and let $\overline{A}$ be the closure of A in $C^0(M, \mathbb{R})$ for the metric d_∞.

(i) $\overline{A}$ is a function algebra.

(ii) If $f \in \overline{A}$ then $|f| \in \overline{A}$.

(iii) If $f, g \in \overline{A}$ then the pointwise maximum $\max(f, g)$ and minimum $\min(f, g)$ lie in $\overline{A}$, and so do the pointwise maximum and minimum of any finitely many members of $\overline{A}$.

***Proof*:**

By theorem 2.2.3 the set $C^0(M, \mathbb{R}) = C_b(M)$ is closed in $\big(B(M), d_\infty\big)$, so by corollary 1.2.10 a bounded function on M that is the d_∞-limit of a sequence in $C^0(M, \mathbb{R})$ lies in $C^0(M, \mathbb{R})$. The closure $\overline{A}$ is a closed subset of $C^0(M, \mathbb{R})$, being an intersection of closed sets as recorded after definition 1.2.7, so $\overline{A}$ is its own closure by corollary 1.2.10.

(i) The set $\overline{A}$ contains A and so is nonempty. Let $f, g \in \overline{A}$ and $c \in \mathbb{R}$. By theorem 1.2.9(iii) there are $f_n, g_n \in A$ with $d_\infty(f_n, f) \to 0$ and $d_\infty(g_n, g) \to 0$, and $f_n + g_n$, cf_n and $f_n g_n$ lie in A. The functions $f + g$, cf and fg are bounded, by $\|f\| + \|g\|$, $|c|\,\|f\|$ and $\|f\|\,\|g\|$, and at every $x \in M$

$$\begin{aligned} |(f_n + g_n)(x) - (f + g)(x)| &\leq d_\infty(f_n, f) + d_\infty(g_n, g) \\ |cf_n(x) - cf(x)| &\leq |c|\, d_\infty(f_n, f) \\ |f_n(x)g_n(x) - f(x)g(x)| &\leq |f_n(x)|\,|g_n(x) - g(x)| + |g(x)|\,|f_n(x) - f(x)| \\ &\leq \big(\|f\| + d_\infty(f_n, f)\big)d_\infty(g_n, g) + \|g\|\, d_\infty(f_n, f) \end{aligned}$$

The right-hand sides do not depend on x and tend to 0, so $f_n + g_n \to f + g$, $cf_n \to cf$ and $f_n g_n \to fg$ in d_∞. By the first paragraph the three limits lie in $C^0(M, \mathbb{R})$, and theorem 1.2.9(iii) puts them in $\overline{A}$.

(ii) The absolute value satisfies $\big||y| - |y'|\big| \leq |y - y'|$, so it is continuous on $\mathbb{R}$ by proposition 1.3.6. Let $f \in \overline{A}$. Then $|f|$ is continuous by proposition 1.3.3, so $|f| \in C^0(M, \mathbb{R})$. Put $r = \|f\|$. If $r = 0$ then $|f| = f \in \overline{A}$, so assume $r > 0$, and let $\epsilon > 0$. The function $y \mapsto |y|$ lies in $C^0([-r, r], \mathbb{R})$, so theorem 3.6.1, read through theorem 1.2.9(ii), gives a polynomial Q with $\big|Q(y) - |y|\big| < \epsilon/2$ for every $y \in [-r, r]$. In particular $|Q(0)| < \epsilon/2$, so $P(y) = Q(y) - Q(0)$ is a polynomial with $P(0) = 0$ and $\big|P(y) - |y|\big| < \epsilon$ for every $y \in [-r, r]$. Write $P(y) = \sum_{i=1}^m c_i y^i$. By (i) every power f^i lies in $\overline{A}$, and then so does $P \circ f = \sum_{i=1}^m c_i f^i$, the empty

sum being the zero function $0 \cdot f \in \overline{A}$. Every value $f(x)$ lies in $[-r, r]$, so $d_\infty(P \circ f, |f|) \leq \epsilon$. Since ϵ was arbitrary, every ball about $|f|$ in $C^0(M, \mathbb{R})$ meets $\overline{A}$, and theorem 1.2.9(ii) puts $|f|$ in the closure of $\overline{A}$, which is $\overline{A}$.

(iii) For real numbers s and t, checking the cases $s \geq t$ and $s < t$ gives

$$\max(s,t) = \tfrac{1}{2}\big(s + t + |s - t|\big) \qquad \min(s,t) = \tfrac{1}{2}\big(s + t - |s - t|\big)$$

So $\max(f,g) = \frac{1}{2}(f + g + |f - g|)$ and $\min(f,g) = \frac{1}{2}(f + g - |f - g|)$, and these lie in $\overline{A}$ by (i) and (ii). For members $h_1, \ldots, h_n$ of $\overline{A}$, induction on n using $\max(h_1, \ldots, h_n) = \max\big(\max(h_1, \ldots, h_{n-1}), h_n\big)$, and the same identity for min, completes the proof.

Theorem 3.6.4: Stone-Weierstrass Theorem

If M is a compact metric space and A is a function-algebra in $C^0(M, \mathbb{R})$ that vanishes nowhere and separates points, then A is dense in $C^0(M, \mathbb{R})$

Taking $M = [a, b]$ and A the polynomials recovers theorem 3.6.1, since the polynomials contain the constant function 1 and the identity function, and so vanish nowhere and separate points. The general theorem rests on that special case only through lemma 3.6.3(ii), which applies it to the absolute value on a closed interval.

***Proof*:**

If M is empty then $C^0(M, \mathbb{R})$ has exactly one element, the empty function, and A contains it because A is nonempty. So assume $M \neq \emptyset$, write d for the metric of M, and let $\overline{A}$ be the closure of A in $C^0(M, \mathbb{R})$ for the metric d_∞. By definition 1.2.11 the claim is $\overline{A} = C^0(M, \mathbb{R})$. Fix $f \in C^0(M, \mathbb{R})$ and $\epsilon > 0$.

Step 1: a member of $\overline{A}$ that agrees with f at a given point and exceeds $f - \epsilon$. Fix $x \in M$. For each $y \in M$ choose $h_y \in A$ with $h_y(x) = f(x)$ and $h_y(y) = f(y)$. For $y \neq x$ such an h_y exists by lemma 3.6.2. For $y = x$, the algebra A vanishes nowhere, so some $k \in A$ has $k(x) \neq 0$, and $h_x = \frac{f(x)}{k(x)}\, k$ serves. Continuity of h_y and of f at y gives a $\rho_y > 0$ with $|h_y(t) - h_y(y)| < \epsilon/2$ and $|f(t) - f(y)| < \epsilon/2$ whenever $d(t, y) < \rho_y$, and for such t

$$h_y(t) - f(t) > \big(h_y(y) - \tfrac{\epsilon}{2}\big) - \big(f(y) + \tfrac{\epsilon}{2}\big) = -\epsilon$$

The balls $B_{\rho_y}(y)$ for $y \in M$ are open by proposition 1.2.3 and cover M, so compactness gives $y_1, \ldots, y_n \in M$ with $M = \bigcup_{j \leq n} B_{\rho_{y_j}}(y_j)$. Put $g_x = \max(h_{y_1}, \ldots, h_{y_n})$, which lies in $\overline{A}$ by lemma 3.6.3(iii). Each $t \in M$ lies in some $B_{\rho_{y_j}}(y_j)$, so $g_x(t) \geq h_{y_j}(t) > f(t) - \epsilon$. Also $g_x(x) = f(x)$, because every $h_{y_j}(x) = f(x)$.

Step 2: a member of $\overline{A}$ within ϵ of f. For each $x \in M$, continuity of g_x and of f at x gives a $\sigma_x > 0$ with $|g_x(t) - g_x(x)| < \epsilon/2$ and $|f(t) - f(x)| < \epsilon/2$ whenever $d(t, x) < \sigma_x$, and for such t, using $g_x(x) = f(x)$,

$$g_x(t) - f(t) < \big(g_x(x) + \tfrac{\epsilon}{2}\big) - \big(f(x) - \tfrac{\epsilon}{2}\big) = \epsilon$$

The balls $B_{\sigma_x}(x)$ for $x \in M$ are open and cover M, so compactness gives $x_1, \ldots, x_m \in M$ with $M = \bigcup_{i \leq m} B_{\sigma_{x_i}}(x_i)$. Put $h = \min(g_{x_1}, \ldots, g_{x_m})$, which lies in $\overline{A}$ by lemma 3.6.3(iii). For every

$t \in M$, each $g_{x_i}(t)$ exceeds $f(t) - \epsilon$ by Step 1, so $h(t) > f(t) - \epsilon$. Choosing i with $d(t, x_i) < \sigma_{x_i}$ gives $h(t) \leq g_{x_i}(t) < f(t) + \epsilon$. Hence $|h(t) - f(t)| < \epsilon$ for every $t \in M$, and $d_\infty(h, f) \leq \epsilon$.

Step 3: conclusion. Since $\epsilon > 0$ was arbitrary, every ball about f in $C^0(M, \mathbb{R})$ meets $\overline{A}$, so theorem 1.2.9(ii) puts f in the closure of $\overline{A}$. The set $\overline{A}$ is closed, so that closure is $\overline{A}$ itself by corollary 1.2.10. Since f was arbitrary, $\overline{A} = C^0(M, \mathbb{R})$ and A is dense in $C^0(M, \mathbb{R})$, as we sought to show.

3.7 Nowhere Differentiable Continuous Functions

Theorem 3.7.1: Nowhere Differentiable Continuous Functions

There exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ that has a derivative at no point whatsoever

***Proof*:**
Let the letters j, k, m, n denote integers. Start with the sawtooth function $\sigma_0 : \mathbb{R} \to \mathbb{R}$ defined as

$$\sigma_0(x) = \begin{cases} x - 2n & \text{if } 2n \leq x \leq 2n + 1 \\ 2n + 2 - x & \text{if } 2n + 1 \leq x \leq 2n + 2 \end{cases}$$

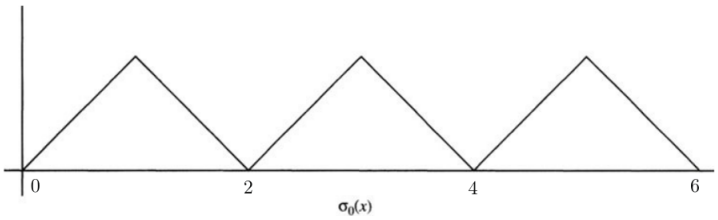


Figure 3.3: σ_0

For $2n \leq x \leq 2n + 1$ every even integer $2j$ satisfies $|x - 2j| \geq x - 2n$: for $j \leq n$ because $x - 2j \geq x - 2n$, and for $j > n$ because $2j - x \geq 2n + 2 - x \geq 1 \geq x - 2n$. For $2n + 1 \leq x \leq 2n + 2$ every even integer $2j$ satisfies $|x - 2j| \geq 2n + 2 - x$: for $j > n$ because $2j - x \geq 2n + 2 - x$, and for $j \leq n$ because $x - 2j \geq x - 2n \geq 1 \geq 2n + 2 - x$. So $\sigma_0(x)$ is the distance $d(x, 2\mathbb{Z})$ from x to the set of even integers, in the sense of definition 1.3.9, and in particular the two formulas agree where the intervals meet. Three properties follow. First, $\sigma_0(x + 2) = \sigma_0(x)$, because translation by 2 maps the even integers onto themselves, and $0 \leq \sigma_0(x) \leq 1$ for every x, by the two formulas. Second, $|\sigma_0(s) - \sigma_0(t)| \leq |s - t|$ for all $s, t \in \mathbb{R}$, by proposition 1.3.10. Third, on each interval $[j, j + 1]$, σ_0 is affine with slope 1 if j is even and slope -1 if j is odd.

For $k \geq 0$ define

$$\sigma_k(x) = \left(\frac{3}{4}\right)^k \sigma_0(4^k x)$$

so that $0 \leq \sigma_k \leq (3/4)^k$ and, by the second property,

$$|\sigma_k(s) - \sigma_k(t)| \leq \left(\frac{3}{4}\right)^k |4^k s - 4^k t| = 3^k |s - t| \qquad \text{for all } s, t \in \mathbb{R}$$

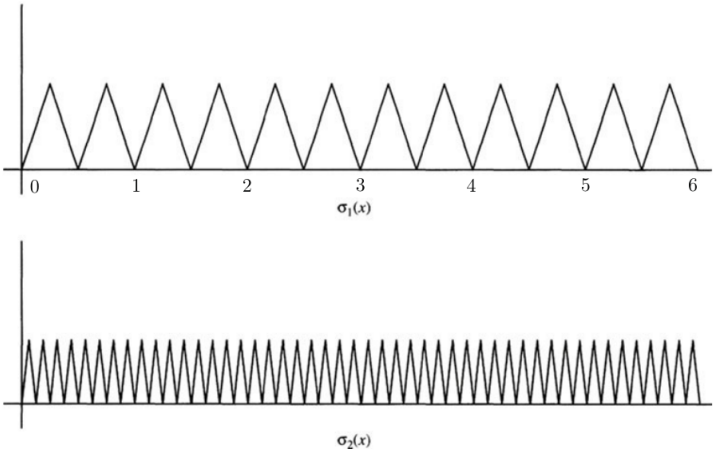


Figure 3.4: Two stages of the iteration

Step 1: the sum is continuous. For each x the terms of $\sum_{k\geq 0} \sigma_k(x)$ lie in $[0, (3/4)^k]$, and $\sum_{k>0}(3/4)^k = 4$ by [1, The Geometric Series], so the series converges by [1, The Comparison Test]. Put

$$f(x) = \sum_{k=0}^{\infty} \sigma_k(x) \qquad F_m(x) = \sum_{k=0}^{m} \sigma_k(x)$$

For every x, $0 \leq f(x) - F_m(x) = \sum_{k>m} \sigma_k(x) \leq \sum_{k>m}(3/4)^k = 4(3/4)^{m+1}$, a bound that does not depend on x and tends to 0, so $F_m \rightrightarrows f$ on $\mathbb{R}$ in the sense of definition 3.3.2. The estimate for σ_k and the triangle inequality give $|F_m(s) - F_m(t)| \leq \left(\sum_{k=0}^{m} 3^k\right)|s - t|$, so each F_m is Lipschitz and hence continuous by proposition 1.3.6. Then f is continuous at every point by theorem 3.3.3. Its graph is sketched in fig. 3.5.

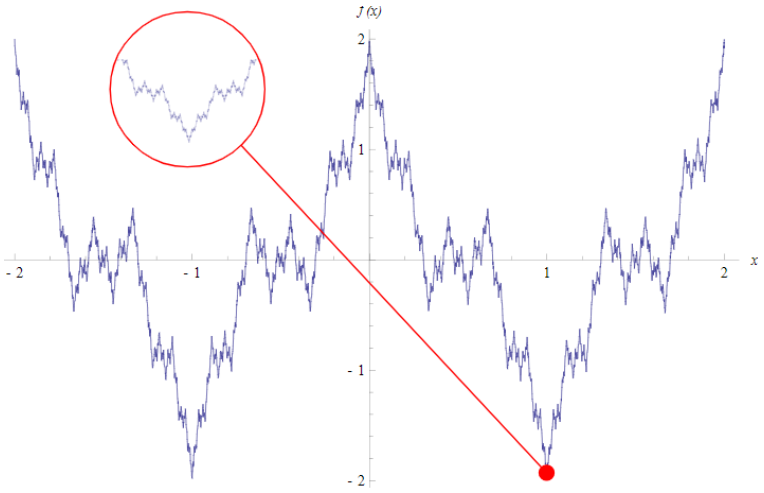


Figure 3.5: A nowhere differentiable continuous function

Step 2: no derivative at a given point. Fix $x \in \mathbb{R}$ and an integer $m \geq 0$. Let j be the integer with $j \leq 4^m x < j+1$. Put $\delta_m = \frac{1}{2}4^{-m}$ if $4^m x \leq j+\frac{1}{2}$, and $\delta_m = -\frac{1}{2}4^{-m}$ otherwise. Then $4^m|\delta_m| = \frac{1}{2}$, and both $4^m x$ and $4^m(x+\delta_m)$ lie in $[j, j+1]$: in the first case $j \leq 4^m x < 4^m x + \frac{1}{2} \leq j+1$, and in the second $j < 4^m x - \frac{1}{2} < 4^m x < j+1$. For each $k \geq 0$ put

$$\gamma_k = \frac{\sigma_k(x+\delta_m) - \sigma_k(x)}{\delta_m}$$

($k > m$) The number $4^k\delta_m = \pm\frac{1}{2}4^{k-m} = \pm 2 \cdot 4^{k-m-1}$ is an even integer, so the first property gives $\sigma_0(4^k x + 4^k\delta_m) = \sigma_0(4^k x)$, and $\gamma_k = 0$.

($k = m$) By the third property σ_0 is affine with slope ± 1 on $[j, j+1]$, which contains $4^m x$ and $4^m x + 4^m\delta_m$, so $|\sigma_0(4^m x + 4^m\delta_m) - \sigma_0(4^m x)| = 4^m|\delta_m|$ and

$$|\gamma_m| = \left(\frac{3}{4}\right)^m \frac{4^m|\delta_m|}{|\delta_m|} = 3^m$$



Figure 3.6: Demonstration of monotonicity

$(k < m)$ The estimate for σ_k gives $|\gamma_k| \leq 3^k$.

The terms with $k > m$ vanish, so $F_N(x + \delta_m) - F_N(x) = \delta_m \sum_{k=0}^{m} \gamma_k$ for every $N \geq m$, and letting $N \to \infty$,

$$\left| \frac{f(x+\delta_m) - f(x)}{\delta_m} \right| = \left| \sum_{k=0}^{m} \gamma_k \right| \geq |\gamma_m| - \sum_{k=0}^{m-1} |\gamma_k| \geq 3^m - \frac{3^m - 1}{2} = \frac{3^m + 1}{2}$$

using $\sum_{k=0}^{m-1} 3^k = (3^m - 1)/2$. Each δ_m is nonzero, and $\delta_m \to 0$ as $m \to \infty$. If f were differentiable at x with derivative L, then [1, The Sequential Criterion for Functional Limits](1), applied to the difference quotient of f at x along the points $x + \delta_m \to x$, would give $(f(x+\delta_m) - f(x))/\delta_m \to L$. A convergent real sequence is Cauchy by proposition 2.1.2 and hence bounded by proposition 2.1.3, while these quotients have absolute value at least $(3^m + 1)/2$, which is unbounded in m. So f is differentiable at no point, as we sought to show.

3.8 Baire Theory

The function constructed above is not exceptional. In $C^0([0,1], \mathbb{R})$ the nowhere differentiable functions form a residual set, meaning that their complement is a countable union of nowhere dense sets [5, Chapter 5, Exercise 42], stated there for complex scalars and with a hint that treats real functions, so in the sense of Baire category almost every continuous function is nowhere differentiable. A probabilistic counterpart also holds: the sample paths of the Wiener process are almost surely nowhere differentiable [5, Section 10.6].

A set D is dense in M if $\overline{D} = M$ (definition 1.2.11). Equivalently, D is dense in M if $W \cap D \neq \emptyset$ for every nonempty open $W \subseteq M$: by theorem 1.2.9(ii) density says that every ball meets D, every ball is a nonempty open set by proposition 1.2.3, and every nonempty open set contains a ball by definition 1.2.2. The intersection of two dense sets need not be dense: $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are both dense in $\mathbb{R}$ ([1, $\mathbb{Q}$ Is Dense in $\mathbb{R}$] and [1, The Irrationals Are Dense in $\mathbb{R}$]), and they are disjoint.

However, if U, V are *open* dense sets in M, then $U \cap V$ is also open and dense in M. It is open by theorem 1.2.4(iii). For density, let $W \subseteq M$ be a nonempty open set. Then $U \cap W$ is open, and it is nonempty because U is dense, so by density of V the set $V \cap (U \cap W)$ is nonempty. Rearranging, $(U \cap V) \cap W = V \cap (U \cap W) \neq \emptyset$, so $U \cap V$ meets every nonempty open set and is dense.

3.9 Cardinality quick

Proposition 3.9.1: C^0 Has the Cardinality of $\mathbb{R}$

For real numbers $a < b$ there is a bijection between $C^0([a,b], \mathbb{R})$ and $\mathbb{R}$.

***Proof*:**
Write C^0 for $C^0([a,b], \mathbb{R})$ and $\mathbb{N} = \{1, 2, 3, \dots\}$. The constant functions give an injection $\mathbb{R} \to C^0$.

Step 1: an injection $C^0 \to \mathbb{R}^{\mathbb{N}}$. Let $D = \mathbb{Q} \cap [a,b]$. Given $x \in [a,b]$ and $\epsilon > 0$, the open interval with endpoints $\max(a, x-\epsilon)$ and $\min(b, x+\epsilon)$ is nonempty because $a < b$, so it contains a rational number [1, $\mathbb{Q}$ Is Dense in $\mathbb{R}$], which lies in D within ϵ of x. So every ball about a point of $[a,b]$ meets D, and theorem 1.2.9, with (ii) implying (iii), makes every $x \in [a,b]$ the limit of a sequence in D. The set D is countable, being a subset of the countable set $\mathbb{Q}$ [5, Corollary 0.11], and it is nonempty, so there is a surjection $n \mapsto q_n$ from $\mathbb{N}$ onto D [5, Proposition 0.6]. If $f, g \in C^0$ satisfy $f(q_n) = g(q_n)$ for every n, then $f = g$ on D, and for each $x \in [a,b]$ a sequence (d_i) in D converging to x gives $f(x) = \lim f(d_i) = \lim g(d_i) = g(x)$ by theorem 1.3.2(ii). So $f \mapsto (f(q_n))_{n \geq 1}$ is an injection $C^0 \to \mathbb{R}^{\mathbb{N}}$.

Step 2: an injection $\mathbb{R}^{\mathbb{N}} \to \mathbb{R}$. Let $\beta \colon \mathbb{R} \to \mathcal{P}(\mathbb{N})$ be a bijection onto the set of subsets of $\mathbb{N}$ [5, Proposition 0.12], and let $\tau \colon \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ be a bijection, which exists because $\mathbb{N} \times \mathbb{N}$ is countable [5, Proposition 0.10(a)] and infinite, hence of the cardinality of $\mathbb{N}$ [5, Proposition 0.10(c)]. Send a sequence (x_n) to the subset $\{\tau(n,m) \mid n \in \mathbb{N} \text{ and } m \in \beta(x_n)\}$ of $\mathbb{N}$. Because τ is a bijection, this subset determines every set $\beta(x_n)$, and hence every x_n, so the map $\mathbb{R}^{\mathbb{N}} \to \mathcal{P}(\mathbb{N})$ is injective. Composing with β^{-1} gives an injection $\mathbb{R}^{\mathbb{N}} \to \mathbb{R}$.

Step 3: conclusion. Steps 1 and 2 give an injection $C^0 \to \mathbb{R}$, the constants give an injection $\mathbb{R} \to C^0$, and the Schröder-Bernstein theorem [5, Theorem 0.8] turns the two into a bijection, completing the proof.

4

What Next?

The theorems of this book are tools, and each one is sharpest in a subject that lies just past its statement.

Dropping the metric. Completeness and the contraction principle need a distance, but compactness, connectedness and continuity do not: they can be stated using open sets alone, and in that generality they apply to spaces no metric describes. *Topology* [4] makes that move and is the point-set foundation of most of the series. Munkres' *Topology* is the standard companion.

Adding linear structure. The space $C(K)$ studied here is a normed vector space that happens to be complete, and that combination has a name: a Banach space. Taking it as the primitive object gives the Hahn-Banach theorem, the open mapping and closed graph theorems (both proved from Baire category), and the spectral theory of operators. That is *Functional Analysis* [6]. Rudin's *Functional Analysis* and Brezis' *Functional Analysis, Sobolev Spaces and Partial Differential Equations* are the two standard routes.

Fixed points as solutions. The contraction principle is the engine behind the existence and uniqueness theorem for differential equations: Picard iteration is a contraction on a space of continuous functions, and its fixed point is the solution. *Ordinary Differential Equations* [3, Picard-Lindelöf Existence and Uniqueness] carries this out. Hartman's *Ordinary Differential Equations* is the thorough reference.

Measuring instead of approximating. $C(K)$ is complete under the sup norm, but not under the integral norms that analysis actually wants, and repairing that requires rebuilding integration from measure theory. *Real Analysis* [7] does so; Folland's *Real Analysis* is the standard text.

Metric spaces as geometry. This book treats a metric as a tool for limits. Treating it instead as geometry, and asking which spaces look like non-positively curved manifolds at large scale, opens metric geometry and geometric group theory. Burago, Burago and Ivanov's *A Course in Metric Geometry* and Bridson and Haefliger's *Metric Spaces of Non-Positive Curvature* are the entry points.

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